Nuclear Quantum Effects on the Phonons of Solid Hydrogen



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This work builds on Daria Kondratyeva's thesis, *A Study of Anharmonic Phonons and Thermal Conductivity*, which investigated anharmonic phonons in a one-dimensional crystal model. Here, we extend this line of investigation to solid para-hydrogen. I am grateful for her work, which provided a useful foundation for the present study.

Summary

In this thesis, we show how nuclear quantum effects (NQEs) can be incorporated into phonon calculations for solid para-hydrogen. We demonstrate how the isomorphism between the partition function of a quantum particle and a classical "ring polymer" leads to path integral molecular dynamics (PIMD). This method is used to obtain the centroid potential of mean force (CPMF), which then serves as an effective interaction in lattice-dynamical calculations of harmonic phonon properties. We find that the CPMF gives substantially better agreement with experimental phonon dispersion relations and heat capacities than the classical Silvera-Goldman potential. We then use the CPMF to calculate the lattice thermal conductivity. Because the relaxation-time approximation (RTA) fails at low temperatures, we construct and diagonalise the full phonon scattering matrix, which leads to an alternative quasi-particle picture of heat transport mediated by "relaxons". Thermal conductivity calculated in this relaxon basis gives much better agreement with experiment.

Contents

List of Figures **vii**

1 Introduction **1**

2 Theory **5**
2.1 Path Integral Molecular Dynamics 6
2.2 Centroid Potential of Mean Force 10
2.3 Harmonic Phonons 14
2.4 Boltzmann Transport Equation 17
2.5 Linearisation 20
2.6 Relaxation Time Approximation 22
2.7 Relaxons 24

3 Results and Discussion **31**
3.1 Pair Potential 32
3.2 Radial Distribution Function 37
3.3 Dispersion Relation 40
3.4 Heat Capacity 43
3.5 Thermal Conductivity 46

4 Conclusion **51**

Bibliography **53**

Appendices

A The Ring Polymer **58**

B Centroid Potential of Mean Force **61**

C Crystal Momentum **68**

D Phonons **72**

E Scattering Matrix **86**

F Boltzmann Transport Equation **94**

G Second Law of Thermodynamics **100**

H Symmetry of Thermal Conductivity Tensor **109**

List of Figures

2.1 A quantum particle represented by a classical ring polymer of 6 beads, the neighbouring beads are connected by harmonic springs of stiffness $m\omega_n^2$. [10] 8
2.2 The interactions between two ring polymers, with 6 beads each. Beads with the same label (k) have the same colour, interaction due to the physical potential $v(r)$ is illustrated by dashed lines. [10] 13

3.1 Comparison between the classical pair potential and the centroid potentials of mean force at two different temperatures. 36
3.2 Radial distribution functions calculated using the bare classical potential (classical) and the centroid potential of mean force (CMD), compared with the quantum mechanically exact PIMD centroid radial distribution function. 39
3.3 Phonon dispersions calculated using the bare classical potential and the centroid potential of mean force, compared with neutron-scattering measurements. [37] 41
3.4 Hexagonal-prism Brillouin zone of the hcp structure and the high-symmetry path used for the dispersion calculation. [53] 42
3.5 Constant-volume heat capacity calculated using the classical potential and the centroid potential of mean force, compared with experiment. [38] 44
3.6 Comparison of the present thermal conductivity calculations with the results of Hill *et al.* [39] and Bohn *et al.* [40] for solid p-H_2 between 5 and 11 K. It is also worth noting that the early thermal conductivity measurements of solid p-H_2 by Hill and Schneidmesser were carried out at the Clarendon Laboratory, University of Oxford. This provides a pleasing local historical connection between the experimental data used here and the present thesis. 47

H.1 The character table for D_{6h} was taken from the Oxford Honour School of Chemistry character tables. [61] 111

Take care of the sense, and the sounds will take care of themselves.

— Lewis Carroll, *Alice's Adventures in Wonderland*

Lewis Carroll, the pen name of Charles Lutwidge Dodgson, was first a student and later a mathematics tutor at Christ Church.

1 Introduction

Solid hydrogen (H_2) is one of the simplest molecular crystals, yet due to the combination of light mass and low temperature, it exhibits strong quantum effects. At $T = 0\,\mathrm{K}$, the p-H_2 molecules are not precisely located at their lattice sites due to the large zero-point motion from Heisenberg's uncertainty principle

$$\Delta q \Delta p \geq \frac{\hbar}{2}. \tag{1.1}$$

The large zero-point motion renders the classical point-like particle picture inadequate. When the atomic nuclei in a physical system cannot be accurately modelled as classical point particles governed by Newtonian mechanics, we refer to the resulting phenomena as nuclear quantum effects (NQEs). NQEs modify the effective intermolecular interaction and, therefore, could have an impact on both harmonic and anharmonic interatomic force constants (IFCs). The harmonic IFCs determine phonon normal modes and phonon frequencies, while the anharmonic IFCs determine phonon linewidths, phonon scattering rates, and hence the lattice thermal conductivity. As a consequence, we may expect classical molecular dynamics and lattice dynamics that regard nuclei as point-like particles to give inaccurate predictions of both the harmonic and anharmonic phonon properties. [4]

Although nuclear quantum effects in solid hydrogen may be pronounced, the system remains simple enough to study in detail. Molecular hydrogen has two nuclear spin isomers, which are ortho-hydrogen (o-H_2) with total nuclear spin $I = 1$ and para-hydrogen (p-H_2) with $I = 0$. Transitions between these nuclear-spin isomers are forbidden, and as a result, samples of very high purity for one species may be prepared. In this thesis, we focus on p-H_2. Since protons are fermions, the Pauli principle requires the total wavefunction of p-H_2 to be antisymmetric with respect to the permutation of its protons, and the $I = 0$ nuclear spin state is itself antisymmetric, so only even rotational states are allowed. Furthermore, since the melting point $T_\mathrm{m} \simeq 13.8\,\mathrm{K}$ is substantially lower than the rotational temperature $\theta_\mathrm{rot} \simeq 87.6\,\mathrm{K}$ of the molecules, the vast majority of p-H_2 molecules occupy the $J = 0$ ground rotational state in the solid phase. Since the $J = 0$ rotational wavefunctions are spherically symmetric, this implies that the interaction between the molecules can be modelled to a good approximation by an isotropic pair potential, which further simplifies the calculations. [5]

Solid p-H_2 crystallises in the hexagonal close-packed (hcp) structure with two molecules in the primitive cell and intermolecular nearest neighbour distance of $a \simeq 3.79\,\text{Å}$. The free rigid rotor that describes the rotational states is almost undistorted by the intermolecular interactions, thus the solid may be seen as an assembly of molecules at their lattice sites, but freely rotating even at $T = 0\,\mathrm{K}$. [4] Finally, high-quality experimental data for phonon dispersion relations, heat capacities and thermal conductivities are available for comparison. All of these factors make solid p-H_2 an ideal testing ground for methods that include quantum mechanical effects in molecular and lattice dynamics simulations. This work also naturally builds on previous work in our group on the diffusion and thermal conductivity of liquid p-H_2, [6, 7] where NQEs were shown to be important for these dynamical properties. Exact quantum dynamical simulation is not feasible for either the

liquid or the solid, as it is computationally too costly to solve the time-dependent Schrödinger equation (TDSE)

$$i\hbar\frac{\partial|\psi\rangle}{\partial t} = \hat{H}|\psi\rangle \tag{1.2}$$

for these condensed phase systems exactly. References [6, 7] therefore used imaginary time path integral methods to study the properties of liquid p-H_2, and we shall do the same here for the solid.

The thesis is organised as follows. Chapter 2 presents the necessary background theory for this work, beginning with the isomorphism between the quantum partition function of a particle and the classical partition function of a "ring polymer", and showing how this leads to a method to incorporate NQEs into an effective potential (the centroid potential of mean force). We then move on to the theory of phonons and explain how to obtain the phonon normal modes, and how to calculate harmonic phonon properties including phonon frequencies and phonon group velocities. Then the theory of thermal conductivity is covered, beginning with the non-equilibrium classical thermodynamics of macroscopic heat transfer and its statistical mechanical origin from scattering of phonons by the anharmonicity in the potential. We then use the Boltzmann transport equation (BTE) to connect the macroscopic picture of heat transport described by Fourier's law with the microscopic phonon quasi-particle picture, and explaining why the usual relaxation-time approximation (RTA) becomes unsuitable when normal phonon scattering dominates. Chapter 3 presents the results in the same logical order. The classical potential is first compared with the centroid potential of mean force (CPMF), with an emphasis on the differences between the two that arise from NQEs. The radial distribution functions (RDFs) calculated from classical molecular dynamics and (centroid) path integral molecular dynamics are then compared, and the comparison is used to justify a pairwise approximation to the full many-body CPMF. The chapter then moves on to comparing phonon dispersion

relations and heat capacities calculated from both the classical potential and the CPMF with experiment. Finally, the thermal conductivities calculated using both the RTA and the direct solution of the linearised Boltzmann transport equation (LBTE) in the "relaxon" basis on CPMFs are compared to experimental data. Chapter 4 summarises the main findings of the thesis and suggests directions for future work.

2
Theory

Contents

2.1 Path Integral Molecular Dynamics 6
2.2 Centroid Potential of Mean Force 10
2.3 Harmonic Phonons 14
2.4 Boltzmann Transport Equation 17
2.5 Linearisation 20
2.6 Relaxation Time Approximation 22
2.7 Relaxons 24

Our calculations of the thermal conductivity for solid p-H_2 have several components. First, we have to use path integral molecular dynamics (PIMD) simulations to construct the centroid potential of mean force (CPMF) at each temperature and density of interest, as described in sections 2.1 and 2.2. Then we have to construct a harmonic phonon basis on the CPMF as described in section 2.3. The thermal conductivity is obtained by solving the linearised Boltzmann transport equation (LBTE) in this phonon basis, which involves anharmonic phonon scattering processes as described in sections 2.4 and 2.5. A commonly used approximation to the LBTE is the relaxation-time approximation (RTA), which we describe in section 2.6. This approximation breaks down at low temperatures for reasons explained

in section 2.6, so we must instead solve the fully coupled LBTE, which we do in a relaxon basis as described in section 2.7.

2.1 Path Integral Molecular Dynamics

The canonical partition function of a single quantum particle is

$$Q \equiv \mathrm{tr}\left[e^{-\beta\hat{H}}\right], \tag{2.1}$$

where $\beta = 1/k_B T$, with k_B the Boltzmann constant and T the thermodynamic temperature. The trace tr is taken over any complete set. The Hamiltonian operator $\hat{H}$ is Hermitian, so its eigenstates $\{|n\rangle\}$ form a complete set

$$Q = \sum_n \langle n|e^{-\beta\hat{H}}|n\rangle = \sum_n e^{-\beta E_n}. \tag{2.2}$$

This is the familiar expression for the partition function. However, it has little practical use for a condensed-phase system such as solid hydrogen because diagonalising the full many-body Hamiltonian scales exponentially with the number of particles and is computationally impossible. We therefore seek an alternative form of the partition function that may be evaluated without explicitly diagonalising $\hat{H}$.

To show how an alternative form can be obtained, it again suffices to consider a single quantum particle moving in one dimension. (The generalisation to more particles is straightforward and will be given in section 2.2.) The position operator $\hat{q}$ is also Hermitian, and its eigenstates form a complete set. Hence, we may write the following

$$Q = \int \mathrm{d}q\, \langle q|e^{-\beta\hat{H}}|q\rangle. \tag{2.3}$$

This form does not yet appear easier to evaluate, but we can make progress by splitting the quantum Boltzmann operator into n pieces and inserting $n-1$ complete sets of position eigenstates between them:

$$Q = \int \mathrm{d}q_1 \cdots \int \mathrm{d}q_n\, \langle q_1|e^{-\beta_n\hat{H}}|q_2\rangle \cdots \langle q_n|e^{-\beta_n\hat{H}}|q_1\rangle, \tag{2.4}$$

where

$$\beta_n \equiv \frac{\beta}{n}. \tag{2.5}$$

This expression can be further manipulated using the Trotter factorisation, [8,9]

$$e^{-\beta_n(\hat{T}+\hat{V})} \simeq e^{-\beta_n\hat{V}/2}e^{-\beta_n\hat{T}}e^{-\beta_n\hat{V}/2}, \tag{2.6}$$

where $\hat{T} = \hat{p}^2/2m$ and $\hat{V} = V(\hat{q})$, and brought into the form (see Appendix A)

$$Q = \lim_{n\to\infty} \frac{1}{(2\pi\hbar)^n} \int \mathrm{d}\mathbf{p} \int \mathrm{d}\mathbf{q}\, e^{-\beta_n H_n(\mathbf{p},\mathbf{q})} \equiv \lim_{n\to\infty} Q_n. \tag{2.7}$$

This is now a $2n$-dimensional integral over an extended classical phase space. The quantity $H_n(\mathbf{p},\mathbf{q})$ is no longer a quantum mechanical operator but a classical Hamiltonian:

$$H_n(\mathbf{p},\mathbf{q}) = \sum_{j=1}^{n}\left[\frac{p_j^2}{2m} + \frac{1}{2}m\omega_n^2(q_j - q_{j+1})^2 + V(q_j)\right] \equiv \frac{|\mathbf{p}|^2}{2m} + V_n(\mathbf{q}) \equiv H_0(\mathbf{p},\mathbf{q}) + V(\mathbf{q}), \tag{2.8}$$

where $q_{n+1} \equiv q_1$,

$$\omega_n \equiv \frac{1}{\beta_n\hbar}. \tag{2.9}$$

This Hamiltonian describes a classical *ring polymer* with n beads. Each bead feels the physical potential $V(q)$, and the neighbouring beads are connected by harmonic springs. The ring polymer Hamiltonian may also be expressed in the alternative forms shown on the right-hand side of Eq. (2.8). Here, $V_n(\mathbf{q})$ incorporates both the physical potential $V(\mathbf{q})$ and the harmonic spring potential between adjacent beads, whereas $H_0(\mathbf{p},\mathbf{q})$ represents the Hamiltonian of the free ring polymer in the absence of this physical potential. The approximation of Q by Q_n becomes exact in the limit $n \to \infty$, although in practical calculations a finite number of beads is sufficient once the convergence is reached. In the one-bead limit, the ring polymer partition function reduces to the classical partition function of a single particle.

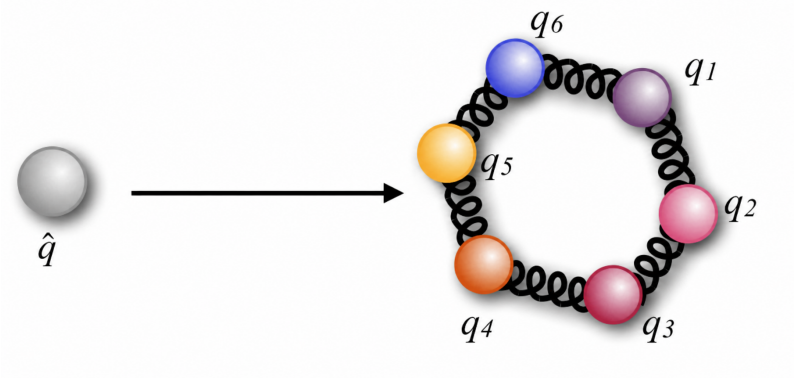


Figure 2.1: A quantum particle represented by a classical ring polymer of 6 beads, the neighbouring beads are connected by harmonic springs of stiffness $m\omega_n^2$. [10]

We have therefore shown that there is an *isomorphism* [11, 12] between the quantum partition function of a particle at inverse physical temperature β and the classical partition function of a ring polymer at inverse temperature β_n. The quantum thermal expectation value of an observable in the canonical ensemble is

$$\langle A \rangle = \frac{1}{Q} \operatorname{tr}\left[e^{-\beta \hat{H}} \hat{A}\right]. \tag{2.10}$$

If $\hat{A} = A(\hat{q})$, then a similar argument gives

$$\langle A \rangle = \lim_{n\to\infty} \frac{1}{(2\pi\hbar)^n Q_n} \int \mathrm{d}\mathbf{p} \int \mathrm{d}\mathbf{q}\, e^{-\beta_n H_n(\mathbf{p},\mathbf{q})} A_n(\mathbf{q}), \tag{2.11}$$

where the ring polymer estimator is

$$A_n(\mathbf{q}) \equiv \frac{1}{n} \sum_{j=1}^{n} A(q_j). \tag{2.12}$$

Thus, a position-dependent observable is first averaged over the beads of the ring polymer, then averaged over the canonical ensemble of the ring polymer. If the system is assumed to be ergodic, the ensemble average can be replaced by a time average over the trajectories generated by the ring polymer Hamiltonian. This is known as path integral molecular dynamics (PIMD). [13] The sampling dynamics are generated by

$$\dot{\mathbf{q}} = \frac{\partial H_n(\mathbf{q},\mathbf{p})}{\partial \mathbf{p}} = \frac{\mathbf{p}}{m}, \tag{2.13}$$

and

$$\dot{\mathbf{p}} = -\frac{\partial H_n(\mathbf{q},\mathbf{p})}{\partial \mathbf{q}} = -\frac{\partial V_n(\mathbf{q})}{\partial \mathbf{q}}. \tag{2.14}$$

However, this dynamics conserves $H_n(\mathbf{p},\mathbf{q})$, so a single trajectory samples the micro-canonical NVE ensemble rather than the canonical NVT ensemble. Furthermore, the trajectories are not ergodic and one would need to run a long simulation to see the energy flow between the normal modes of the ring polymer. [14] These problems are usually addressed by attaching a path integral Langevin equation (PILE) thermostat to the dynamics. [15] One proceeds by first transforming the harmonic part $H_0(\mathbf{p},\mathbf{q})$ into the normal mode representation $H_0(\mathbf{P},\mathbf{Q})$, via

$$P_k = \sum_{j=1}^{n} p_j C_{jk}, \qquad Q_k = \sum_{j=1}^{n} q_j C_{jk}, \tag{2.15}$$

where

$$C_{jk} = \begin{cases} \sqrt{1/n}, & k = 0, \\ \sqrt{2/n}\cos(2\pi jk/n), & k = 1,\ldots,n/2-1, \\ \sqrt{1/n}(-1)^j, & k = n/2, \\ \sqrt{2/n}\sin(2\pi jk/n), & k = n/2+1,\ldots,n-1. \end{cases} \tag{2.16}$$

Therefore,

$$H_0(\mathbf{P},\mathbf{Q}) = \sum_{k=0}^{n-1}\Big[\frac{P_k^2}{2m} + \frac{1}{2}m\omega_k^2 Q_k^2\Big] \tag{2.17}$$

with

$$\omega_k = 2\omega_n \sin\frac{k\pi}{n}. \tag{2.18}$$

Then the PILE algorithm attaches a separate Langevin thermostat to each internal mode of the free ring polymer,

$$\dot{Q}_k = \frac{P_k}{m} \tag{2.19}$$

$$\dot{P}_k = -m\omega_k^2 Q_k - \gamma_k P_k + \sqrt{\frac{2m\gamma_k}{\beta_n}}\xi_k(t) \tag{2.20}$$

where,

$$\gamma_k = \omega_k \tag{2.21}$$

and $\xi_k(t)$ is a Gaussian white noise,

$$\langle \xi_k(t)\rangle = 0, \qquad \langle \xi_k(t)\xi_{k'}(t')\rangle = \delta_{kk'}\delta(t-t'). \tag{2.22}$$

Since $\omega_0 = 0$, this does not apply a thermostat to the $k = 0$ mode of the ring polymer, which has to be thermostatted separately. In practice we use a global canonical sampling through velocity rescaling (CSVR) thermostat for this with time constant $\tau = 0$. [16]

The ring polymer isomorphism is an equilibrium statistical-mechanical one: it allows quantum thermal expectation values to be computed from a classical phase space expectation value of the ring polymer. The fictitious ring polymer dynamics used in PIMD should therefore be understood primarily as a sampling device, rather than as the exact real-time quantum dynamics of the particle [13]. The static equilibrium properties are independent of the choice of the mass of the beads of the ring polymer. Nevertheless, treating ring polymer dynamics as an approximate real-time quantum dynamics and setting m to the physical mass of the particle leads to ring polymer molecular dynamics (RPMD), which is widely used to approximate quantum time-correlation functions. [17]

2.2 Centroid Potential of Mean Force

The most classical-like coordinate of a ring polymer is its *centroid*, defined as the average position of all the beads,

$$\bar{q} \equiv \frac{1}{n} \sum_{j=1}^{n} q_j. \tag{2.23}$$

Note the following relationship between the centroid and the zero-frequency $\omega_0 = 0$ mode of the free ring polymer,

$$\bar{q} = \frac{Q_0}{\sqrt{n}}. \tag{2.24}$$

The centroid distribution is defined as, [12]

$$\rho_c(q_c) \equiv \left(\frac{m}{2\pi\beta_n\hbar^2}\right)^{n/2} \int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})} \delta(\bar{q} - q_c). \tag{2.25}$$

The quantum partition function can then be obtained by integrating over all possible centroid positions

$$Q = \int \mathrm{d}q_c \, \rho_c(q_c). \tag{2.26}$$

As shown in Appendix B.I, this is equivalent to Eq. (2.7). The *centroid potential of mean force* $V_c(q_c)$ is defined through the centroid distribution by

$$\rho_c(q_c) = e^{-\beta V_c(q_c)}. \tag{2.27}$$

The corresponding centroid mean force is

$$F_c(q_c) \equiv -\frac{\partial V_c(q_c)}{\partial q_c}. \tag{2.28}$$

In Appendix B.II we show that this force may also be written as [18]

$$F_c(q_c) = \left\langle \frac{1}{n} \sum_{j=1}^{n} \left(-\frac{\partial V(q_j)}{\partial q_j} \right) \right\rangle_{\bar{q}=q_c}. \tag{2.29}$$

The expectation value is taken over the ring polymer configurations constrained to have centroid position q_c. Importantly, this is an average over the physical forces due to the potential $V(q)$. The fictitious spring forces between neighbouring beads do not contribute.

We now generalise this expression to N interacting particles in three dimensions. Let $\mathbf{r}_i^{(k)}$ be the position of bead k on molecule i, and let

$$\bar{\mathbf{r}}_i = \frac{1}{n} \sum_{k=1}^{n} \mathbf{r}_i^{(k)} \tag{2.30}$$

be the centroid of the ring polymer i. The centroid mean force on the molecule i is then

$$\mathbf{F}_i^c(\mathbf{r}^c) \equiv -\nabla_{\mathbf{r}_i^c} V_c(\mathbf{r}^c) = \left\langle \frac{1}{n} \sum_{k=1}^{n} \left[-\nabla_{\mathbf{r}_i^{(k)}} V(\mathbf{r}_1^{(k)}, \ldots, \mathbf{r}_N^{(k)}) \right] \right\rangle_{\bar{\mathbf{r}}=\mathbf{r}_i^c}, \tag{2.31}$$

where

$$\mathbf{r} \equiv \{\mathbf{r}_i\}, \qquad \bar{\mathbf{r}} \equiv \{\bar{\mathbf{r}}_i\}. \tag{2.32}$$

The exact centroid potential of mean force is, in general, a many-body Helmholtz free energy. As discussed in chapter 1, for solid p-H_2, the intermolecular interaction may be modelled to a good approximation by an isotropic pair potential $v(r)$. This motivates the following pairwise approximations:

$$V(\mathbf{r}) \simeq \frac{1}{2}\sum_{i=1}^{N}\sum_{j\neq i}^{N} v(r_{ij}), \tag{2.33}$$

and

$$V_c(\mathbf{r}^c) \simeq \frac{1}{2}\sum_{i=1}^{N}\sum_{j\neq i}^{N} v_c(r^c_{ij}), \tag{2.34}$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$. The factor of $1/2$ prevents double counting, and the condition $j \neq i$ excludes self-interactions. The validity of these approximations will be examined in section 3.2. In Appendix B.III, we show that these approximations lead to the following procedure for the calculation of the pairwise centroid potential of mean force $v_c(r)$. [7] The instantaneous pairwise centroid force between molecules i and j is computed by averaging the physical forces between the molecules with the same bead label:

$$\mathbf{f}^c_{ij} = \frac{1}{n}\sum_{k=1}^{n}\left[-v'(r^{(k)}_{ij})\hat{\mathbf{r}}^{(k)}_{ij}\right], \tag{2.35}$$

where $\hat{\mathbf{r}}^{(k)}_{ij}$ is a unit vector directed from molecule j to molecule i for the bead index k. The physical potential only couples beads with the same bead index. The radial component of this centroid force is obtained by the projection,

$$f^c_{ij} = \hat{\mathbf{r}}^c_{ij}\cdot\mathbf{f}^c_{ij}, \tag{2.36}$$

where $\hat{\mathbf{r}}^c_{ij}$ is a unit vector directed from the centroid of the ring polymer j to the centroid of the ring polymer i. The pairwise centroid potential of mean force is then obtained from [7]

$$v_c(r) = \int_r^\infty \mathrm{d}r'\,\langle f^c_{ij}\rangle_{r^c_{ij}=r'}. \tag{2.37}$$

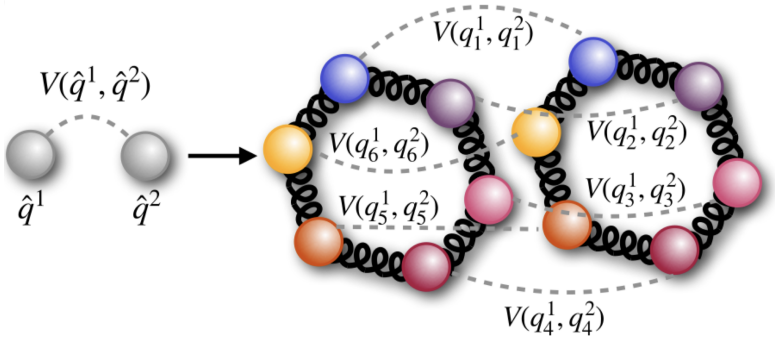


Figure 2.2: The interactions between two ring polymers, with 6 beads each. Beads with the same label (k) have the same colour, interaction due to the physical potential $v(r)$ is illustrated by dashed lines. [10]

In practice, the upper limit is replaced by a cutoff r_{cut}, chosen such that the intermolecular force has effectively decayed to zero.

At the end of section 2.1, we mentioned that using the classical dynamics generated by the ring polymer Hamiltonian $H_n(\mathbf{p}, \mathbf{q})$ to approximate the exact quantum dynamics leads to RPMD. Alternatively, one can approximate the exact quantum dynamics by propagating only the centroid on the centroid potential of mean force:

$$H_c(\mathbf{p}, \mathbf{q}) = \frac{|\mathbf{p}|^2}{2m} + V_c(\mathbf{q}) \tag{2.38}$$

$$\dot{\mathbf{q}} = \frac{\partial H_c(\mathbf{p}, \mathbf{q})}{\partial \mathbf{p}} = \frac{\mathbf{p}}{m} \tag{2.39}$$

$$\dot{\mathbf{p}} = -\frac{\partial H_c(\mathbf{p}, \mathbf{q})}{\partial \mathbf{q}} = -\frac{\partial V_c(\mathbf{q})}{\partial \mathbf{q}}, \tag{2.40}$$

this leads to another widely used method known as centroid molecular dynamics (CMD). [19–21]

Sutherland *et al.* have recently used this method to calculate the thermal conductivity of liquid p-H_2. [7] Here, we shall instead perform lattice dynamics (i.e. solve the linearised Boltzmann transport equation) on the centroid potential of mean force, which is a more appropriate way to calculate the thermal conductivity of *solid* p-H_2.

2.3 Harmonic Phonons

The first part of this chapter described path integral molecular dynamics (PIMD) methods. We now turn to lattice dynamics (LD), where the central objects are the normal modes of a crystal. In lattice dynamics, normal-mode coordinates replace atomic Cartesian coordinates as natural degrees of freedom. We should begin by finding out what these normal modes are. The properties and dynamics of a crystal are governed by its Hamiltonian,

$$\hat{H} = \hat{H}_0 + \hat{V}', \tag{2.41}$$

where the full Hamiltonian has been split into a harmonic part $\hat{H}_0$ and an anharmonic part $\hat{V}'$, which is treated as a weak perturbation. The ring polymer may be regarded as a one-dimensional crystal with periodic boundary conditions. We were able to obtain the normal modes of the one-dimensional crystal in Eq. (2.15)–(2.17) with relative ease. This section extends the procedure to a three-dimensional crystal.

The harmonic Hamiltonian $\hat{H}_0$ couples many atomic degrees of freedom. The aim of lattice dynamics is to first transform the coupled displacements from the equilibrium geometry into independent normal modes, so that

$$\hat{H}_0 = \sum_{\mathbf{q}s} \hat{H}_{\mathbf{q}s}. \tag{2.42}$$

Here, the normal mode is labelled by a branch index s and a wavevector $\mathbf{q}$ inside the first Brillouin zone. The corresponding crystal momentum is defined modulo a reciprocal lattice vector, as discussed in Appendix C. Appendix D derives the formulae used below. Once the atomic degrees of freedom have been decoupled, each normal mode behaves as a quantum harmonic oscillator:

$$\hat{H}_{\mathbf{q}s} = \frac{1}{2}\Big(\hat{p}_{-\mathbf{q}s}\hat{p}_{\mathbf{q}s} + \omega_{\mathbf{q}s}^2 \hat{u}_{-\mathbf{q}s}\hat{u}_{\mathbf{q}s}\Big) = \hbar\omega_{\mathbf{q}s}\Big(\hat{a}_{\mathbf{q}s}^\dagger \hat{a}_{\mathbf{q}s} + \frac{1}{2}\Big). \tag{2.43}$$

Here, $\hat{u}_{\mathbf{q}s}$ and $\hat{p}_{\mathbf{q}s}$ are the normal-mode displacements and momenta, which satisfy

$$\hat{u}_{\mathbf{q}s}^\dagger = \hat{u}_{-\mathbf{q}s}, \qquad \hat{p}_{\mathbf{q}s}^\dagger = \hat{p}_{-\mathbf{q}s}, \tag{2.44}$$

the operator $\hat{a}_{\mathbf{q}s}$ is the phonon annihilation operator for normal mode $(\mathbf{q}, s)$, defined as

$$\hat{a}_{\mathbf{q}s} \equiv \frac{1}{\sqrt{2\hbar}} \left(\omega_{\mathbf{q}s}^{1/2} \hat{u}_{\mathbf{q}s} + i\omega_{\mathbf{q}s}^{-1/2} \hat{p}_{\mathbf{q}s} \right). \tag{2.45}$$

The time-independent Schrödinger equation (TISE) for each normal mode under the harmonic approximation is

$$\hat{H}_{\mathbf{q}s}|N_{\mathbf{q}s}\rangle = \hbar\omega_{\mathbf{q}s}\left(N_{\mathbf{q}s} + \frac{1}{2}\right)|N_{\mathbf{q}s}\rangle. \tag{2.46}$$

Thus, when we say that the mode $(\mathbf{q}, s)$ contains $N_{\mathbf{q}s}$ phonons, we mean that the normal mode is in its $N_{\mathbf{q}s}$-th harmonic oscillator excitation state. We now proceed to find expressions for $\hat{u}_{\mathbf{q}s}$, $\hat{p}_{\mathbf{q}s}$ and $\omega_{\mathbf{q}s}$.

We first decompose the molecular label as $i \equiv (l, b)$, where l labels the lattice point $\mathbf{R}_l$, which is defined as an integer linear combination of primitive lattice vectors,

$$\mathbf{R}_l = l_1\mathbf{a}_1 + l_2\mathbf{a}_2 + l_3\mathbf{a}_3, \tag{2.47}$$

and b labels the basis within the primitive cell; for the hcp structure considered here, each primitive cell contains two p-H_2 molecules, $b = 1, 2$. The equilibrium position of molecule b within primitive cell l is

$$\mathbf{R}_{lb}^0 = \mathbf{R}_l + \boldsymbol{\tau}_b, \tag{2.48}$$

where $\boldsymbol{\tau}_b$ is the basis position relative to the lattice point. The displacement from equilibrium is

$$\mathbf{u}_{lb} = \mathbf{R}_{lb} - \mathbf{R}_{lb}^0, \tag{2.49}$$

where $\mathbf{R}_{lb}$ is a general non-equilibrium position of the molecule. The Hamiltonian under the harmonic approximation to the potential is [22, 23]

$$\hat{H}_0 = \sum_{lb\alpha} \frac{\hat{p}_{lb\alpha}^2}{2m_b} + \frac{1}{2} \sum_{lb\alpha} \sum_{l'b'\alpha'} \hat{u}_{lb\alpha} \Phi_{lb\alpha,l'b'\alpha'} \hat{u}_{l'b'\alpha'}. \tag{2.50}$$

The harmonic intermolecular force constants (FC2) are defined as

$$\Phi_{lb\alpha,l'b'\alpha'} \equiv \left.\frac{\partial^2 V}{\partial u_{lb\alpha}\partial u_{l'b'\alpha'}}\right|_{u=0}. \tag{2.51}$$

By symmetry of mixed second derivatives for a well-behaved function,

$$\Phi_{lb\alpha,l'b'\alpha'} = \Phi_{l'b'\alpha',lb\alpha}. \tag{2.52}$$

Translational invariance also imposes the acoustic sum rule,

$$\sum_{l'b'} \Phi_{lb\alpha,l'b'\alpha'} = 0, \quad \forall\, l, b, \alpha, \alpha', \tag{2.53}$$

which ensures that the three acoustic phonon branches satisfy $\omega_{\mathbf{q}s} \to 0$ as $\mathbf{q} \to 0$. Direct diagonalisation of the full force-constant matrix Φ is computationally expensive. However, because the crystal has periodic translational symmetry along lattice vectors $\mathbf{R}_l$, the full mass-weighted Hessian $K_{lb\alpha,l'b'\alpha'} = \Phi_{lb\alpha,l'b'\alpha'}/\sqrt{m_b m_{b'}}$ may be block-diagonalised into smaller $\mathbf{q}$-blocks. These blocks are the dynamical matrices,

$$D_{b\alpha,b'\alpha'}(\mathbf{q}) \equiv \frac{1}{\sqrt{m_b m_{b'}}} \sum_{l'} \Phi_{0b\alpha,l'b'\alpha'} e^{i\mathbf{q}\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'}-\boldsymbol{\tau}_b)}. \tag{2.54}$$

For solid p-H_2, each $D(\mathbf{q})$ is a 6×6 matrix. The phonon frequencies $\omega_{\mathbf{q}s}$ and polarisation vectors $\boldsymbol{\epsilon}_s(\mathbf{q})$ are obtained by solving the eigenvalue problem,

$$D(\mathbf{q})\boldsymbol{\epsilon}_s(\mathbf{q}) = \omega^2_{\mathbf{q}s}\boldsymbol{\epsilon}_s(\mathbf{q}). \tag{2.55}$$

The dynamical matrix is Hermitian; thus the eigenvalues $\omega^2_{\mathbf{q}s}$ are real, the phonon normal modes are orthonormal and form a complete set in the mass-weighted displacement space. Positive $\omega^2_{\mathbf{q}s}$ indicates a stable structure whereas negative $\omega^2_{\mathbf{q}s}$, thus imaginary $\omega_{\mathbf{q}s}$ indicates instabilities in the structure. The polarisation vectors $\boldsymbol{\epsilon}_s(\mathbf{q})$ are orthonormal within the subspace that the dynamical matrix $D(\mathbf{q})$ operates on for a given $\mathbf{q}$,

$$\delta_{ss'} = \sum_{b\alpha} \epsilon^*_{s,b\alpha}(\mathbf{q})\epsilon_{s',b\alpha}(\mathbf{q}). \tag{2.56}$$

Once the polarisation vectors are known, the normal mode displacements are

$$\hat{u}_{\mathbf{q}s} = \frac{1}{\sqrt{N_{\mathbf{q}}}} \sum_{lb\alpha} \sqrt{m_b}\, \hat{u}_{lb\alpha} e^{-i\mathbf{q}\cdot(\mathbf{R}_l+\boldsymbol{\tau}_b)} \epsilon^*_{s,b\alpha}(\mathbf{q}), \tag{2.57}$$

and the normal mode momenta are

$$\hat{p}_{\mathbf{q}s} = \frac{1}{\sqrt{N_{\mathbf{q}}}} \sum_{lb\alpha} \frac{\hat{p}_{lb\alpha}}{\sqrt{m_b}} e^{-i\mathbf{q}\cdot(\mathbf{R}_l+\boldsymbol{\tau}_b)} \epsilon^*_{s,b\alpha}(\mathbf{q}). \tag{2.58}$$

The phonon group velocity is defined as

$$\mathbf{v}_{\mathbf{q}s} \equiv \nabla_{\mathbf{q}} \omega_{\mathbf{q}s}. \tag{2.59}$$

which may be calculated using

$$v_{\mathbf{q}s,\alpha} = \frac{1}{2\omega_{\mathbf{q}s}} \sum_{b'\alpha'} \sum_{b''\alpha''} \epsilon^*_{s,b'\alpha'}(\mathbf{q}) \frac{\partial D_{b'\alpha',b''\alpha''}(\mathbf{q})}{\partial q_\alpha} \epsilon_{s,b''\alpha''}(\mathbf{q}). \tag{2.60}$$

In an insulating crystal, heat is transported by phonons. The harmonic phonons obtained from the harmonic interatomic force constants $\Phi_{lb\alpha,l'b'\alpha'}$ therefore provide the microscopic basis for the calculation of thermal conductivity of solid para-hydrogen. Anharmonicity $\hat{V}'$, neglected here, produces phonon-phonon scattering and hence finite thermal conductivity, we shall discuss this further in the following sections.

2.4 Boltzmann Transport Equation

Here, we connect heat conduction to phonons. We begin with a macroscopic picture of heat transfer, starting from classical thermodynamics. The first law of thermodynamics in differential form is

$$\mathrm{d}U = đq + đw. \tag{2.61}$$

We assume that the insulating crystal has negligible thermal expansion. At constant volume no mechanical work is done, so $\mathrm{d}U = đq$. We define the internal energy density $u(\mathbf{r}, t)$ within a volume Ω via [24]

$$U = \int_\Omega u(\mathbf{r}, t)\, \mathrm{d}V. \tag{2.62}$$

The heat flux density $\mathbf{J}$ is defined such that the heat entering the volume Ω during time $\mathrm{d}t$ is

$$đq = -\oint_{\partial\Omega} \mathbf{J}(\mathbf{r},t)\cdot \mathrm{d}\mathbf{S}\,\mathrm{d}t = -\int_{\Omega} \nabla\cdot\mathbf{J}(\mathbf{r},t)\,\mathrm{d}V\,\mathrm{d}t. \tag{2.63}$$

The minus sign appears because $\mathrm{d}\mathbf{S}$ points out of the volume, whereas $đq$ is defined as heat entering the volume. We have used the divergence theorem for the second equality. These definitions, together with the first law, give the continuity equation

$$\frac{\partial u}{\partial t} + \nabla\cdot\mathbf{J} = 0, \tag{2.64}$$

which is a statement of local energy conservation.

Fourier's law relates the heat flux density to the temperature gradient and defines the thermal conductivity tensor $\boldsymbol{\kappa}$,

$$\mathbf{J} = -\boldsymbol{\kappa}\nabla T. \tag{2.65}$$

In general, $\boldsymbol{\kappa}$ is a function of temperature and pressure.

The next goal is to connect this macroscopic classical thermodynamics picture of heat transfer to the microscopic quantum mechanical phonon picture. This will eventually provide us with a way to calculate the thermal conductivities from lattice dynamics.

The statistical mechanical definition of internal energy is

$$U \equiv \sum_i p_i E_i, \tag{2.66}$$

where E_i and $|i\rangle$ are the solutions of the time-independent Schrödinger equation (TISE),

$$\hat{H}|i\rangle = E_i|i\rangle. \tag{2.67}$$

For a many-body crystal, diagonalising $\hat{H}$ is not practical. We proceed with the harmonic approximation $\hat{H} \simeq \hat{H}_0$, where $\hat{H}_0 = \sum_{\mathbf{q}s}\hat{H}_{\mathbf{q}s}$ is a sum of independent

normal-mode Hamiltonians. We use the joint index $k \equiv (\mathbf{q}, s)$ to simplify the notation, so that the energy of a microstate $|i\rangle$ becomes

$$E_i = \sum_k \hbar\omega_k N_{k,i}, \tag{2.68}$$

under the harmonic approximation, where $i \equiv \{N_{k,i}\}$ is a set of good quantum numbers. These are good quantum numbers because the number operator $\hat{a}_k^\dagger \hat{a}_k$ for a given mode always commutes with $\hat{H}_0$. The zero of energy is arbitrary, since we are only concerned with the energy differences in thermodynamics; thus, we choose the summed zero-point energy of all phonon normal modes, $E_0 = \sum_k \hbar\omega_k/2$, as the zero of our energy scale. Therefore we have

$$U = \sum_k \hbar\omega_k n_k, \tag{2.69}$$

where

$$n_k \equiv \sum_i p_i N_{k,i} \tag{2.70}$$

is the mean occupation of mode k. In thermal equilibrium,

$$n_k^0(T) = \frac{1}{e^{\hbar\omega_k/k_B T} - 1}. \tag{2.71}$$

Heat conduction requires the system to be out of global thermal equilibrium. We assume, however, that the departure from local equilibrium is small, so that

$$n_k(\mathbf{r}, t) \simeq n_k^0(T(\mathbf{r}, t)) + \delta n_k(\mathbf{r}, t), \tag{2.72}$$

holds. The corresponding harmonic phonon internal energy density is,

$$u(\mathbf{r}, t) = \frac{1}{\mathcal{V}} \sum_k \hbar\omega_k n_k(\mathbf{r}, t), \tag{2.73}$$

where $\mathcal{V}$ is the crystal volume.

The time evolution of the phonon distribution is governed by the Boltzmann transport equation (BTE), [25]

$$\frac{\partial n_k}{\partial t} + \mathbf{v}_k \cdot \nabla n_k = \left(\frac{\partial n_k}{\partial t}\right)_{\text{scat}}, \tag{2.74}$$

where $(\partial n_k/\partial t)_{\text{scat}}$ is a collision term that accounts for the rate of change of n_k due to scattering processes.

A phonon is an excitation of a normal mode, thus its position in real space is undefined. The Boltzmann equation should therefore be understood semiclassically: it describes phonon wavepackets that are sufficiently localised in real space while remaining sufficiently narrow in reciprocal space. Each phonon wavepacket is constructed from a superposition of phonon eigenstates near $\hbar\mathbf{q}$. A phonon wavepacket may be regarded as a quasiparticle with energy $\hbar\omega_k$ and crystal momentum $\hbar\mathbf{q}$. Multiplying the BTE by $\hbar\omega_k$, summing over k, and dividing by $\mathcal{V}$ gives

$$\frac{\partial}{\partial t}\left[\frac{1}{\mathcal{V}}\sum_k \hbar\omega_k n_k\right] + \nabla\cdot\left[\frac{1}{\mathcal{V}}\sum_k \hbar\omega_k \mathbf{v}_k n_k\right] = \frac{1}{\mathcal{V}}\sum_k \hbar\omega_k \left(\frac{\partial n_k}{\partial t}\right)_{\text{scat}}. \tag{2.75}$$

For phonon-phonon scattering, energy is conserved, so the right-hand side vanishes. Comparing with the energy continuity equation Eq. (2.64) gives the first term as the rate of change of the internal energy density, and allows us to identify the harmonic phonon heat flux density as, [26]

$$\mathbf{J} = \frac{1}{\mathcal{V}}\sum_k \hbar\omega_k \mathbf{v}_k n_k. \tag{2.76}$$

The equilibrium contribution vanishes by symmetry, since

$$\omega_k = \omega_{-k}, \qquad n_k^0 = n_{-k}^0, \qquad \mathbf{v}_{-k} = -\mathbf{v}_k, \tag{2.77}$$

where $-k \equiv (-\mathbf{q}, s)$. Therefore,

$$\mathbf{J}(\mathbf{r},t) = \frac{1}{\mathcal{V}}\sum_k \hbar\omega_k \mathbf{v}_k \delta n_k(\mathbf{r},t). \tag{2.78}$$

2.5 Linearisation

The full BTE is difficult to solve because the scattering term is a nonlinear functional of all phonon occupations. For small linear deviations from local thermal equilibrium,

we may linearise the collision term, [27]

$$\left(\frac{\partial n_k}{\partial t}\right)_{\text{scat}} \simeq -\sum_{k'} \Omega_{kk'} \delta n_{k'}, \tag{2.79}$$

where $\Omega_{kk'}$ is an element of a scattering matrix. Substituting $n_k = n_k^0(T) + \delta n_k$ into the BTE gives

$$\frac{\partial n_k^0}{\partial T}\left(\frac{\partial T}{\partial t} + \mathbf{v}_k \cdot \nabla T\right) + \frac{\partial \delta n_k}{\partial t} + \mathbf{v}_k \cdot \nabla \delta n_k = -\sum_{k'} \Omega_{kk'} \delta n_{k'}. \tag{2.80}$$

Then, we consider the system in steady state,

$$\frac{\partial T(\mathbf{r}, t)}{\partial t} = 0, \qquad \frac{\partial \delta n_k(\mathbf{r}, t)}{\partial t} = 0. \tag{2.81}$$

We also use the fact that $\delta n_k \sim \mathcal{O}(\nabla T)$, [27] so that

$$\nabla \delta n_k \sim \mathcal{O}(\nabla^2 T), \tag{2.82}$$

which is a second-order term and therefore neglected. The linearised Boltzmann transport equation (LBTE) in steady state is therefore

$$\frac{\partial n_k^0}{\partial T} \mathbf{v}_k \cdot \nabla T = -\sum_{k'} \Omega_{kk'} \delta n_{k'}. \tag{2.83}$$

In an insulating crystal, the main scattering mechanisms are boundary scattering, impurity scattering, and phonon-phonon scattering. Since the experimental systems that we will compare our calculated results are large, high-purity crystals, we may neglect boundary and impurity scattering and focus on phonon-phonon scattering. We further restrict attention to three-phonon processes, which arise from the third-order interatomic force constants $\Phi^{(3)}$ and are taken to dominate over higher-order processes in the temperature range considered. The scattering matrix Ω is obtained from the anharmonic part $\hat{V}'$ of the crystal Hamiltonian. Keeping only the leading anharmonic term gives

$$\hat{V}' \simeq \hat{V}^{(3)} = \frac{1}{3!} \sum_{lb\alpha} \sum_{l'b'\alpha'} \sum_{l''b''\alpha''} \Phi^{(3)}_{lb\alpha, l'b'\alpha', l''b''\alpha''} \hat{u}_{lb\alpha} \hat{u}_{l'b'\alpha'} \hat{u}_{l''b''\alpha''}, \tag{2.84}$$

where

$$\Phi^{(3)}_{lb\alpha,l'b'\alpha',l''b''\alpha''} \equiv \left.\frac{\partial^3 V}{\partial u_{lb\alpha}\partial u_{l'b'\alpha'}\partial u_{l''b''\alpha''}}\right|_{u=0}. \tag{2.85}$$

After transforming the displacements into phonon normal mode coordinates we have

$$\hat{V}^{(3)} = \frac{1}{3!}\sum_{kk'k''}\Phi^{(3)}_{kk'k''}\hat{u}_k\hat{u}_{k'}\hat{u}_{k''} \tag{2.86}$$

where

$$\Phi^{(3)}_{kk'k''} = \left.\frac{\partial^3 V}{\partial u_k\partial u_{k'}\partial u_{k''}}\right|_{u=0}. \tag{2.87}$$

Thus, the anharmonicity in the potential couples different phonon normal modes together, and $\hat{V}^{(3)}$ leads to three-phonon scattering. We show in Appendix E that the scattering matrix is given by [28]

$$\Omega_{kk'} = \frac{1}{n^0_{k'}(n^0_{k'}+1)}\left[\sum_{k''k'''}\left(\mathcal{P}^{k''}_{kk'''} + \frac{1}{2}\mathcal{P}^{k}_{k''k'''}\right)\delta_{kk'} + \sum_{k''}\left(\mathcal{P}^{k''}_{kk'} - \mathcal{P}^{k'}_{kk''} - \mathcal{P}^{k}_{k'k''}\right)\right], \tag{2.88}$$

where $\mathcal{P}^{k''}_{kk'}$ is the rate for the coalescence event $k + k' \to k''$,

$$\mathcal{P}^{k''}_{kk'} = \frac{2\pi}{\hbar^2}\left|F^{(3)}_{kk'-k''}\right|^2 n^0_k n^0_{k'}(n^0_{k''}+1)\delta(\omega_{k''} - \omega_k - \omega_{k'}), \tag{2.89}$$

and

$$F^{(3)}_{kk'k''} \equiv \left(\frac{\hbar^3}{8\omega_k\omega_{k'}\omega_{k''}}\right)^{1/2}\Phi^{(3)}_{kk'k''}. \tag{2.90}$$

2.6 Relaxation Time Approximation

The scattering matrix is complicated, both to construct and to use to solve the LBTE in Eq. (2.83). A common simplification is the relaxation-time approximation (RTA), [29, 30]

$$\Omega_{kk'} \simeq \frac{\delta_{kk'}}{\tau^{\mathrm{RTA}}_{k'}}, \tag{2.91}$$

which neglects the off-diagonal elements in the scattering matrix. The steady-state LBTE then gives

$$\delta n_k = -\tau^{\mathrm{RTA}}_k\frac{\partial n^0_k}{\partial T}\mathbf{v}_k\cdot\nabla T. \tag{2.92}$$

Substituting this into the heat flux density expression, Eq. (2.78) gives

$$\mathbf{J} = -\frac{1}{\mathcal{V}} \sum_k C_k \tau_k^{\mathrm{RTA}} \mathbf{v}_k \mathbf{v}_k^\intercal \nabla T, \tag{2.93}$$

where C_k is the mode-dependent harmonic heat capacity given by

$$C_k = \hbar\omega_k \frac{\partial n_k^0}{\partial T} = k_B \left(\frac{\hbar\omega_k}{k_B T}\right)^2 n_k^0 (n_k^0 + 1). \tag{2.94}$$

Comparing Eq. (2.93) with Fourier's law gives the lattice thermal conductivity in closed form,

$$\boldsymbol{\kappa}^{\mathrm{RTA}} = \frac{1}{\mathcal{V}} \sum_k C_k \mathbf{v}_k \mathbf{v}_k^\intercal \tau_k^{\mathrm{RTA}}. \tag{2.95}$$

If the intermolecular potential were purely harmonic, there would be no phonon scattering ($\tau_k^{\mathrm{RTA}} \to \infty$ gives $\boldsymbol{\kappa} \to \infty$), thus finite thermal conductivity requires phonon scattering.

While it is certainly a useful simplification, there are several reasons why one should be cautious when applying the RTA. Three-phonon processes conserve energy. For example, a coalescence process requires

$$\hbar\omega_k + \hbar\omega_{k'} = \hbar\omega_{k''}, \tag{2.96}$$

which is enforced by the δ-function in the scattering matrix. It also conserves crystal momentum,

$$\mathbf{q} + \mathbf{q}' = \mathbf{q}'' + \mathbf{G}, \tag{2.97}$$

which is implicitly enforced by $|\Phi^{(3)}|^2$ in the scattering matrix. If $\mathbf{G} = 0$, the process is a normal (N) process. If $\mathbf{G} \neq 0$, the process is an umklapp (U) process. At low temperature, umklapp processes are suppressed because they require thermally populated phonons with sufficiently large wavevectors to take $\mathbf{q} + \mathbf{q}'$ outside the first Brillouin zone. Normal processes are not suppressed in the same way as they may still occur when normal modes close to $\mathbf{q} = 0$ are populated. At low temperature,

normal processes dominate, in which case the off-diagonal elements in the scattering matrix are not negligible, and the relaxation-time approximation becomes unreliable. To understand why this is, we need to look into the statistical mechanics of the nonequilibrium phonon distribution. This motivates an alternative quasiparticle picture of heat transport, that we shall describe next.

2.7 Relaxons

The definition of entropy in statistical mechanics is

$$S \equiv -k_B \sum_i p_i \ln p_i. \tag{2.98}$$

We adopt a new notation $i \equiv \{j_{k,i}\} \leftarrow \{N_{k,i}\}$ in this section. We first consider a single phonon normal mode k. Omitting the zero-point energy, which does not affect the occupation statistics, the eigenstates satisfy

$$\hat{H}_k|j\rangle = j\hbar\omega_k|j\rangle. \tag{2.99}$$

Maximising

$$S_k = -k_B \sum_j p_j^{(k)} \ln p_j^{(k)} \tag{2.100}$$

subject to the constraints:

$$\sum_j p_j^{(k)} = 1, \qquad \sum_j j p_j^{(k)} = n_k, \tag{2.101}$$

can be done using Lagrange undetermined multipliers, and gives the geometric distribution

$$p_j^{(k)} = \frac{1}{1+n_k}\left(\frac{n_k}{1+n_k}\right)^j. \tag{2.102}$$

Substituting this distribution into Eq. (2.100) gives the entropy of a single bosonic mode,

$$S_k = k_B\Big[(1+n_k)\ln(1+n_k) - n_k \ln n_k\Big]. \tag{2.103}$$

Since the harmonic Hamiltonian $\hat{H}_0 = \sum_k \hat{H}_k$ is a sum of independent normal-mode Hamiltonians, there is no interaction entropy between different modes under the harmonic approximation, thus the total harmonic phonon entropy is, [31, 32]

$$S = k_B \sum_k \Big[(1 + n_k) \ln(1 + n_k) - n_k \ln n_k \Big]. \tag{2.104}$$

This expression may also be used locally for a nonequilibrium phonon gas, provided the distribution is still well described by local mode occupations and the anharmonic perturbation $\hat{V}'$ that couples the phonon normal modes remains weak. At local thermal equilibrium, maximising the entropy subject only to fixed local internal energy Eq. (2.73) gives the local Bose–Einstein distribution Eq. (2.71), as anticipated. We may further verify the harmonic phonon entropy expression in Eq. (2.104) by making another connection back to classical thermodynamics. The most compact statement of the second law is the Clausius inequality,

$$\mathrm{d}S \geq \frac{đq}{T}. \tag{2.105}$$

For a reversible process, the equality holds:

$$\mathrm{d}S \equiv \frac{đq_{\mathrm{rev}}}{T}, \tag{2.106}$$

which serves as the definition of entropy in classical thermodynamics. In Appendix G.II, we show that the phonon entropy above recovers this thermodynamic relation at equilibrium. The entropy analogue of the energy continuity equation is the entropy balance equation [24, 31]

$$\frac{\partial s}{\partial t} + \nabla \cdot \mathbf{J}_s = \sigma, \tag{2.107}$$

where $s(\mathbf{r}, t)$ and $\mathbf{J}_s$ are the local entropy density and the entropy flux density, defined similarly to $u(\mathbf{r}, t)$ and $\mathbf{J}$. σ is the local entropy production rate. In the regime where the local phonon mean occupation number is a *linear* deviation δn_k from the

local equilibrium mean occupation number $n_k^0(T)$, the entropy flux density is

$$\mathbf{J}_s = \frac{\mathbf{J}}{T}, \tag{2.108}$$

and

$$\sigma = \mathbf{J} \cdot \nabla \frac{1}{T} = \frac{1}{T^2} (\nabla T)^\intercal \boldsymbol{\kappa} \nabla T. \tag{2.109}$$

The thermal conductivity tensor is always positive semi-definite in the sense that $(\nabla T)^\intercal \boldsymbol{\kappa} \nabla T \geq 0$. Substituting Eq. (2.108) into Eq. (2.107) and using $\sigma \geq 0$ gives the *local* version of the second law,

$$\frac{\partial s}{\partial t} \geq -\nabla \cdot \left(\frac{\mathbf{J}}{T} \right). \tag{2.110}$$

Thus, heat conduction can redistribute entropy through the entropy flux $\mathbf{J}/T$, but it cannot produce negative entropy locally. We also show in Appendix G.II that the local harmonic phonon entropy density under the evolution of BTE recovers this local version of the second law. This provides us with confidence in Eq. (2.104).

In this near-local-equilibrium regime, one also obtains the following relation

$$\frac{1}{T(\mathbf{r}, t)} = \frac{\partial s(\mathbf{r}, t)}{\partial u(\mathbf{r}, t)}, \tag{2.111}$$

which justifies the use of a local temperature in the preceding sections.

To analyse normal processes, we now maximise the harmonic phonon entropy subject to both fixed harmonic phonon internal energy density and fixed harmonic phonon crystal momentum density, [33]

$$\mathbf{P}(\mathbf{r}, t) = \frac{1}{\mathcal{V}} \sum_k \hbar \mathbf{q} n_k(\mathbf{r}, t). \tag{2.112}$$

The resulting distribution is the drifting Bose–Einstein distribution, [34]

$$n_k^d[T(\mathbf{r}, t), \boldsymbol{\mu}(\mathbf{r}, t)] = \frac{1}{e^{(\hbar\omega_k - \hbar\mathbf{q}\cdot\boldsymbol{\mu}(\mathbf{r},t))/k_B T(\mathbf{r},t)} - 1}. \tag{2.113}$$

Here $\boldsymbol{\mu}(\mathbf{r}, t)$ is a set of Lagrange multipliers identified as a drift velocity associated with conserved $\mathbf{P}$.

If only normal scattering processes are present, this drifting distribution maximises entropy. Unlike the ordinary Bose–Einstein distribution, it is not symmetric under $k \to -k$:

$$n_k^d \neq n_{-k}^d. \tag{2.114}$$

Therefore, the heat flux

$$\mathbf{J} = \frac{1}{\mathcal{V}} \sum_k \hbar\omega_k \mathbf{v}_k n_k^d \tag{2.115}$$

does not vanish in general. Normal processes therefore relax the phonon gas towards a drifting distribution, rather than directly dissipate the heat flux. This explains why the relaxation-time approximation could fail when normal processes dominate: the RTA assumes that each phonon mode independently relaxes to the Bose–Einstein distribution n_k^0. It is therefore reasonable when U-processes dominate, but it is not appropriate when N-processes dominate, especially at low temperature.

A more complete treatment keeps the full linearised scattering matrix. We introduce the symmetrised scattering matrix, [35]

$$\tilde{\Omega}_{kk'} \equiv \Omega_{kk'} \frac{\sqrt{n_{k'}^0(n_{k'}^0+1)}}{\sqrt{n_k^0(n_k^0+1)}}, \tag{2.116}$$

and define its eigenvectors θ_k^α by

$$\sum_{k'} \tilde{\Omega}_{kk'} \theta_{k'}^\alpha = \frac{1}{\tau_\alpha} \theta_k^\alpha. \tag{2.117}$$

These eigenvectors are termed *relaxons*. [36] $\tilde{\Omega}$ is real and symmetric, so its eigenvalues are real and its eigenvectors are orthonormal and form a complete set in the linearised scattering space on which $\tilde{\Omega}$ acts. The scaled deviation

$$\delta\tilde{n}_k \equiv \frac{\delta n_k}{\sqrt{n_k^0(n_k^0+1)}} \tag{2.118}$$

can therefore be expanded as a linear combination of the relaxons,

$$\delta\tilde{n}_k = \sum_\alpha f_\alpha \theta_k^\alpha, \tag{2.119}$$

where the expansion coefficients $\{f_\alpha\}$ are the relaxon occupation numbers. We also introduce the normalised heat-capacity vector

$$\theta_k^0 = \sqrt{\frac{C_k}{C}}, \qquad C = \sum_k C_k. \tag{2.120}$$

The steady-state LBTE may then be recast into the relaxon representation, as shown in Appendix F,

$$\sqrt{\frac{C}{k_B T^2}}\, \mathbf{V}_\alpha \cdot \nabla T = -\frac{f_\alpha}{\tau_\alpha}, \tag{2.121}$$

where

$$\mathbf{V}_\alpha = \sum_k \theta_k^0 \mathbf{v}_k \theta_k^\alpha \tag{2.122}$$

is the relaxon velocity. Although we can define this relaxon velocity, the group velocity matrix

$$\mathbf{v}_{\alpha\alpha'} \equiv \sum_k \theta_k^\alpha \mathbf{v}_k \theta_k^{\alpha'} \tag{2.123}$$

in the relaxon basis is in general not diagonal; thus different relaxon modes are coupled by the group velocity matrix. The harmonic heat flux density can be written as

$$\mathbf{J} = \frac{\sqrt{C k_B T^2}}{\mathcal{V}} \sum_\alpha \mathbf{V}_\alpha f_\alpha. \tag{2.124}$$

Substituting the solution for f_α gives

$$\mathbf{J} = -\frac{C}{\mathcal{V}} \sum_\alpha \mathbf{V}_\alpha \mathbf{V}_\alpha^\intercal \tau_\alpha \nabla T \tag{2.125}$$

and comparison with Fourier's law gives the relaxon expression for the thermal-conductivity tensor in closed form, [36]

$$\boldsymbol{\kappa} = \frac{C}{\mathcal{V}} \sum_\alpha \mathbf{V}_\alpha \mathbf{V}_\alpha^\intercal \tau_\alpha. \tag{2.126}$$

This resembles the relaxation-time expression in Eq. (2.95), but it uses the full scattering matrix rather than independent phonon lifetimes. In this representation,

the heat capacity prefactor is the total crystal heat capacity C, rather than the mode-dependent phonon heat capacity C_k. We show in Appendix G.III that

$$\sigma = \left(\frac{\partial s}{\partial t}\right)_{\text{scat}} = \frac{k_B}{\mathcal{V}} \sum_{kk'} \delta\tilde{n}_k \tilde{\Omega}_{kk'} \delta\tilde{n}_{k'} \geq 0. \tag{2.127}$$

Thus, $\tilde{\Omega}$ is positive semi-definite, so the eigenvalues τ_α^{-1} are non-negative. However, we may still have $\tau_\alpha^{-1} = 0$. It turns out that the θ_k^0 defined in Eq. (2.120) is a special relaxon mode that is annihilated by the symmetrised scattering operator,

$$\sum_{k'} \tilde{\Omega}_{kk'} \theta_{k'}^0 = 0. \tag{2.128}$$

This is due to the conservation of energy for all scattering processes. Thus, θ_k^0 is an eigenmode of $\tilde{\Omega}_{kk'}$ with eigenvalue $\tau_0^{-1} = 0$. The inverse of $\tilde{\Omega}$ is therefore defined only as a pseudoinverse on the subspace excluding this conserved mode. We show in Appendix F.II that the θ_k^0 relaxon mode does not contribute to thermal conductivity, so this mode may be safely excluded as is done when calculating the pseudoinverse.

At low temperature, where normal processes dominate, the relaxation time approximation that assigns each phonon with mean free path $\mathbf{v}_k \tau_k^{\text{RTA}}$ becomes inadequate. The phonon distribution relaxes collectively and the eigenmodes of the full symmetrised scattering matrix $\tilde{\Omega}$ are relaxon modes rather than phonon normal modes. The relaxon picture is therefore especially useful in the hydrodynamic regime of heat transport, where heat flow resembles a collective fluid-like motion of the phonon gas. Each relaxon has a well-defined relaxation time τ_α, effective velocity $\mathbf{V}_\alpha$, and mean free path $\mathbf{V}_\alpha \tau_\alpha$ so the thermal conductivity can be interpreted as a kinetic theory of relaxons.

Mathematically, the phonons and relaxons form a complete set within the same linearised scattering space. Any linear response of the system with respect to a gentle external perturbation ∇T can be expressed as a superposition of relaxons, with expansion coefficients $\{f_\alpha\}$ or just as well as a superposition of phonons, with

expansion coefficients $\{\delta\tilde{n}_k\}$. Observables such as the local entropy production rate σ, the heat flux density $\mathbf{J}$, and the thermal conductivity $\boldsymbol{\kappa}$ are inner products within this vector space and can be evaluated in either representation.

Physically, classical thermodynamics is fundamentally independent of the atomic or molecular hypothesis. Consequently, its macroscopic description of heat transfer is also independent of any specific underlying quasiparticle picture. Although Eq. (2.128) and Eq. (2.127) are currently written in the phonon basis, they can equivalently be cast in the relaxon basis. When expressed in this way, they may be regarded as the first and second laws of thermodynamics within the linearised scattering space, respectively.

3
Results and Discussion

Contents

3.1 Pair Potential 32
3.2 Radial Distribution Function 37
3.3 Dispersion Relation 40
3.4 Heat Capacity 43
3.5 Thermal Conductivity 46

The results in this chapter follow the same logical order as in Chapter 2. Section 3.1 implements the path integral molecular dynamics (PIMD) and centroid potential of mean force (CPMF) methods described in Sections 2.1 and 2.2. Section 3.2 presents the calculated centroid pair radial distribution functions (RDFs) of solid p-H_2 and verifies the validity of the pairwise approximation to the many-body centroid potential of mean force. Sections 3.3 and 3.4 compare the harmonic phonon properties obtained from the classical potential and the CPMFs. Finally, Section 3.5 presents the calculated thermal conductivity by solving the linearised Boltzmann transport equation (LBTE) in the phonon and relaxon bases. We compare our calculated results with the experimental data. [37–40]

3.1 Pair Potential

We begin with the computational details. As described in Chapter 1, the pairwise interaction between $J = 0$ p-H_2 can be modelled by an isotropic pair potential. We use the isotropic part of the Silvera–Goldman pair potential [41], as has been successfully done in the majority of previous work on liquid p-H_2, [6, 7, 42–49]

$$v(r) = e^{\alpha-\beta r-\gamma r^2} - \left[\frac{C_6}{r^6} + \frac{C_8}{r^8} - \frac{C_9}{r^9} + \frac{C_{10}}{r^{10}}\right] f(r), \tag{3.1}$$

where

$$f(r) = \begin{cases} e^{-(r_c/r-1)^2}, & r \leq r_c, \\ 1, & r > r_c. \end{cases} \tag{3.2}$$

α	1.713	C_6	12.14
β	1.5671	C_8	215.2
γ	0.00993	C_9	143.1
r_c	8.32	C_{10}	4813.9

Table 3.1: Parameters of the Silvera–Goldman pair potential in atomic units.

At short range, the first exponential term dominates as the two para-hydrogen molecules experience strong repulsion due to the overlap of their electronic orbitals. At long range, the terms in the square brackets dominate, which are dispersion interactions that arise from quantum mechanical fluctuations of electron density. In particular, r^{-6} is characteristic of the dispersion interaction and r^{-8} and r^{-10} are due to interactions involving multipoles. All of these interactions are attractive. The C_9 term is added to include a pairwise approximation to the three-body tripole-dipole repulsive dispersion interaction. [41]

The centroid potential of mean force $v_c(r)$ is obtained from PIMD simulations through, [7]

$$v_c(r) = \int_r^{r_{\text{cut}}} \mathrm{d}r' \, \langle f_{ij}^c \rangle_{r_{ij}^c = r'}, \tag{3.3}$$

as described in section 2.2. We used a cutoff distance of $r_{\text{cut}} = 12\,\text{Å}$. In the PIMD simulations, we used the experimental lattice parameters $a = 3.79\,\text{Å}$ and $c = 6.19\,\text{Å}$, for the hcp structure, which are measured under the condition of approximately zero external pressure. [4] The simulation cell we used contains 448 p-H_2 molecules. Periodic boundary conditions were applied using the minimum image convention.

At each temperature, the system was first equilibrated for 10 ps by running an NVT trajectory. We then ran a further 1000 ps NVT dynamics with a time step of 2 fs. We found that $n = 128$ ring polymer beads were sufficient to converge all results at 5 K. To check that the code was working correctly, we repeated the calculation with an $n = 1$ bead ring polymer, and found that the resulting CPMF agreed with the classical potential to graphical accuracy. Sampling from the canonical ensemble is achieved by attaching a path integral Langevin equation (PILE) thermostat to the internal modes of the ring polymer, [15] and a canonical sampling via velocity rescaling (CSVR) thermostat to the centroid mode. [16] We used a standard symplectic algorithm to integrate the equations of motion of the ring polymer, which can be written as

$$e^{-\mathcal{L}_n \delta t} \simeq e^{-\mathcal{L}_V \delta t/2} e^{-\mathcal{L}_0 \delta t} e^{-\mathcal{L}_V \delta t/2} \tag{3.4}$$

in the NVE case, and

$$e^{-\mathcal{L} \delta t} \simeq e^{-\mathcal{L}_\gamma \delta t/2} e^{-\mathcal{L}_V \delta t/2} e^{-\mathcal{L}_0 \delta t} e^{-\mathcal{L}_V \delta t/2} e^{-\mathcal{L}_\gamma \delta t/2} \tag{3.5}$$

in the NVT case, where $\mathcal{L}_n = \mathcal{L}_0 + \mathcal{L}_V$ is the Liouvillian associated with the ring polymer Hamiltonian H_n, $\mathcal{L}_\gamma$ is that associated with the thermostat (see below), and $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_V + \mathcal{L}_\gamma$. Here, $e^{-\mathcal{L}_0 \delta t}$ consists of first transforming the ring polymer to its normal modes

$$P_k = \sum_{j=1}^{n} p_j C_{jk}, \qquad Q_k = \sum_{j=1}^{n} q_j C_{jk}, \tag{3.6}$$

then evolving each individual normal mode by δt,

$$\begin{pmatrix} P_k(t+\delta t) \\ Q_k(t+\delta t) \end{pmatrix} = \begin{pmatrix} \cos\omega_k\delta t & -m\omega_k \sin\omega_k\delta t \\ (1/m\omega_k)\sin\omega_k\delta t & \cos\omega_k\delta t \end{pmatrix} \begin{pmatrix} P_k(t) \\ Q_k(t) \end{pmatrix}, \tag{3.7}$$

then transforming back to the atomic Cartesian variables,

$$p_j = \sum_{k=0}^{n-1} C_{jk} P_k, \qquad q_j = \sum_{k=0}^{n-1} C_{jk} Q_k. \tag{3.8}$$

Evolution under the physical potential, $e^{-\mathcal{L}_V \delta t/2}$, is understood to be

$$p_j(t+\delta t/2) = p_j(t) - \frac{\delta t}{2}\frac{\partial V(q_j(t))}{\partial q_j(t)}, \tag{3.9}$$

and the thermostatting steps $e^{-\mathcal{L}_\gamma \delta t/2}$ are understood to be,

$$P_k(t+\delta t/2) = e^{-\gamma_k \delta t/2} P_k(t) + \sqrt{\frac{m(1-e^{-\gamma_k \delta t})}{\beta_n}}\xi_k(t) \tag{3.10}$$

in the normal mode representation. C_{jk}, γ_k, ξ_k are given in Eq. (2.16), Eq. (2.21) and Eq. (2.22) respectively. Note that transforming into normal modes of the ring polymer allows a longer time step δt than would be possible by propagating all degrees of freedom directly in Cartesian bead coordinates. This is because the high-frequency ring polymer modes with $\omega_k > \omega_n = n/(\beta\hbar)$, may be integrated analytically in the normal-mode representation.

At each time step during the NVT production run, the vector $\mathbf{r}_{ij}^c$ connecting the centroids between a pair of ring polymers is calculated. If the distance $r_{ij}^c < r_{\text{cut}}$, it is placed within a histogram bin of width $0.01\,a_0$ ($1\,a_0 = 0.529\,\text{Å}$), between 0 and r_{cut}. The centroid radial forces f_{ij}^c are then accumulated within each histogram bin, and averaged over the length of the simulation. [50] The resulting mean forces are integrated inwards from r_{cut} to r to obtain the centroid potential of mean force $v_c(r)$. Since statistics may be poor in regions between the coordination shells, the data $\{r_i, v_c(r_i)\}$ generated from the PIMD simulation are fitted to a smooth function of the form

$$v_c^{\text{fit}}(r) = \sum_{k=1}^{m} c_k \left[\left(\frac{r_{\text{cut}}}{r}\right)^{k+5} - 1\right] \tag{3.11}$$

up to a maximum value of $v_c = 10\,k_B T$, assuming $v(r_{\mathrm{cut}}) = 0$. In practice, $m = 10$ was found to be sufficiently accurate to reach graphical agreement with the data from the PIMD simulation. The fitting is a linear least squares problem of finding the $\mathbf{x}$ that minimises the norm

$$\min_{\mathbf{x}} \|\mathbf{b} - \mathbf{A}\mathbf{x}\|, \tag{3.12}$$

where

$$b_j = v_j(r_j), \qquad A_{jk} = \left(\frac{r_{\mathrm{cut}}}{r_j}\right)^{k+5} + 1, \qquad x_k = c_k. \tag{3.13}$$

Once the set of expansion coefficients $\{c_k\}$ is determined, the forces can be evaluated analytically, by differentiating Eq. (3.11) with respect to r,

$$f_c(r) = -v_c'(r) = \frac{1}{r}\sum_{k=1}^{m} c_k(k+5)\left(\frac{r_{\mathrm{cut}}}{r}\right)^{k+5}. \tag{3.14}$$

We chose to use this fitting form, which is infinitely differentiable, instead of the more commonly used cubic spline. This is because the calculation of third-order force constants (FC3) requires smooth and continuous derivatives of the potential up to the third derivative. This is not the case for a cubic spline, which has discontinuous third derivatives at its nodes. The code for PIMD and fitting was provided by Prof. Manolopoulos.

Figure 3.1 compares the classical Silvera–Goldman (SG) pair potential with the centroid potential of mean force (CPMF) at 10 K and 5.4 K. The first thing to notice is that the classical potential is temperature independent, whereas the centroid potentials of mean force are temperature dependent. At the two temperatures shown, the centroid potential of mean force differs significantly from the classical potential. [50] It is more repulsive at short range and more attractive at long range, and both effects become more pronounced as the temperature is lowered. This behaviour may be rationalised using the ring polymer picture. The radius

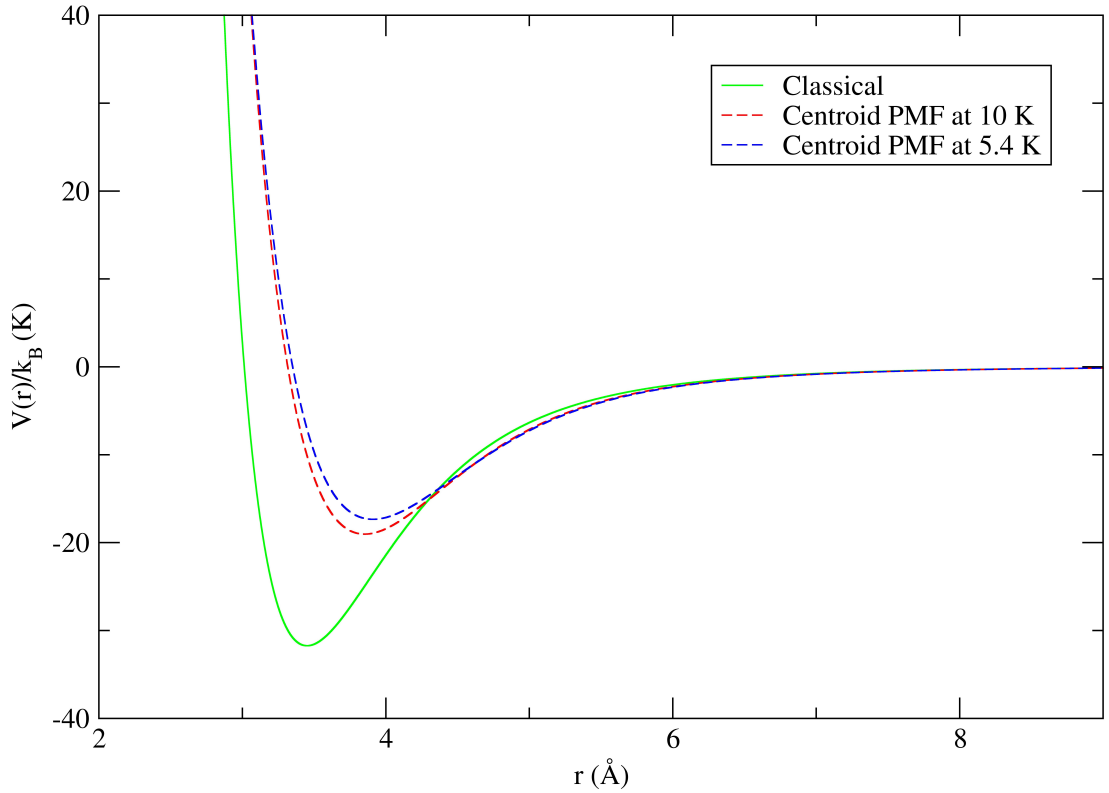


Figure 3.1: Comparison between the classical pair potential and the centroid potentials of mean force at two different temperatures.

of gyration of a free ring polymer is [6]

$$\Delta q = r_G = \frac{\hbar}{2\sqrt{mk_BT}}. \tag{3.15}$$

For p-H_2, this gives $r_G \simeq 0.75\,\text{Å}$ at 10 K and $r_G \simeq 1.02\,\text{Å}$ at 5.4 K. The swelling of the ring polymer is a physical representation of quantum delocalisation of the nuclei, and it becomes more pronounced at lower temperature, where quantum effects are expected to be more important. The ring polymer also has uncertainty in momentum from its phase space distribution

$$\Delta p = \sqrt{mk_BT}, \tag{3.16}$$

which together with Δq gives

$$\Delta q \Delta p = \frac{\hbar}{2}. \tag{3.17}$$

This is reminiscent of the Heisenberg uncertainty principle. A subtlety is that the

expression for the Heisenberg uncertainty principle for a ground state harmonic oscillator is $\Delta q \Delta p = \hbar/2$ in one dimension, but r_G given in Eq. (3.15) is in three dimensions, so the connection is not exact. Nevertheless, this provides a useful physical picture for interpreting the results.

The centroid potentials of mean force are shallower compared to the classical potential. This may be understood as the ring polymer incorporates zero-point energy, whereas the classical potential does not. At short range, this delocalisation increases the effective overlap between neighbouring ring polymers, leading to enhanced repulsion. At longer range, some bead pairs are closer than all of the centroids of the ring polymers to which they belong, which enhances the attractive part of the interaction. The classical potential may be regarded as the high-temperature limit of the CPMF, where the ring polymer has collapsed onto its centroid, and the quantum partition function reduces to the classical partition function.

3.2 Radial Distribution Function

The radial distribution function is defined as

$$g(r) \equiv \frac{\rho(r)}{\rho}, \tag{3.18}$$

where $\rho(r)$ is the radially averaged number density at distance r, and ρ is the bulk number density. The average number of molecules $\mathrm{d}N(r)$ in a spherical shell between r and $r + \mathrm{d}r$ is therefore

$$\mathrm{d}N(r) = 4\pi r^2 \rho g(r)\, \mathrm{d}r. \tag{3.19}$$

In principle, the centroid radial distribution function may be computed from

$$g_c(r) = \frac{1}{4\pi r^2 N \rho} \left\langle \sum_{i=1}^{N} \sum_{j \neq i}^{N} \delta(r - r_{ij}^c) \right\rangle. \tag{3.20}$$

Here, N is the total number of molecules, r_{ij}^c is the distance between the centroids of the molecules i and j, $\delta(x)$ is the Dirac delta function, and the average is over

the canonical ensemble, as described in Chapter 2. The equation above may be computed by counting pairs of centroids separated by a distance between r and $r+\Delta r$, where Δr is the width of the histogram bin for a given configuration. Then it is averaged over the configurations generated by the molecular dynamics simulation. In practice, we used the force-based method of Borgis and Rotenberg, [51, 52]

$$g_c(r) = \frac{\beta}{4\pi N\rho}\left\langle \sum_{i=1}^{N}\sum_{j\neq i}^{N} \frac{1}{2(r_{ij}^c)^3}(\mathbf{F}_i^c - \mathbf{F}_j^c)\cdot(\mathbf{r}_j^c - \mathbf{r}_i^c)\theta(r - r_{ij}^c)\right\rangle, \tag{3.21}$$

where $\mathbf{F}_i^c$ is the total force on the centroid of molecule i, as described in Chapter 2, and $\theta(r - r_{ij}^c)$ is the Heaviside step function, defined as

$$\theta(x) \equiv \begin{cases} 1, & x \geq 0, \\ 0, & x < 0. \end{cases} \tag{3.22}$$

This method gives faster convergence than direct histogram binning of pair separations. The delta function is replaced by a step function for the force-based method; thus, for large r, all samples with $r_{ij}^c < r$ contribute to the average. This is likely the reason for the fast convergence of the force-based method compared to direct histogram binning. The centroid radial distribution functions are calculated simultaneously with the centroid potential of mean force in the PIMD simulation.

Figure 3.2 compares the centroid radial distribution functions calculated from PIMD, CMD and classical molecular dynamics simulation at 5.4 K. The RDFs computed from PIMD and CMD agree to graphical accuracy, which supports the use of the pairwise approximation to the many-body interaction potential

$$V_c(\mathbf{r}^c) \simeq \frac{1}{2}\sum_{i=1}^{N}\sum_{j\neq i}^{N} v_c(r_{ij}^c). \tag{3.23}$$

The peaks in Figure 3.2 indicate the coordination shells, and the troughs indicate empty spaces between the coordination shells. The persistent oscillations of centroid RDFs indicate long-range order, which suggests that both classical molecular dynamics and CMD are simulating a crystalline solid at 5.4 K. In the classical molecular dynamics simulation, the first peak is at about 3.35 Å. In the

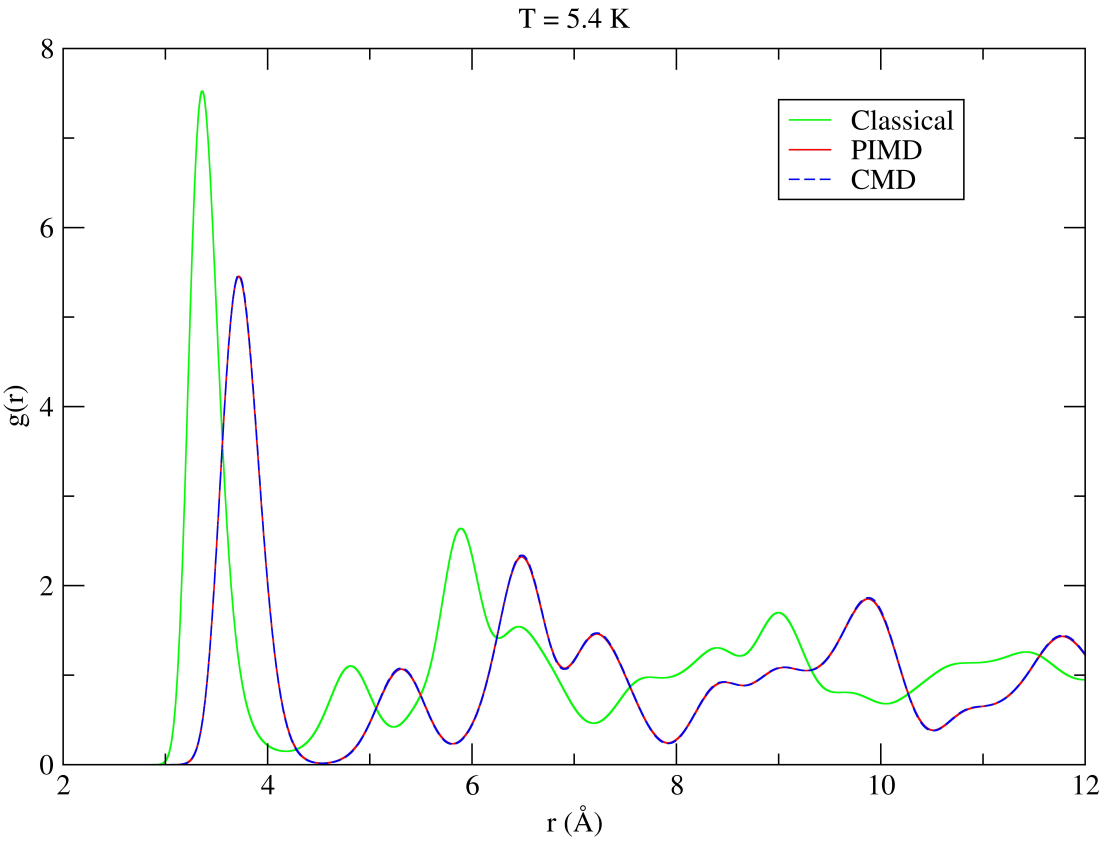


Figure 3.2: Radial distribution functions calculated using the bare classical potential (classical) and the centroid potential of mean force (CMD), compared with the quantum mechanically exact PIMD centroid radial distribution function.

CMD simulation, the first peak shifts to about 3.75 Å. This suggests the short-range repulsion in the centroid potential of mean force has pushed the centroids apart. The first peak is also lower in intensity and slightly broader in the CMD simulation. This is a reflection of thermal quantum fluctuations: the centroids of the p-H_2 molecules are not so sharply localised at their equilibrium position at 5.4 K. Overall, the entire profile of RDF for CMD is shifted to larger distances. Notice that $g(r)$ vanishes at roughly 4.5Å between the first and second coordination shells, which implies that there were few samples of the average radial centroid mean force $\langle f^c \rangle$ close to this centroid separation. This was one of the reasons to fit the data from the simulation to a smooth v_c^{fit} curve.

3.3 Dispersion Relation

The dispersion relations are obtained by diagonalising the dynamical matrix,

$$D(\mathbf{q})\boldsymbol{\epsilon}_s(\mathbf{q}) = \omega^2_{\mathbf{q}s}\boldsymbol{\epsilon}_s(\mathbf{q}), \tag{3.24}$$

where

$$D_{b\alpha,b'\alpha'}(\mathbf{q}) \equiv \frac{1}{\sqrt{m_b m_{b'}}} \sum_{l'} \Phi_{0b\alpha,l'b'\alpha'} e^{i\mathbf{q}\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'}-\boldsymbol{\tau}_b)}. \tag{3.25}$$

A custom Fortran code was used to calculate the second-order force constants from finite displacements,

$$\Phi_{l'b'\alpha',0b\alpha} = -\frac{\partial F_{l'b'\alpha'}}{\partial u_{0b\alpha}} \simeq -\frac{F_{l'b'\alpha'}(+\delta u_{0b\alpha}) - F_{l'b'\alpha'}(-\delta u_{0b\alpha})}{2\delta u_{0b\alpha}}. \tag{3.26}$$

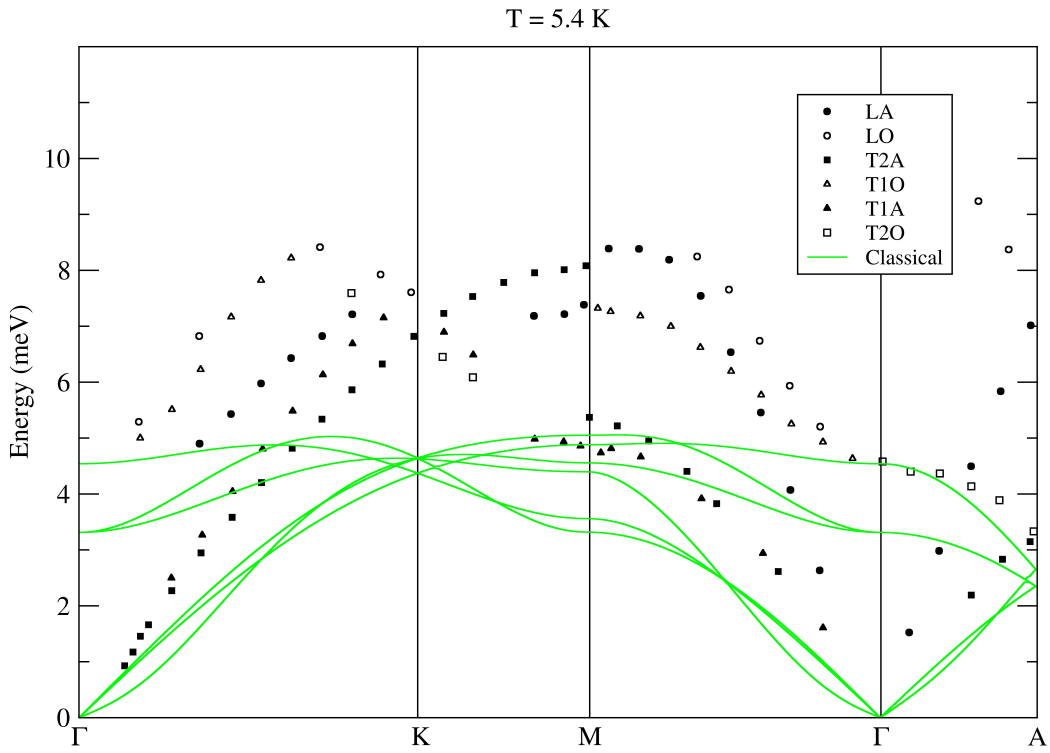
with a displacement of $\delta u = 8.08 \times 10^{-6}$ Å, a force cutoff distance of 12 Å, and a $7 \times 7 \times 7$ supercell containing 686 p-H_2 molecules. Periodic boundary conditions were applied, and the mass-weighted Hessian was symmetrised before diagonalising along the chosen path in the Brillouin zone. The calculation was also repeated using an external Python package `phonopy` [2, 3] with a $6 \times 6 \times 3$ supercell containing 216 p-H_2 molecules, $\delta u = 0.01$ Å, and a force cutoff distance of 9 Å. The phonon dispersion curves calculated with the Python code and the Fortran code were found to agree to graphical accuracy.

The first Brillouin zone is defined as the region of reciprocal space closer to a central reciprocal lattice point (the Γ-point) than to any other. For the hcp structure, the Brillouin zone is a hexagonal prism. The primitive lattice vectors used here are [54]

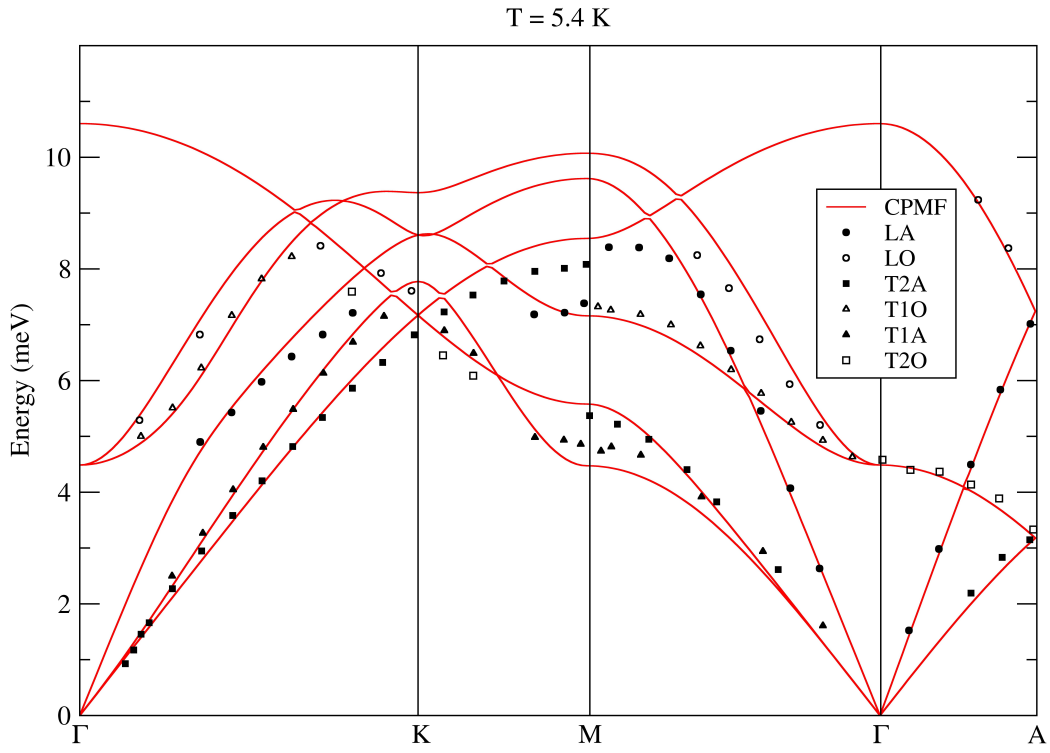
$$\mathbf{a}_1 = a\hat{\mathbf{x}}, \qquad \mathbf{a}_2 = \frac{a}{2}\hat{\mathbf{x}} + \frac{\sqrt{3}a}{2}\hat{\mathbf{y}}, \qquad \mathbf{a}_3 = c\hat{\mathbf{z}}, \tag{3.27}$$

where $\{\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}\}$ are basis vectors in real space. The corresponding reciprocal lattice vectors are

$$\mathbf{b}_1 = \frac{2\pi}{\sqrt{3}a}\left(\sqrt{3}\hat{\mathbf{k}}_x - \hat{\mathbf{k}}_y\right), \qquad \mathbf{b}_2 = \frac{4\pi}{\sqrt{3}a}\hat{\mathbf{k}}_y, \qquad \mathbf{b}_3 = \frac{2\pi}{c}\hat{\mathbf{k}}_z, \tag{3.28}$$



(a) Classical Silvera–Goldman potential.



(b) Centroid potential of mean force.

Figure 3.3: Phonon dispersions calculated using the bare classical potential and the centroid potential of mean force, compared with neutron-scattering measurements. [37]

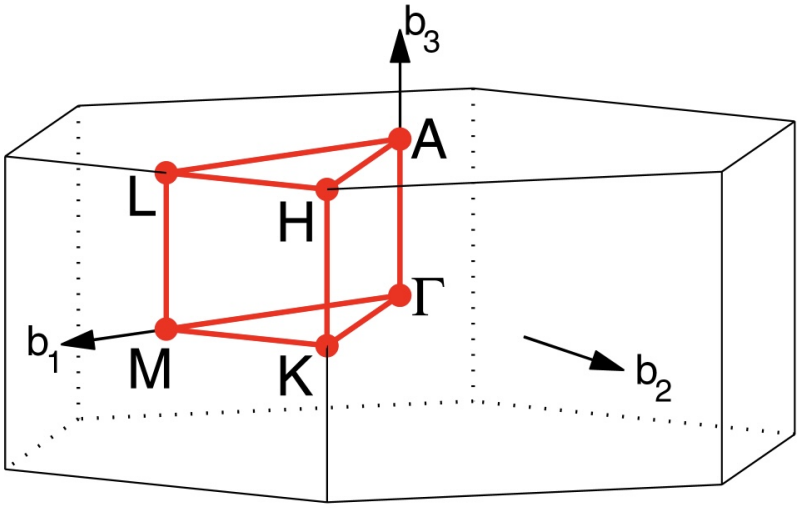


Figure 3.4: Hexagonal-prism Brillouin zone of the hcp structure and the high-symmetry path used for the dispersion calculation. [53]

where $\{\hat{\mathbf{k}}_x, \hat{\mathbf{k}}_y, \hat{\mathbf{k}}_z\}$ are basis vectors in reciprocal space. The reciprocal lattice vectors satisfy

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}. \tag{3.29}$$

The path used for a direct comparison with the neutron scattering experimental data is

$$\Gamma(0,0,0) \xrightarrow{\Lambda} K(2/3,1/3,0) \xrightarrow{T} M(1/2,0,0) \xrightarrow{\Sigma} \Gamma \xrightarrow{\Delta} A(0,0,1/2). \tag{3.30}$$

The high-symmetry lines are parametrised as

$$\Lambda(2\nu,\nu,0), \qquad \nu : 0 \to \frac{1}{3}, \tag{3.31}$$

$$T(1/2+\nu/2,\nu,0), \qquad \nu : \frac{1}{3} \to 0, \tag{3.32}$$

$$\Sigma(u,0,0), \qquad u : \frac{1}{2} \to 0, \tag{3.33}$$

$$\Delta(0,0,w), \qquad w : 0 \to \frac{1}{2}. \tag{3.34}$$

The two molecules in the primitive cell are placed at fractional coordinates $(0,0,0)$ and $(1/3,1/3,1/2)$. Since each molecule has three displacement directions, there are six phonon branches at each $\mathbf{q}$-point. Three are acoustic branches, whose

frequencies vanish linearly near the Γ-point, and three are optical branches, whose frequencies remain finite at Γ.

The calculated phonon dispersion relations are compared with neutron-scattering experiments in Figures 3.3a and 3.3b. Figure 3.3a shows the result obtained using the classical potential, while Figure 3.3b shows the result obtained using the CPMF, both at 5.4 K. The phonon dispersion calculated from the centroid potential of mean force shows much better agreement with experiment than the dispersion calculated from the classical potential. The classical potential severely underestimates the phonon frequencies across almost all branches, whereas the centroid potential gives quantitative agreement for several acoustic modes.

The higher phonon frequencies calculated from CPMF are explained by the centroid potential being more repulsive at short range. The stronger long-range attraction increases the phonon bandwidth. The remaining discrepancies between the phonon dispersion calculated from CPMF and the neutron scattering data are most visible in the optical branches. These discrepancies are acceptable for low-temperature heat capacity and thermal conductivity calculations, because at 5.4 K one has $k_B T \simeq 0.47\,\mathrm{meV}$, so the high-energy optical modes are negligibly populated.

3.4 Heat Capacity

The harmonic heat capacity of a phonon mode is

$$C_{\mathbf{q}s} = k_B \left(\frac{\hbar \omega_{\mathbf{q}s}}{k_B T} \right)^2 n^0_{\mathbf{q}s}(n^0_{\mathbf{q}s} + 1), \tag{3.35}$$

with

$$n^0_{\mathbf{q}s} = \frac{1}{e^{\hbar \omega_{\mathbf{q}s}/k_B T} - 1}. \tag{3.36}$$

The total harmonic heat capacity is

$$C = \sum_{\mathbf{q}s} C_{\mathbf{q}s}, \tag{3.37}$$

with the appropriate normalisation applied when compared with molar experimental data.

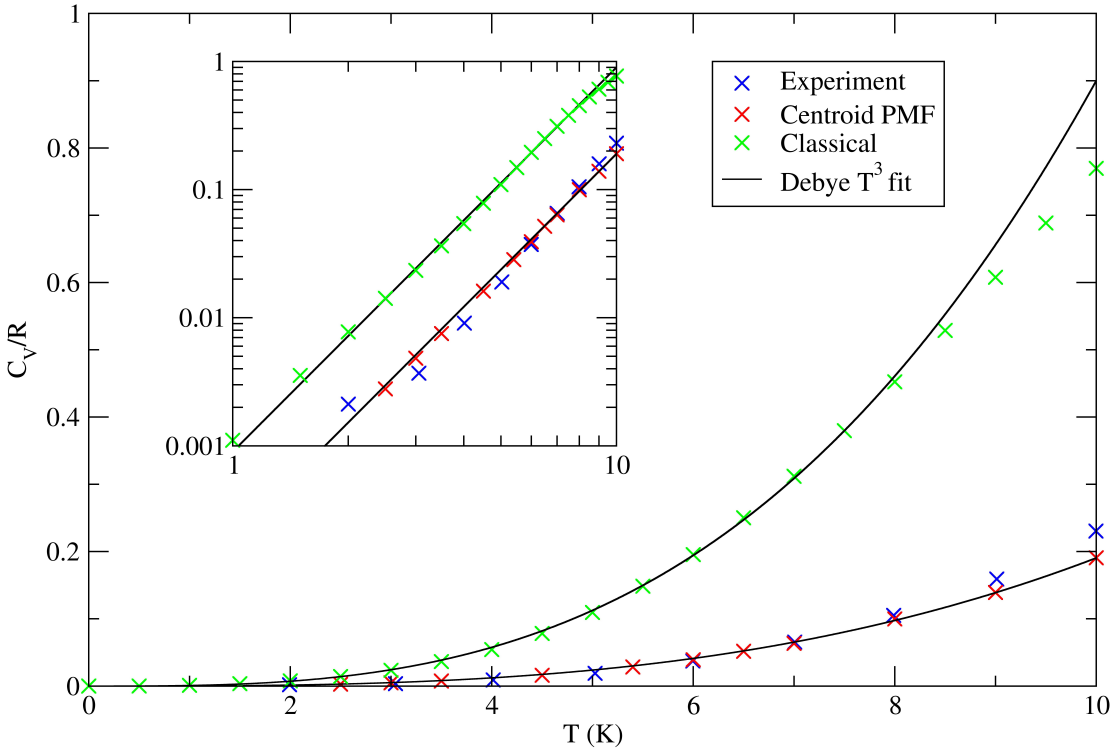


Figure 3.5: Constant-volume heat capacity calculated using the classical potential and the centroid potential of mean force, compared with experiment. [38]

The constant-volume heat capacities calculated using the Silvera–Goldman potential and the CPMF are shown in Figure 3.5. Both were calculated using the Python package `Phonopy`, [2,3] and found to converge using a **q**-mesh of $25 \times 25 \times 15$ points in the Brillouin zone. At low temperature, both the classical and CPMF calculations approximately follow the Debye law $C_V \sim T^3$ [54,55] as can be seen both from the main panel and the inset panel. This is expected at low temperature when only acoustic modes are significantly populated. The heat capacity calculated from the classical potential begins to deviate from the T^3 fit above about 8 K as it moves too rapidly towards the classical limit of $C_V = 3\,R$. This is because the higher frequency acoustic and even some optical modes are starting to be occupied.

The heat capacity calculated using the CPMF clearly agrees much better with

experiment than the heat capacity calculated using the classical SG potential. This is anticipated since the dispersion relations calculated from the CPMF were also in much better agreement with experiment, especially in the acoustic branches. The remaining discrepancy between the heat capacity calculated from CPMF and experiment may arise from inaccuracies in the SG potential and/or anharmonic contributions to the experimental heat capacity. The heat capacity formula in Eq. (3.35) contains harmonic phonon frequencies $\omega_{\mathbf{q}s}$, and thus ignores any anharmonic frequency shifts. Nevertheless, the comparisons of both phonon dispersion relations and heat capacities with experiment strongly suggest that the centroid potential of mean force provides a much better starting point for thermal conductivity calculations than the classical potential. We should therefore abandon the classical potential from this point onward.

The Silvera–Goldman potential is a semiempirical pair potential. Since it was fitted using experimental solid-state properties, one might expect some quantum effects of the real solid to be implicitly absorbed into the fitted parameters. However, it remains a temperature-independent bare interaction potential between molecules with well-defined separation. It does not explicitly include the thermal quantum fluctuation of the nuclei. In contrast, the centroid potential of mean force is a temperature-dependent free energy obtained from a NVT PIMD simulation.

The importance of including quantum effects in condensed-phase dynamical properties was already demonstrated in earlier studies of quantum diffusion in liquid p-H_2. [6, 7, 42–49] At a physical state point corresponding to the quantum liquid, a purely classical molecular dynamics simulation using the SG potential lies in the liquid-vapour coexistence region of the phase diagram. As such, when performing a classical molecular dynamics simulation at these state points, a bubble will spontaneously form within the simulation. [6] By contrast, ring polymer molecular dynamics (RPMD) simulations lie correctly in the liquid phase close to the saturated vapour line of the quantum thermodynamic phase diagram. RPMD incorporates the

quantum Boltzmann distribution of the nuclei and gives self-diffusion coefficients in good agreement with experiments. [6] The inclusion of quantum effects is expected to be necessary for lattice thermal conductivity calculations of solid p-H_2 as well.

3.5 Thermal Conductivity

We calculate lattice thermal conductivity via lattice dynamics by solving the linearised Boltzmann transport equation (LBTE). The relaxation-time approximation (RTA) results in a phonon-based kinetic theory expression for the lattice thermal conductivity,

$$\boldsymbol{\kappa}^{\mathrm{RTA}} = \frac{1}{\mathcal{V}} \sum_k C_k \mathbf{v}_k \mathbf{v}_k^{\intercal} \tau_k^{\mathrm{RTA}}, \tag{3.38}$$

where $\mathbf{v}_k$ is the phonon group velocity, τ_k^{RTA} is the phonon lifetime and $k \equiv (\mathbf{q}, s)$. Diagonalising the symmetrised full phonon scattering matrix leads to the relaxon-based kinetic theory expression,

$$\boldsymbol{\kappa} = \frac{C}{\mathcal{V}} \sum_\alpha \mathbf{V}_\alpha \mathbf{V}_\alpha^{\intercal} \tau_\alpha, \tag{3.39}$$

where $\mathbf{V}_\alpha$ is the relaxon velocity, and τ_α is the relaxon lifetime. The scattering matrix is highly nontrivial to construct from Eq. (2.88) so we used an external Python package `Phono3py` [1, 3, 35] for this purpose. The third-order force constants (FC3) were computed using a finite displacement of 0.01 Å in a $6 \times 6 \times 3$ supercell containing 216 p-H_2 molecules, with periodic boundary conditions and a force cutoff distance of 9 Å. The harmonic force constants (FC2) obtained using this setup were checked against the custom Fortran calculation and gave the same phonon dispersions to graphical accuracy. The energy-conserving Dirac δ-functions were evaluated using the tetrahedron method, [3] which avoids the ambiguity associated with choosing a smearing width for the Gaussian/Lorentzian representation of the pre-limit δ-function. The Brillouin zone was sampled using a $29 \times 29 \times 19$ $\mathbf{q}$-mesh for temperatures greater than or equal to 5.4 K. Fewer grid points were used along the c-axis to account for

the larger real-space lattice parameter in that direction, so that the reciprocal-space resolution was similar in all three directions. At 5 K, a denser $35 \times 35 \times 21$ mesh was required for convergence. Both the relaxons and the scattering operators have a well-defined parity, with only the odd-parity relaxons contributing to the thermal conductivity. [33] The use of an odd-sized Γ-centered **q**-mesh in the Brillouin zone is necessary to account for the parity symmetry of the scattering operator. [56]

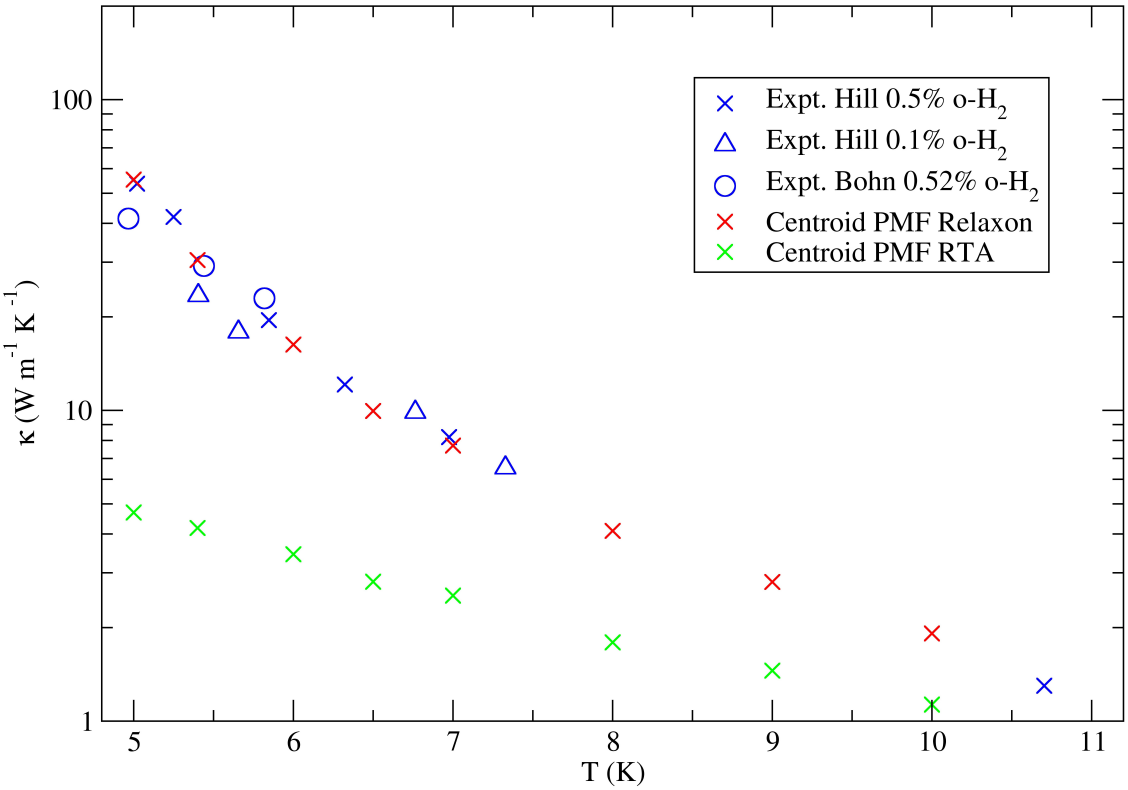


Figure 3.6: Comparison of the present thermal conductivity calculations with the results of Hill *et al.* [39] and Bohn *et al.* [40] for solid p-H_2 between 5 and 11 K. It is also worth noting that the early thermal conductivity measurements of solid p-H_2 by Hill and Schneidmesser were carried out at the Clarendon Laboratory, University of Oxford. This provides a pleasing local historical connection between the experimental data used here and the present thesis.

The experimental thermal conductivities reported by Bohn *et al.* [40] and Hill *et al.* [39] agree with each other within their experimental uncertainties. The diameters of the crystal reported by Bohn *et al.* and Hill *et al.* are 6 mm and 5 cm, respectively. Therefore, boundary scattering is unlikely to be significant in the temperature range considered here. Furthermore, the small concentration of o-H_2 impurities does not

appear to have a significant effect on thermal conductivity in this temperature range either. We show in Appendix H, by symmetry, that the thermal conductivity tensor of an hcp crystal has the form

$$\boldsymbol{\kappa} = \begin{pmatrix} \kappa_\perp & 0 & 0 \\ 0 & \kappa_\perp & 0 \\ 0 & 0 & \kappa_\parallel \end{pmatrix}. \tag{3.40}$$

Since the orientations of the experimental crystals were not determined, we compare experiment with the isotropic average

$$\kappa = \frac{1}{3} \operatorname{tr} \boldsymbol{\kappa}. \tag{3.41}$$

Figure 3.6 shows the thermal conductivity obtained using the CPMFs appropriate to each temperature in the range from 5 to 11 K. We compare the relaxation-time approximation (RTA) with the result obtained from diagonalising the full symmetrised phonon scattering matrix, as is done in the direct solution of LBTE in `phono3py`. [35] The direct solution of LBTE can be shown to be equivalent to the kinetic theory of relaxon framework discussed in Chapter 2. In the temperature range that we have considered, the thermal conductivity decreases as the temperature increases. This is because the phonon occupation number of a given mode increases as temperature increases. This increases the intensity of phonon-phonon scattering events, and therefore decreases the thermal conductivity. The phonon lifetime τ_k and hence the phonon mean free path $\mathbf{v}_k \tau_k$ become increasingly ill-defined at lower temperatures, where the off-diagonal elements of the scattering matrix become more significant and the RTA breaks down. The kinetic theory of phonons therefore does not provide a complete description of the thermal conductivity at low temperature. The relevant heat carriers in this regime are not individual phonons, but relaxons, which are eigenvectors of the symmetrised scattering matrix $\tilde{\Omega}$. Remarkably, the relaxons all carry the same heat capacity, which is equivalent to that of the bulk crystal (see Eq. (3.39)).

The RTA severely underestimates the thermal conductivity, especially at the lowest temperatures, where it is lower than the experimental values by an order of magnitude. This is because normal processes dominate in this regime, and RTA is expected to be inaccurate for the reasons explained in Chapter 2. As the temperature of the simulation is increased, the RTA approximation gets closer to both experiment and the relaxon-based kinetic theory expression using the full scattering matrix. The thermal conductivity calculated from the direct solution of LBTE reaches a quantitative agreement with the experiment from $5 - 7$ K where the experimental data are available. It also bridges the gap in the experiment from $8 - 10$ K. This demonstrates that including normal processes is crucial for capturing the correct thermal conductivity at low temperature.

We were unable to extend the calculation to lower temperatures. We know that the symmetrised scattering matrix must be positive semi-definite for the second law of thermodynamics to be satisfied (see Appendix G.III). At lower temperatures, we found that the scattering matrix developed spurious large negative eigenvalues, breaking its positive semi-definiteness. These negative eigenvalues are numerical artefacts that prevented us from completing the calculations. We believe they are due to the use of the tetrahedron method for the numerical implementation of the pre-limit Dirac delta functions in Eq. (2.89). No large spurious negative eigenvalues were observed for temperatures greater than or equal to 5 K when using sufficiently dense $\mathbf{q}$-meshes. Although the experimental crystals were large, they were reported to be polycrystalline, so boundary or grain-boundary scattering may also play some role at low temperature. Lastly, impurity scattering from residual o-H_2 becomes increasingly important at low temperatures too. This is difficult to model, since o-H_2 has the same mass as p-H_2, and the scattering arises instead from its rotational quadrupole interaction with phonons. [57, 58]

Finally, the anisotropy of the calculated thermal conductivity from both the RTA

and in the relaxon-basis is consistent with the symmetry of the thermal conductivity tensor in Eq. (3.40). At 5 K, our calculation gives

$$\kappa_\perp = 5.1\,\kappa_\parallel, \tag{3.42}$$

in good agreement with the experimental value, [59]

$$\kappa_\perp = 5\,\kappa_\parallel. \tag{3.43}$$

4
Conclusion

In this thesis, we have demonstrated two main results. First, incorporating nuclear quantum effects (NQEs) through the centroid potential of mean force (CPMF) leads to much better agreement with experiment for harmonic phonon properties, including the phonon dispersion and heat capacity, of solid p-H_2. Second, we have shown that normal (N) processes must be included in low-temperature thermal conductivity calculations. In this regime, heat transport is better understood in the relaxon picture, where the relevant heat carriers are the eigenvectors of the symmetrised full scattering matrix $\tilde{\Omega}$ rather than individual phonons with independent relaxation times. Relaxons have well-defined lifetimes τ_α and mean free paths $\mathbf{V}_\alpha\tau_\alpha$, and each carry a mode-independent heat capacity equivalent to that of the bulk crystal C.

In this work, we have focused on para-hydrogen. A natural extension would be to repeat the calculation for other isotopic or spin species, such as $J = 0$ ortho-deuterium (o-D_2). Finally, the present scattering matrix was constructed from perturbation theory and included only three-phonon processes. Although this treatment includes the correct quantum occupation numbers, the calculation becomes increasingly expensive when higher-order phonon scattering processes are included. Moreover, the numerical treatment of energy conservation (i.e. the need to use

pre-limit delta functions in the perturbation theory) remains a significant practical challenge. To overcome the numerical limitations observed at low temperatures, our group has recently developed a self-consistent Lorentzian linewidth and frequency shift approach that eliminates the artefacts associated with pre-limit delta functions. We intend to apply this method to extend the thermal conductivity calculations to temperatures lower than 5 K. It would also be desirable to develop a method for constructing the scattering matrix beyond perturbation theory, allowing higher-order scattering processes to be incorporated in systems with non-negligible nuclear quantum effects on the phonons.

Bibliography

[1] Togo, A., Chaput, L. & Tanaka, I. Distributions of phonon lifetimes in brillouin zones. *Phys. Rev. B* **91**, 094306 (2015).

[2] Togo, A. First-principles phonon calculations with phonopy and phono3py. *J. Phys. Soc. Jpn.* **92**, 012001 (2023).

[3] Togo, A., Chaput, L., Tadano, T. & Tanaka, I. Implementation strategies in phonopy and phono3py. *J. Phys. Condens. Matter* **35**, 353001 (2023).

[4] Silvera, I. F. The solid molecular hydrogens in the condensed phase: Fundamentals and static properties. *Reviews of Modern Physics* **52**, 393 (1980).

[5] Scharf, D., Martyna, G. J. & Klein, M. L. Structure and energetics of fluid para-hydrogen. *Low Temperature Physics* **19**, 364–367 (1993).

[6] Miller, T. F. & Manolopoulos, D. E. Quantum diffusion in liquid para-hydrogen from ring-polymer molecular dynamics. *The Journal of chemical physics* **122** (2005).

[7] Sutherland, B. J., Moore, W. H. & Manolopoulos, D. E. Nuclear quantum effects in thermal conductivity from centroid molecular dynamics. *The Journal of Chemical Physics* **154** (2021).

[8] Trotter, H. F. On the product of semi-groups of operators. *Proceedings of the American Mathematical Society* **10**, 545–551 (1959).

[9] Schulman, L. S. *Techniques and applications of path integration* (Courier Corporation, 2012).

[10] Ceriotti, M., Manolopoulos, D. E., Markland, T. E. & Rossi, M. Path integral methods in atomistic modelling: An introduction (2026). URL `https://arxiv.org/abs/2603.28588`. `2603.28588`.

[11] Chandler, D. & Wolynes, P. G. Exploiting the isomorphism between quantum theory and classical statistical mechanics of polyatomic fluids. *The Journal of Chemical Physics* **74**, 4078–4095 (1981).

[12] Feynman, R. P., Hibbs, A. R. & Styer, D. F. *Quantum mechanics and path integrals* (Courier Corporation, 2010).

[13] Parrinello, M. & Rahman, A. Study of an f center in molten kcl. *The Journal of chemical physics* **80**, 860–867 (1984).

[14] Hall, R. W. & Berne, B. J. Nonergodicity in path integral molecular dynamics. *The Journal of chemical physics* **81**, 3641–3643 (1984).

[15] Ceriotti, M., Parrinello, M., Markland, T. E. & Manolopoulos, D. E. Efficient stochastic thermostatting of path integral molecular dynamics. *The Journal of chemical physics* **133** (2010).

[16] Bussi, G., Donadio, D. & Parrinello, M. Canonical sampling through velocity rescaling. *The Journal of chemical physics* **126** (2007).

[17] Craig, I. R. & Manolopoulos, D. E. Quantum statistics and classical mechanics: Real time correlation functions from ring polymer molecular dynamics. *The Journal of chemical physics* **121**, 3368–3373 (2004).

[18] Cao, J. & Voth, G. A. The formulation of quantum statistical mechanics based on the feynman path centroid density. i. equilibrium properties. *The Journal of chemical physics* **100**, 5093–5105 (1994).

[19] Cao, J. & Voth, G. A. The formulation of quantum statistical mechanics based on the feynman path centroid density. ii. dynamical properties. *The Journal of chemical physics* **100**, 5106–5117 (1994).

[20] Cao, J. & Voth, G. A. The formulation of quantum statistical mechanics based on the feynman path centroid density. iii. phase space formalism and analysis of centroid molecular dynamics. *The Journal of chemical physics* **101**, 6157–6167 (1994).

[21] Cao, J. & Voth, G. A. The formulation of quantum statistical mechanics based on the feynman path centroid density. iv. algorithms for centroid molecular dynamics. *The Journal of chemical physics* **101**, 6168–6183 (1994).

[22] Born, M. & Huang, K. *Dynamical theory of crystal lattices* (Oxford university press, 1996).

[23] Dove, M. T. *Introduction to lattice dynamics*. 4 (Cambridge university press, 1993).

[24] De Groot, S. R. & Mazur, P. *Non-equilibrium thermodynamics* (Courier Corporation, 2013).

[25] Peierls, R. E. *Zur kinetischen theorie der wärmeleitung in kristallen* (JA Barth, 1929).

[26] Hardy, R. J. Energy-flux operator for a lattice. *Physical Review* **132**, 168 (1963).

[27] Ziman, J. M. *Electrons and phonons: the theory of transport phenomena in solids* (Oxford university press, 2001).

[28] Srivastava, G. P. *The physics of phonons* (CRC press, 2022).

[29] Klemens, P. The thermal conductivity of dielectric solids at low temperatures (theoretical). *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences* **208**, 108–133 (1951).

[30] McGaughey, A. J. & Kaviany, M. Quantitative validation of the boltzmann transport equation phonon thermal conductivity model under the single-mode relaxation time approximation. *Physical Review B* **69**, 094303 (2004).

[31] Landau, L. D. & Lifshitz, E. M. *Statistical Physics, Part 1*, vol. 5 of *Course of Theoretical Physics* (Butterworth-Heinemann, Oxford, 1980).

[32] Schekochihin, A. A. Lectures on kinetic theory of gases and statistical physics. *Lecture Notes for the Oxford Physics course* (2020).

[33] Simoncelli, M., Marzari, N. & Cepellotti, A. Generalization of fourier's law into viscous heat equations. *Physical Review X* **10**, 011019 (2020).

[34] Callaway, J. Model for lattice thermal conductivity at low temperatures. *Phys. Rev.* **113**, 1046–1051 (1959). URL `https://link.aps.org/doi/10.1103/PhysRev.113.1046`.

[35] Chaput, L. Direct solution to the linearized phonon boltzmann equation. *Phys. Rev. Lett.* **110**, 265506 (2013). URL `https://link.aps.org/doi/10.1103/PhysRevLett.110.265506`.

[36] Cepellotti, A. & Marzari, N. Thermal transport in crystals as a kinetic theory of relaxons. *Phys. Rev. X* **6**, 041013 (2016). URL `https://link.aps.org/doi/10.1103/PhysRevX.6.041013`.

[37] Nielsen, M. Phonons in solid hydrogen and deuterium studied by inelastic coherent neutron scattering. *Phys. Rev. B* **7**, 1626–1635 (1973). URL `https://link.aps.org/doi/10.1103/PhysRevB.7.1626`.

[38] Ahlers, G. Lattice heat capacity of solid hydrogen. *The Journal of Chemical Physics* **41**, 86–94 (1964).

[39] Hill, B. W. & Schneidmesser, B. The thermal conductivity of solid hydrogen. *Zeitschrift für Physikalische Chemie* **16**, 257–266 (1958). URL `https://api.semanticscholar.org/CorpusID:102024536`.

[40] Bohn, R. G. & Mate, C. Thermal conductivity of solid hydrogen. *Physical Review B* **2**, 2121 (1970).

[41] Silvera, I. F. & Goldman, V. V. The isotropic intermolecular potential for h2 and d2 in the solid and gas phases. *The Journal of Chemical Physics* **69**, 4209–4213 (1978).

[42] Rabani, E., Reichman, D. R., Krilov, G. & Berne, B. J. The calculation of transport properties in quantum liquids using the maximum entropy numerical analytic continuation method: Application to liquid para-hydrogen. *Proceedings of the National Academy of Sciences* **99**, 1129–1133 (2002).

[43] Reichman, D. R. & Rabani, E. A self-consistent mode-coupling theory for dynamical correlations in quantum liquids: Application to liquid para-hydrogen. *The Journal of chemical physics* **116**, 6279–6285 (2002).

[44] Pavese, M. & Voth, G. A. Pseudopotentials for centroid molecular dynamics: application to self-diffusion in liquid para-hydrogen. *Chemical physics letters* **249**, 231–236 (1996).

[45] Yonetani, Y. & Kinugawa, K. Transport properties of liquid para-hydrogen: The path integral centroid molecular dynamics approach. *The Journal of chemical physics* **119**, 9651–9660 (2003).

[46] Hone, T. D. & Voth, G. A. A centroid molecular dynamics study of liquid para-hydrogen and ortho-deuterium. *The Journal of chemical physics* **121**, 6412–6422 (2004).

[47] Nakayama, A. & Makri, N. Forward–backward semiclassical dynamics for quantum fluids using pair propagators: Application to liquid para-hydrogen. *The Journal of chemical physics* **119**, 8592–8605 (2003).

[48] Poulsen, J. A., Nyman, G. & Rossky, P. J. Quantum diffusion in liquid parahydrogen: An application of the feynman- kleinert linearized path integral approximation. *The Journal of Physical Chemistry B* **108**, 19799–19808 (2004).

[49] Martyna, G. J., Hughes, A. & Tuckerman, M. E. Molecular dynamics algorithms for path integrals at constant pressure. *The Journal of chemical physics* **110**, 3275–3290 (1999).

[50] Sutherland, B. *Inclusion of nuclear quantum effects in thermal conductivity prediction from path integral techniques*. Ph.D. thesis, University of Oxford (2022).

[51] Borgis, D., Assaraf, R., Rotenberg, B. & Vuilleumier, R. Computation of pair distribution functions and three-dimensional densities with a reduced variance principle. *Molecular Physics* **111**, 3486–3492 (2013).

[52] Rotenberg, B. Use the force! reduced variance estimators for densities, radial distribution functions, and local mobilities in molecular simulations. *The Journal of Chemical Physics* **153** (2020).

[53] Setyawan, W. & Curtarolo, S. High-throughput electronic band structure calculations: Challenges and tools. *Computational materials science* **49**, 299–312 (2010).

[54] Kittel, C. & McEuen, P. *Introduction to solid state physics* (John Wiley & Sons, 2018).

[55] Debye, P. Zur theorie der spezifischen wärmen. *Annalen der Physik* **344**, 789–839 (1912).

[56] Simoncelli, M. *Thermal transport beyond Fourier, and beyond Boltzmann*. Ph.D. thesis, EPFL (2021).

[57] Ebner, C. & Sung, C. C. Dependence of the thermal conductivity of dilute solid mixtures of orthohydrogen and parahydrogen on orthohydrogen concentration. *Phys. Rev. B* **2**, 2115–2120 (1970). URL `https://link.aps.org/doi/10.1103/PhysRevB.2.2115`.

[58] Harris, A. B., Amstutz, L. I., Meyer, H. & Myers, S. M. Observation of pair interaction between ortho molecules in solid h_2. *Phys. Rev.* **175**, 603–609 (1968). URL `https://link.aps.org/doi/10.1103/PhysRev.175.603`.

[59] Korolyuk, O. A., Gorodilov, B. Y., Krivchikov, A. I., Pirogov, A. S. & Dudkin, V. V. Anisotropy of the thermal conductivity of parahydrogen crystals. *Journal of Low Temperature Physics* **111**, 515–520 (1998). URL `https://api.semanticscholar.org/CorpusID:116438546`.

[60] Nye, J. F. *Physical properties of crystals: their representation by tensors and matrices* (Oxford university press, 1985).

[61] Atkins, P. W., Child, M. S. & Phillips, C. S. G. *Tables for group theory*, vol. 6 (Oxford University Press Oxford, 1970).

Appendices

A
The Ring Polymer

The quantum canonical partition function is

$$Q \equiv \mathrm{tr}\Big[e^{-\beta\hat{H}}\Big]. \tag{A.1}$$

We split the Boltzmann operator into n factors,

$$e^{-\beta\hat{H}} = \Big(e^{-\beta_n\hat{H}}\Big)^n, \tag{A.2}$$

where

$$\beta_n \equiv \frac{\beta}{n}. \tag{A.3}$$

Evaluating the trace in the position basis and inserting complete sets of position eigenstates gives

$$Q = \int \mathrm{d}q_1 \cdots \int \mathrm{d}q_n \, \langle q_1|e^{-\beta_n\hat{H}}|q_2\rangle \cdots \langle q_n|e^{-\beta_n\hat{H}}|q_1\rangle. \tag{A.4}$$

Now consider a single matrix element. For sufficiently small β_n, the symmetric Trotter factorisation gives

$$e^{-\beta_n(\hat{T}+\hat{V})} \simeq e^{-\beta_n\hat{V}/2}e^{-\beta_n\hat{T}}e^{-\beta_n\hat{V}/2}, \tag{A.5}$$

where $\hat{T} = \hat{p}^2/2m$ and $\hat{V} = V(\hat{q})$. Therefore

$$\langle q|e^{-\beta_n\hat{H}}|q'\rangle \simeq e^{-\beta_n[V(q)+V(q')]/2}\langle q|e^{-\beta_n\hat{T}}|q'\rangle. \tag{A.6}$$

To evaluate the matrix element of $e^{-\beta_n \hat{T}}$ in position basis, we insert a complete set of momentum eigenstates, so we have

$$\langle q|e^{-\beta_n \hat{T}}|q'\rangle = \int dp\, \langle q|p\rangle\langle p|e^{-\beta_n \hat{T}}|q'\rangle = \int dp\, \langle q|p\rangle e^{-\beta_n p^2/2m}\langle p|q'\rangle \tag{A.7}$$

we note that a state of well-defined momentum in the position representation is

$$\langle q|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{iqp/\hbar} = \langle p|q\rangle^* \tag{A.8}$$

to arrive at,

$$\langle q|e^{-\beta_n \hat{T}}|q'\rangle = \frac{1}{2\pi\hbar}\int dp\, e^{-\beta_n p^2/2m} e^{ip(q-q')/\hbar} \tag{A.9}$$

then note the following standard result,

$$\int_{-\infty}^{\infty} \mathrm{d}x\, e^{-(b^2x^2+ax)} = \frac{\sqrt{\pi}}{b} e^{a^2/4b^2} \tag{A.10}$$

so we have

$$\langle q|e^{-\beta_n \hat{T}}|q'\rangle = \frac{1}{2\pi\hbar}\sqrt{\frac{2\pi m}{\beta_n}} e^{-(q-q')^2 m/2\beta_n\hbar^2} = \frac{1}{2\pi\hbar}\sqrt{\frac{2\pi m}{\beta_n}} e^{-\beta_n m\omega_n^2 (q-q')^2/2} \tag{A.11}$$

where we have defined,

$$\omega_n \equiv \frac{1}{\beta_n \hbar} \tag{A.12}$$

and clearly we also have,

$$\frac{1}{2\pi\hbar}\int dp\, e^{-\beta_n p^2/2m} e^{-\beta_n m\omega_n^2 (q-q')^2/2} = \frac{1}{2\pi\hbar}\sqrt{\frac{2\pi m}{\beta_n}} e^{-\beta_n m\omega_n^2 (q-q')^2/2} \tag{A.13}$$

and,

$$\langle q|e^{-\beta_n \hat{T}}|q'\rangle = \frac{1}{2\pi\hbar}\int dp\, e^{-\beta_n [p^2/2m + m\omega_n^2 (q-q')^2/2]} \tag{A.14}$$

thus, the matrix element of a single slice of the Boltzmann operator is approximately

$$\langle q|e^{-\beta_n \hat{H}}|q'\rangle \simeq \frac{1}{2\pi\hbar}\int dp\, e^{-\beta_n [p^2/2m + m\omega_n^2 (q-q')^2/2 + V(q)/2 + V(q')/2]} \tag{A.15}$$

substituting back into the full partition function integral Eq. (A.4) yields

$$Q \simeq \frac{1}{(2\pi\hbar)^n} \int d\mathbf{p} \int d\mathbf{q}\, e^{-\beta_n H_n(\mathbf{p},\mathbf{q})} \tag{A.16}$$

where the ring polymer Hamiltonian is,

$$H_n(\mathbf{p},\mathbf{q}) = \sum_{j=1}^{n} \left[\frac{p_j^2}{2m} + \frac{1}{2} m\omega_n^2 (q_j - q_{j+1})^2 + V(q_j) \right] \tag{A.17}$$

the approximation above has an error of $\mathcal{O}(n^{-2})$.

B

Centroid Potential of Mean Force

I

The centroid coordinate of a ring polymer is defined as the average position of its beads,

$$\bar{q} \equiv \frac{1}{n} \sum_{j=1}^{n} q_j. \tag{B.1}$$

The centroid distribution is defined as

$$\rho_c(q_c) \equiv \left(\frac{m}{2\pi\beta_n\hbar^2}\right)^{n/2} \int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})} \delta\Big(q_c - \bar{q}\Big). \tag{B.2}$$

Consider

$$\begin{aligned} Q &= \int \mathrm{d}q_c\, \rho_c(q_c) \\ &= \left(\frac{m}{2\pi\beta_n\hbar^2}\right)^{n/2} \int \mathrm{d}q_c \int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})} \delta\Big(q_c - \bar{q}\Big) \\ &= \left(\frac{m}{2\pi\beta_n\hbar^2}\right)^{n/2} \int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})} \int \mathrm{d}q_c\, \delta\Big(q_c - \bar{q}\Big) \\ &= \left(\frac{m}{2\pi\beta_n\hbar^2}\right)^{n/2} \int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})}. \end{aligned} \tag{B.3}$$

This is the original integral over the ring-polymer distribution after the kinetic-energy degrees of freedom have been integrated out. We will drop this cumbersome prefactor from now on and leave it understood.

II

The centroid potential of mean force is defined through the centroid distribution,

$$\rho_c(q_c) = e^{-\beta V_c(q_c)}, \tag{B.4}$$

where an arbitrary normalisation constant has been absorbed into an additive constant in V_c. Thus

$$V_c(q_c) = -\frac{1}{\beta}\ln \rho_c(q_c). \tag{B.5}$$

The centroid force is then

$$\begin{aligned} F_c(q_c) &\equiv -\frac{\partial V_c(q_c)}{\partial q_c} \\ &= -\frac{\partial}{\partial q_c}\Big(-\frac{1}{\beta}\ln \rho_c(q_c)\Big) \\ &= \frac{1}{\beta}\frac{\partial}{\partial q_c}\ln \rho_c(q_c) \\ &= \frac{1}{\beta}\frac{1}{\rho_c}\frac{\partial \rho_c(q_c)}{\partial q_c} \\ &= \frac{1}{\beta}\frac{1}{\rho_c}\frac{\partial}{\partial q_c}\int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})}\delta\Big(q_c - \bar{q}\Big) \\ &= \frac{1}{\beta}\frac{1}{\rho_c}\int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})}\frac{\partial}{\partial q_c}\delta\Big(q_c - \bar{q}\Big). \end{aligned} \tag{B.6}$$

The derivative of the Dirac delta is difficult to deal with. To proceed, consider

$$\begin{aligned} -i\hbar\int \mathrm{d}x'\,\psi(x')\frac{\partial}{\partial x}\delta(x-x') &= \int \mathrm{d}x'\,\langle x|\hat{p}|x'\rangle\langle x'|\psi\rangle \\ &= \langle x|\hat{p}|\psi\rangle \\ &= \int \mathrm{d}x'\,\langle x|x'\rangle\langle x'|\hat{p}|\psi\rangle \\ &= -i\hbar\int \mathrm{d}x'\,\delta(x-x')\frac{\partial\psi(x')}{\partial x'}. \end{aligned} \tag{B.7}$$

Therefore,

$$\begin{aligned} F_c(q_c) &= \frac{1}{\beta}\frac{1}{\rho_c}\int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})}\frac{\partial}{\partial q_c}\delta(q_c-\bar{q}) \\ &= \frac{1}{\beta}\frac{1}{\rho_c}\int \mathrm{d}\mathbf{q}\,\delta(q_c-\bar{q})\frac{\partial}{\partial \bar{q}}e^{-\beta_n V_n(\mathbf{q})}. \end{aligned} \tag{B.8}$$

We now decompose each bead coordinate into the centroid and an internal coordinate,

$$q_j = q_c + \xi_j, \tag{B.9}$$

with

$$\sum_{j=1}^{n} \xi_j = 0. \tag{B.10}$$

This follows from the definition of the centroid. Holding the internal coordinates fixed gives

$$\frac{\partial q_j}{\partial \bar{q}} = 1, \tag{B.11}$$

and therefore

$$\frac{\partial}{\partial \bar{q}} = \sum_{j=1}^{n} \frac{\partial}{\partial q_j}. \tag{B.12}$$

Thus

$$\begin{aligned} F_c(q_c) &= \frac{1}{\beta}\frac{1}{\rho_c}\int \mathrm{d}\mathbf{q}\, \delta(q_c - \bar{q}) \sum_{j=1}^{n} \frac{\partial}{\partial q_j} e^{-\beta_n V_n(\mathbf{q})} \\ &= \frac{1}{\beta}\frac{1}{\rho_c}\int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})} \left[-\beta_n \sum_{j=1}^{n} \frac{\partial V_n(\mathbf{q})}{\partial q_j}\right] \delta(q_c - \bar{q}) \\ &= \frac{1}{\rho_c}\int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})} \left[\frac{1}{n}\sum_{j=1}^{n} -\frac{\partial V_n(\mathbf{q})}{\partial q_j}\right] \delta(q_c - \bar{q}). \end{aligned} \tag{B.13}$$

The ring-polymer potential is

$$V_n(\mathbf{q}) = \sum_{k=1}^{n} \left[\frac{1}{2} m\omega_n^2 (q_k - q_{k+1})^2 + V(q_k)\right], \tag{B.14}$$

with $q_{n+1} = q_1$. The spring contribution vanishes under the centroid derivative:

$$\begin{aligned} \sum_{j=1}^{n} \frac{\partial}{\partial q_j} \sum_{k=1}^{n} \frac{1}{2} m\omega_n^2 (q_k - q_{k+1})^2 &= \sum_{j=1}^{n}\sum_{k=1}^{n} m\omega_n^2 (q_k - q_{k+1}) \left(\frac{\partial q_k}{\partial q_j} - \frac{\partial q_{k+1}}{\partial q_j}\right) \\ &= \sum_{j=1}^{n}\sum_{k=1}^{n} m\omega_n^2 (q_k - q_{k+1})(\delta_{jk} - \delta_{j,k+1}) \\ &= m\omega_n^2 \sum_{j=1}^{n} (q_j - q_{j+1}) - m\omega_n^2 \sum_{j=1}^{n} (q_{j-1} - q_j) \\ &= 0. \end{aligned} \tag{B.15}$$

Hence

$$\begin{aligned}\sum_{j=1}^{n}\frac{\partial V_n(\mathbf{q})}{\partial q_j} &= \sum_{j=1}^{n}\frac{\partial}{\partial q_j}\sum_{k=1}^{n}V(q_k)\\ &= \sum_{j=1}^{n}\sum_{k=1}^{n}\frac{\partial V(q_k)}{\partial q_k}\frac{\partial q_k}{\partial q_j}\\ &= \sum_{j=1}^{n}\sum_{k=1}^{n}\frac{\partial V(q_k)}{\partial q_k}\delta_{jk}\\ &= \sum_{j=1}^{n}\frac{\partial V(q_j)}{\partial q_j}. \end{aligned} \tag{B.16}$$

Therefore

$$F_c(q_c) = \frac{1}{\rho_c}\int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})}\left[\frac{1}{n}\sum_{j=1}^{n}-\frac{\partial V(q_j)}{\partial q_j}\right]\delta(q_c-\bar{q}). \tag{B.17}$$

This is the average force on the beads of the ring polymer, constrained to configurations with centroid position q_c. The factor $1/\rho_c$ provides the normalisation of this constrained average. More explicitly, for any observable $A(\mathbf{q})$,

$$\langle A\rangle_{\bar{q}=q_c} = \frac{\int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})}A_n(\mathbf{q})\delta(q_c-\bar{q})}{\int \mathrm{d}\mathbf{q}\, e^{-\beta_n V_n(\mathbf{q})}\delta(q_c-\bar{q})}. \tag{B.18}$$

Thus

$$F_c(q_c) = \left\langle \frac{1}{n}\sum_{j=1}^{n}-\frac{\partial V(q_j)}{\partial q_j}\right\rangle_{\bar{q}=q_c}. \tag{B.19}$$

The centroid potential of mean force is obtained by integrating the centroid force:

$$\int_{q'}^{q_c}\mathrm{d}q\,\frac{\partial V_c(q)}{\partial q} = -\int_{q'}^{q_c}\mathrm{d}q\,F_c(q). \tag{B.20}$$

By the fundamental theorem of calculus,

$$V_c(q_c) = V_c(q') - \int_{q'}^{q_c}\mathrm{d}q\,F_c(q). \tag{B.21}$$

If we let $q'\to\infty$, where $V_c(\infty)=0$, then

$$V_c(q_c) = -\int_{\infty}^{q_c}\mathrm{d}q\,F_c(q). \tag{B.22}$$

Reversing the limits gives

$$V_c(q_c) = \int_{q_c}^{\infty}\mathrm{d}q\,F_c(q). \tag{B.23}$$

III

For interactions among N particles in three dimensions, the centroid force is generalised to

$$\mathbf{F}_i^c(\mathbf{r}^c) \equiv -\nabla_{\mathbf{r}_i^c} V_c(\mathbf{r}^c) = \left\langle \frac{1}{n} \sum_{k=1}^{n} -\nabla_{\mathbf{r}_i^{(k)}} V(\mathbf{r}_1^{(k)}, \ldots, \mathbf{r}_N^{(k)}) \right\rangle_{\bar{\mathbf{r}}=\mathbf{r}^c}. \tag{B.24}$$

where,

$$\mathbf{r} \equiv \{\mathbf{r}_i\}, \qquad \bar{\mathbf{r}} \equiv \{\bar{\mathbf{r}}_i\}. \tag{B.25}$$

Under the pairwise interaction approximation,

$$V(\mathbf{r}) \simeq \frac{1}{2} \sum_{a=1}^{N} \sum_{b \neq a}^{N} v(r_{ab}), \tag{B.26}$$

where $r_{ab} = |\mathbf{r}_a - \mathbf{r}_b|$. Then

$$\nabla_{\mathbf{r}_l} V(\mathbf{r}) \simeq \frac{1}{2} \sum_{a=1}^{N} \sum_{b \neq a}^{N} \nabla_{\mathbf{r}_l} v(r_{ab}) = \frac{1}{2} \sum_{a=1}^{N} \sum_{b \neq a}^{N} \frac{\partial v(r_{ab})}{\partial r_{ab}} \nabla_{\mathbf{r}_l} r_{ab}. \tag{B.27}$$

We now evaluate the derivative in Cartesian index notation,

$$\nabla_{\mathbf{r}_l} \equiv \sum_{\alpha} \hat{\mathbf{e}}_\alpha \frac{\partial}{\partial r_{l\alpha}}, \tag{B.28}$$

where

$$\hat{\mathbf{e}}_\alpha \cdot \hat{\mathbf{e}}_\beta = \delta_{\alpha\beta}. \tag{B.29}$$

Then

$$\begin{aligned}
\frac{\partial r_{ab}}{\partial r_{l\alpha}} &= \frac{\partial}{\partial r_{l\alpha}} \left[\sum_{\beta} (r_{a\beta} - r_{b\beta})^2 \right]^{1/2} \\
&= \frac{1}{2} \left[\sum_{\beta} (r_{a\beta} - r_{b\beta})^2 \right]^{-1/2} \sum_{\beta} 2(r_{a\beta} - r_{b\beta}) \left(\frac{\partial r_{a\beta}}{\partial r_{l\alpha}} - \frac{\partial r_{b\beta}}{\partial r_{l\alpha}} \right) \\
&= r_{ab}^{-1} \sum_{\beta} (r_{a\beta} - r_{b\beta})(\delta_{al} - \delta_{bl}) \delta_{\alpha\beta} \\
&= r_{ab}^{-1} (r_{a\alpha} - r_{b\alpha})(\delta_{al} - \delta_{bl}).
\end{aligned} \tag{B.30}$$

Thus

$$\nabla_{\mathbf{r}_l} r_{ab} = \hat{\mathbf{r}}_{ab} (\delta_{al} - \delta_{bl}), \tag{B.31}$$

where

$$\hat{\mathbf{r}}_{ab} \equiv \frac{\mathbf{r}_a - \mathbf{r}_b}{r_{ab}}. \tag{B.32}$$

Substituting back gives

$$\begin{aligned}
\nabla_{\mathbf{r}_l} V(\mathbf{r}) &\simeq \frac{1}{2}\sum_{a=1}^{N}\sum_{b\neq a}^{N} v'(r_{ab})\hat{\mathbf{r}}_{ab}(\delta_{al}-\delta_{bl}) \\
&= \frac{1}{2}\sum_{b\neq l}^{N} v'(r_{lb})\hat{\mathbf{r}}_{lb} - \frac{1}{2}\sum_{a\neq l}^{N} v'(r_{al})\hat{\mathbf{r}}_{al} \\
&= \frac{1}{2}\sum_{j\neq l}^{N} v'(r_{lj})\hat{\mathbf{r}}_{lj} - \frac{1}{2}\sum_{j\neq l}^{N} v'(r_{jl})\hat{\mathbf{r}}_{jl} \\
&= \frac{1}{2}\sum_{j\neq l}^{N} v'(r_{lj})\hat{\mathbf{r}}_{lj} + \frac{1}{2}\sum_{j\neq l}^{N} v'(r_{lj})\hat{\mathbf{r}}_{lj} \\
&= \sum_{j\neq l}^{N} v'(r_{lj})\hat{\mathbf{r}}_{lj}.
\end{aligned} \tag{B.33}$$

Therefore

$$\nabla_{\mathbf{r}_i} V(\mathbf{r}) \simeq \sum_{j\neq i}^{N} v'(r_{ij})\hat{\mathbf{r}}_{ij}. \tag{B.34}$$

The total centroid force is then

$$\begin{aligned}
-\nabla_{\mathbf{r}_i^c} V_c &\simeq \left\langle \frac{1}{n}\sum_{k=1}^{n}\sum_{j\neq i}^{N} -v'(r_{ij}^{(k)})\hat{\mathbf{r}}_{ij}^{(k)} \right\rangle_{\mathbf{r}=\mathbf{r}^c} \\
&= \sum_{j\neq i}^{N} \left\langle \frac{1}{n}\sum_{k=1}^{n} -v'(r_{ij}^{(k)})\hat{\mathbf{r}}_{ij}^{(k)} \right\rangle_{\mathbf{r}=\mathbf{r}^c}.
\end{aligned} \tag{B.35}$$

We therefore define the pairwise centroid force as

$$\mathbf{f}_{ij}^c(r_{ij}^c) \equiv \frac{1}{n}\sum_{k=1}^{n} -v'(r_{ij}^{(k)})\hat{\mathbf{r}}_{ij}^{(k)}, \tag{B.36}$$

so that

$$\mathbf{F}_i^c(\mathbf{r}^c) = \sum_{j\neq i}^{N} \langle \mathbf{f}_{ij}^c \rangle_{\bar{\mathbf{r}}=\mathbf{r}^c} \tag{B.37}$$

We now also make the pairwise approximation to the total centroid potential of mean force,

$$V_c(\mathbf{r}^c) \simeq \frac{1}{2}\sum_{a=1}^{N}\sum_{b\neq a}^{N} v_c(r_{ab}^c). \tag{B.38}$$

By the same argument as above,

$$\nabla_{\mathbf{r}_i^c} \frac{1}{2} \sum_{a=1}^{N} \sum_{b \neq a}^{N} v_c(r_{ab}^c) = \sum_{j \neq i}^{N} v_c'(r_{ij}^c) \hat{\mathbf{r}}_{ij}^c. \tag{B.39}$$

Thus

$$\mathbf{F}_i^c(\mathbf{r}^c) = -\sum_{j \neq i}^{N} v_c'(r_{ij}^c) \hat{\mathbf{r}}_{ij}^c = \sum_{j \neq i}^{N} \langle \mathbf{f}_{ij}^c \rangle_{\bar{\mathbf{r}} = \mathbf{r}^c} = \sum_{j \neq i}^{N} \langle \mathbf{f}_{ij}^c \rangle_{\bar{r}_{ij} = r_{ij}^c}. \tag{B.40}$$

where

$$\bar{r}_{ij} \equiv |\bar{\mathbf{r}}_i - \bar{\mathbf{r}}_j|. \tag{B.41}$$

Matching the pairwise contributions gives

$$-v_c'(r_{ij}^c) \hat{\mathbf{r}}_{ij}^c = \langle \mathbf{f}_{ij}^c \rangle_{\bar{r}_{ij} = r_{ij}^c}. \tag{B.42}$$

Projecting along the centroid–centroid direction,

$$-v_c'(r_{ij}^c) = \langle \mathbf{f}_{ij}^c \cdot \hat{\mathbf{r}}_{ij}^c \rangle_{\bar{r}_{ij} = r_{ij}^c}. \tag{B.43}$$

We now define the radial pairwise centroid force as

$$f_{ij}^c \equiv \mathbf{f}_{ij}^c \cdot \hat{\mathbf{r}}_{ij}^c. \tag{B.44}$$

Therefore

$$-v_c'(r) = \left\langle f_{ij}^c \right\rangle_{\bar{r}_{ij} = r}. \tag{B.45}$$

Finally, choosing the zero of energy such that $v_c(\infty) = 0$, the pairwise centroid potential of mean force is

$$v_c(r) = \int_r^\infty \mathrm{d}r' \left\langle f_{ij}^c \right\rangle_{\bar{r}_{ij} = r'}. \tag{B.46}$$

C
Crystal Momentum

Let $\mathbf{R}$ be a Bravais lattice vector. By the periodic translational symmetry of the lattice,

$$V(\mathbf{r}) = V(\mathbf{r} + \mathbf{R}). \tag{C.1}$$

In one dimension, the translation operator is defined as

$$\hat{T}(a) \equiv e^{-ia\hat{p}/\hbar}. \tag{C.2}$$

Consider the commutator

$$\begin{aligned}
[\hat{x}, \hat{T}(a)] &= [\hat{x}, e^{-ia\hat{p}/\hbar}] \\
&= -\frac{ia}{\hbar} e^{-ia\hat{p}/\hbar} [\hat{x}, \hat{p}] \\
&= -\frac{ia}{\hbar} e^{-ia\hat{p}/\hbar} i\hbar \\
&= a e^{-ia\hat{p}/\hbar} \\
&= a\hat{T}(a).
\end{aligned} \tag{C.3}$$

Now consider

$$\begin{aligned}
\hat{x}\hat{T}(a)|x\rangle &= \hat{T}(a)\hat{x}|x\rangle + [\hat{x}, \hat{T}(a)]|x\rangle \\
&= x\hat{T}(a)|x\rangle + a\hat{T}(a)|x\rangle \\
&= (x + a)\hat{T}(a)|x\rangle.
\end{aligned} \tag{C.4}$$

Thus $\hat{T}(a)|x\rangle$ is another position eigenstate, with eigenvalue $x + a$. Therefore

$$\hat{T}(a)|x\rangle = |x + a\rangle. \tag{C.5}$$

Since the translation operator is unitary by construction, the translated state is already normalised:

$$\langle x + a|x' + a\rangle = \langle x|\hat{T}^\dagger(a)\hat{T}(a)|x'\rangle = \langle x|x'\rangle = \delta(x - x'). \tag{C.6}$$

The result generalises to three dimensions:

$$\hat{T}(\mathbf{a}) \equiv e^{-i\mathbf{a}\cdot\hat{\mathbf{p}}/\hbar}, \qquad \hat{T}(\mathbf{a})|\mathbf{r}\rangle = |\mathbf{r} + \mathbf{a}\rangle. \tag{C.7}$$

Now consider the commutator,

$$\begin{aligned}
[\hat{T}(\mathbf{R}), V(\hat{\mathbf{r}})] &= \int d^3\mathbf{r}\, V(\mathbf{r})\Big[\hat{T}(\mathbf{R}), |\mathbf{r}\rangle\langle\mathbf{r}|\Big] \\
&= \int d^3\mathbf{r}\, V(\mathbf{r})\Big(|\mathbf{r} + \mathbf{R}\rangle\langle\mathbf{r}| - |\mathbf{r}\rangle\langle\mathbf{r} - \mathbf{R}|\Big) \\
&= \int d^3\mathbf{r}\, V(\mathbf{r})|\mathbf{r} + \mathbf{R}\rangle\langle\mathbf{r}| - \int d^3\mathbf{r}\, V(\mathbf{r})|\mathbf{r}\rangle\langle\mathbf{r} - \mathbf{R}| \\
&= \int d^3\mathbf{r}\, V(\mathbf{r})|\mathbf{r} + \mathbf{R}\rangle\langle\mathbf{r}| - \int d^3\mathbf{r}'\, V(\mathbf{r}' + \mathbf{R})|\mathbf{r}' + \mathbf{R}\rangle\langle\mathbf{r}'| \\
&= 0
\end{aligned} \tag{C.8}$$

where we have used the spectral expansion,

$$V(\hat{\mathbf{r}}) = \int d^3\mathbf{r}\, |\mathbf{r}\rangle V(\mathbf{r})\langle\mathbf{r}| \tag{C.9}$$

and $\mathbf{r}' = \mathbf{r} - \mathbf{R}$ and Eq. (C.1).

Since both the translation operator and kinetic part of the crystal Hamiltonian $\hat{H}$ are functions of the momentum operator, we have,

$$[\hat{H}, \hat{T}(\mathbf{R})] = 0 \tag{C.10}$$

when two operators commute, we can find a complete set of simultaneous eigenstates of both operators, furthermore, we have identified a good quantum number/constant of motion which is associated with the eigenvalue of the translation operator,

$$\hat{H}|\mathbf{q}\rangle = E_{\mathbf{q}}|\mathbf{q}\rangle \tag{C.11}$$

$$\hat{T}(\mathbf{R})|\mathbf{q}\rangle = \lambda_{\mathbf{q}}(\mathbf{R})|\mathbf{q}\rangle \tag{C.12}$$

since,

$$\hat{T}(\mathbf{R}+\mathbf{R}') = e^{-i(\mathbf{R}+\mathbf{R}')\cdot\hat{\mathbf{p}}/\hbar} = e^{-i\mathbf{R}\cdot\hat{\mathbf{p}}/\hbar}e^{-i\mathbf{R}'\cdot\hat{\mathbf{p}}/\hbar} = \hat{T}(\mathbf{R})\hat{T}(\mathbf{R}') \tag{C.13}$$

thus,

$$\lambda_{\mathbf{q}}(\mathbf{R}+\mathbf{R}') = \lambda_{\mathbf{q}}(\mathbf{R})\lambda_{\mathbf{q}}(\mathbf{R}') \tag{C.14}$$

a function that satisfies this is the exponential function,

$$e^{-i\mathbf{q}\cdot(\mathbf{R}+\mathbf{R}')} = e^{-i\mathbf{q}\cdot\mathbf{R}}e^{-i\mathbf{q}\cdot\mathbf{R}'} \tag{C.15}$$

the minus sign is for convention, thus,

$$\hat{T}(\mathbf{R})|\mathbf{q}\rangle = e^{-i\mathbf{q}\cdot\mathbf{R}}|\mathbf{q}\rangle \tag{C.16}$$

Then recall the reciprocal lattice vector $\mathbf{G}$ by construction satisfy,

$$e^{i\mathbf{G}\cdot\mathbf{R}} = 1 \tag{C.17}$$

thus $\mathbf{q}$ is only defined up to modulo $\mathbf{G}$, i.e. The first Brillouin zone. State with $\mathbf{q}$ and $\mathbf{q}+\mathbf{G}$ are physically identical. $\mathbf{q}\hbar$ is known as crystal momentum, and this is a conserved quantity. Now consider,

$$\langle\mathbf{r}|\hat{T}(\mathbf{R})|\mathbf{q}\rangle = \langle\mathbf{r}-\mathbf{R}|\mathbf{q}\rangle \tag{C.18}$$

using Eq. (C.16) yields,

$$\langle\mathbf{r}-\mathbf{R}|\mathbf{q}\rangle = e^{-i\mathbf{q}\cdot\mathbf{R}}\langle\mathbf{r}|\mathbf{q}\rangle \tag{C.19}$$

then letting $\mathbf{R}' = -\mathbf{R}$ then removing the prime mark yields,

$$\langle\mathbf{r}+\mathbf{R}|\mathbf{q}\rangle = e^{i\mathbf{q}\cdot\mathbf{R}}\langle\mathbf{r}|\mathbf{q}\rangle. \tag{C.20}$$

Now let,

$$u_{\mathbf{q}}(\mathbf{r}) \equiv e^{-i\mathbf{q}\cdot\mathbf{r}}\langle\mathbf{r}|\mathbf{q}\rangle \tag{C.21}$$

therefore,

$$\begin{aligned} u_{\mathbf{q}}(\mathbf{r}+\mathbf{R}) &\equiv e^{-i\mathbf{q}\cdot(\mathbf{r}+\mathbf{R})}\langle \mathbf{r}+\mathbf{R}|\mathbf{q}\rangle \\ &= e^{-i\mathbf{q}\cdot\mathbf{r}}e^{-i\mathbf{q}\cdot\mathbf{R}}\langle \mathbf{r}|\hat{T}(-\mathbf{R})|\mathbf{q}\rangle \\ &= e^{-i\mathbf{q}\cdot\mathbf{r}}e^{-i\mathbf{q}\cdot\mathbf{R}}e^{+i\mathbf{q}\cdot\mathbf{R}}\langle \mathbf{r}|\mathbf{q}\rangle \\ &= e^{-i\mathbf{q}\cdot\mathbf{r}}\langle \mathbf{r}|\mathbf{q}\rangle \\ &= u_{\mathbf{q}}(\mathbf{r}) \end{aligned} \tag{C.22}$$

thus $u_{\mathbf{q}}(\mathbf{r})$ is periodic, along translations along lattice vector. Thus,

$$\langle \mathbf{r}|\mathbf{q}\rangle = e^{i\mathbf{q}\cdot\mathbf{r}}e^{-i\mathbf{q}\cdot\mathbf{r}}\langle \mathbf{r}|\mathbf{q}\rangle = e^{i\mathbf{q}\cdot\mathbf{r}}u_{\mathbf{q}}(\mathbf{r}) \tag{C.23}$$

which is Bloch's theorem: an eigenstate of the Hamiltonian with periodic translation symmetry can be written as a plane wave $e^{i\mathbf{q}\cdot\mathbf{r}}$ multiplied into a periodic function with same periodicity as lattice $u_{\mathbf{q}}(\mathbf{r})$.

D
Phonons

I

Suppose the molecules have equilibrium positions $\{\mathbf{R}^0\}$. We Taylor expand the potential energy about the equilibrium geometry up to second order, and then call the harmonic part V:

$$V\{\mathbf{R}\} = V\{\mathbf{R}^0\} + \sum_{i\alpha} \frac{\partial V\{\mathbf{R}^0\}}{\partial R_{i\alpha}}(R_{i\alpha} - R^0_{i\alpha}) + \frac{1}{2}\sum_{i\alpha j\alpha'} \frac{\partial^2 V\{\mathbf{R}^0\}}{\partial R_{i\alpha}\partial R_{j\alpha'}}(R_{i\alpha} - R^0_{i\alpha})(R_{j\alpha'} - R^0_{j\alpha'}) \tag{D.1}$$

where $\partial V\{\mathbf{R^0}\}/\partial R_{i\alpha}$ is the derivative with respect to the displacement of atom i in direction α, evaluated at the equilibrium geometry. Using $i \equiv (l, b)$, $u_{lb\alpha} \equiv R_{lb\alpha} - R^0_{lb\alpha}$, and the chain rule,

$$\frac{\partial}{\partial u_{lb\alpha}} = \frac{\partial R_{lb\alpha}}{\partial u_{lb\alpha}}\frac{\partial}{\partial R_{lb\alpha}} = \frac{\partial}{\partial R_{lb\alpha}} \tag{D.2}$$

and defining the zero of energy by $V\{\mathbf{R}^0\} = 0$, together with the fact that there is no force at equilibrium, we obtain

$$V_0 = \frac{1}{2}\sum_{lb\alpha}\sum_{l'b'\alpha'} u_{lb\alpha}\Phi_{lb\alpha,l'b'\alpha'}u_{l'b'\alpha'} \tag{D.3}$$

where we have used the definition

$$\Phi_{lb\alpha,l'b'\alpha'} \equiv \left.\frac{\partial^2 V}{\partial u_{lb\alpha}\partial u_{l'b'\alpha'}}\right|_{u=0} \tag{D.4}$$

The crystal Hamiltonian under the harmonic approximation is

$$H_0 = \sum_{lb\alpha} \frac{p_{lb\alpha}^2}{2m_b} + \frac{1}{2} \sum_{lb\alpha} \sum_{l'b'\alpha'} u_{lb\alpha} \Phi_{lb\alpha,l'b'\alpha'} u_{l'b'\alpha'} \tag{D.5}$$

In the rest of this subsection, we introduce the notation used throughout Appendix D. If we have N_l lattice points, and hence N_l primitive cells in the supercell, with n_b atoms in each primitive cell, then the force-constant matrix Φ, which couples the displacement directions of the atoms, is a $3n_bN_l$ by $3n_bN_l$ square matrix. The unit operator of the mode space can be written as

$$\tilde{1} = \sum_{lb\alpha} |lb\alpha\rangle\langle lb\alpha| \tag{D.6}$$

$$\langle lb\alpha|l'b'\alpha'\rangle = \delta_{ll'}\delta_{bb'}\delta_{\alpha\alpha'} \tag{D.7}$$

where $|lb\alpha\rangle$ denotes the α-direction degree of freedom of atom b in primitive cell l, whose lattice point is $\mathbf{R}_l$. Let $l = 0$ be the reference cell and let $\mathbf{R}_0 = 0$ be the origin of our coordinate system.

We also know that $N_{\mathbf{q}} = N_l$ and $N_s = 3n_b$, where $N_{\mathbf{q}}$ is the number of $\mathbf{q}$-points in the Brillouin zone and N_s is the number of phonon branches. Thus,

$$\tilde{1} = \sum_{\mathbf{q}s} |\mathbf{q}s\rangle\langle\mathbf{q}s| \tag{D.8}$$

$$\langle\mathbf{q}s|\mathbf{q}'s'\rangle = \delta_{\mathbf{q}\mathbf{q}'}\delta_{ss'} \tag{D.9}$$

We shall reserve tildes for operators in mode space and hats for operators in the quantum state space. We define the mass-weighted quantities as

$$K_{lb\alpha,l'b'\alpha'} \equiv \frac{\Phi_{lb\alpha,l'b'\alpha'}}{\sqrt{m_b m_{b'}}} = \langle lb\alpha|\tilde{K}|l'b'\alpha'\rangle \tag{D.10}$$

$$\hat{\pi}_{lb\alpha} \equiv \frac{\hat{p}_{lb\alpha}}{\sqrt{m_b}} = \langle lb\alpha|\hat{\pi}\rangle \tag{D.11}$$

$$\hat{\mu}_{lb\alpha} \equiv \sqrt{m_b}\hat{u}_{lb\alpha} = \langle lb\alpha|\hat{\mu}\rangle \tag{D.12}$$

Thus, the harmonic crystal Hamiltonian can be written as

$$\hat{H}_0 = \frac{1}{2} \sum_{lb\alpha} \langle\hat{\pi}|lb\alpha\rangle\langle lb\alpha|\hat{\pi}\rangle + \frac{1}{2} \sum_{lb\alpha} \sum_{l'b'\alpha'} \langle\hat{\mu}|lb\alpha\rangle\langle lb\alpha|\tilde{K}|l'b'\alpha'\rangle\langle l'b'\alpha'|\hat{\mu}\rangle \tag{D.13}$$

Removing the inserted unit operators gives

$$\hat{H}_0 = \frac{1}{2}\langle\hat{\pi}|\hat{\pi}\rangle + \frac{1}{2}\langle\hat{\mu}|\tilde{K}|\hat{\mu}\rangle \tag{D.14}$$

Now note that $\tilde{K}$ has the spectral representation

$$\tilde{K} = \sum_{\mathbf{q}s} |\mathbf{q}s\rangle\omega_{\mathbf{q}s}^2\langle\mathbf{q}s| \tag{D.15}$$

Reinserting this spectral representation and the unit operator in the normal-mode representation gives

$$\hat{H}_0 = \frac{1}{2}\sum_{\mathbf{q}s}\langle\hat{\pi}|\mathbf{q}s\rangle\langle\mathbf{q}s|\hat{\pi}\rangle + \frac{1}{2}\sum_{\mathbf{q}s}\langle\hat{\mu}|\mathbf{q}s\rangle\omega_{\mathbf{q}s}^2\langle\mathbf{q}s|\hat{\mu}\rangle \tag{D.16}$$

We then define the normal-mode coordinates and momenta as

$$\hat{P}_{\mathbf{q}s} \equiv \langle\mathbf{q}s|\hat{\pi}\rangle = \langle\hat{\pi}|\mathbf{q}s\rangle^* \tag{D.17}$$

$$\hat{Q}_{\mathbf{q}s} \equiv \langle\mathbf{q}s|\hat{\mu}\rangle = \langle\hat{\mu}|\mathbf{q}s\rangle^* \tag{D.18}$$

Thus,

$$\hat{H}_0 = \frac{1}{2}\sum_{\mathbf{q}s}\left(\hat{P}_{\mathbf{q}s}^\dagger\hat{P}_{\mathbf{q}s} + \omega_{\mathbf{q}s}^2\hat{Q}_{\mathbf{q}s}^\dagger\hat{Q}_{\mathbf{q}s}\right) \tag{D.19}$$

Now let us introduce the frequency operator

$$\tilde{\omega} \equiv \sum_{\mathbf{q}s} |\mathbf{q}s\rangle\omega_{\mathbf{q}s}\langle\mathbf{q}s| \tag{D.20}$$

and the new ket

$$|\hat{a}\rangle \equiv \frac{1}{\sqrt{2\hbar}}\left(\tilde{\omega}^{1/2}|\hat{\mu}\rangle + i\tilde{\omega}^{-1/2}|\hat{\pi}\rangle\right) \tag{D.21}$$

where

$$\langle\mathbf{q}s|\hat{a}\rangle = \hat{a}_{\mathbf{q}s} \tag{D.22}$$

$$\langle\hat{a}|\mathbf{q}s\rangle = \hat{a}_{\mathbf{q}s}^\dagger \tag{D.23}$$

Consider

$$\begin{aligned}\langle\hat{a}|\hbar\tilde{\omega}|\hat{a}\rangle &= \hbar\frac{1}{2\hbar}\Big(\langle\hat{\mu}|\tilde{\omega}^{1/2} - i\langle\hat{\pi}|\tilde{\omega}^{-1/2}\Big)\tilde{\omega}\Big(\tilde{\omega}^{1/2}|\hat{\mu}\rangle + i\tilde{\omega}^{-1/2}|\hat{\pi}\rangle\Big)\\ &= \frac{1}{2}\Big(\langle\hat{\mu}|\tilde{\omega}^2|\hat{\mu}\rangle + \langle\hat{\pi}|\hat{\pi}\rangle + i(\langle\hat{\mu}|\tilde{\omega}|\hat{\pi}\rangle - \langle\hat{\pi}|\tilde{\omega}|\hat{\mu}\rangle)\Big)\\ &= \frac{1}{2}\Big(\langle\hat{\mu}|\tilde{K}|\hat{\mu}\rangle + \langle\hat{\pi}|\hat{\pi}\rangle + i(\langle\hat{\mu}|\tilde{\omega}|\hat{\pi}\rangle - \langle\hat{\mu}|\tilde{\omega}|\hat{\pi}\rangle^*)\Big)\\ &= H_0 + \frac{i}{2}2i\mathrm{Im}\langle\hat{\mu}|\tilde{\omega}|\hat{\pi}\rangle\\ &= H_0 - \mathrm{Im}\langle\hat{\mu}|\tilde{\omega}|\hat{\pi}\rangle \end{aligned} \tag{D.24}$$

Thus,

$$\hat{H}_0 = \langle\hat{a}|\hbar\tilde{\omega}|\hat{a}\rangle + \mathrm{Im}\langle\hat{\mu}|\tilde{\omega}|\hat{\pi}\rangle \tag{D.25}$$

II

We will derive the form of the dynamical matrix in this section. Consider the translation operator defined so that,

$$\tilde{T}(\mathbf{R}_{l'})|lb\alpha\rangle = |(l+l')b\alpha\rangle \tag{D.26}$$

and notice that the force constant matrix only depends on the relative cell positions between the two molecules rather than absolute position, we have,

$$K_{lb\alpha,l'b'\alpha'} = K_{(l-l')b\alpha,0b'\alpha'} \tag{D.27}$$

in new notation,

$$\langle lb\alpha|\tilde{K}|l'b'\alpha'\rangle = \langle(l-l')b\alpha|\tilde{K}|0b'\alpha'\rangle \tag{D.28}$$

using the definition of the translation operator,

$$\langle lb\alpha|\tilde{K}\tilde{T}(\mathbf{R}_{l'})|0b'\alpha'\rangle = \langle lb\alpha|\tilde{T}(\mathbf{R}_{l'})\tilde{K}|0b'\alpha'\rangle \tag{D.29}$$

therefore,

$$[\tilde{K}, \tilde{T}(\mathbf{R}_l)] = 0 \tag{D.30}$$

thus there exists a complete set of simultaneous eigenstates of both operators so that,

$$\tilde{T}(\mathbf{R}_l)|\mathbf{q}s\rangle = e^{-i\mathbf{q}\cdot\mathbf{R}_l}|\mathbf{q}s\rangle \tag{D.31}$$

$$\tilde{K}|\mathbf{q}s\rangle = \omega^2_{\mathbf{q}s}|\mathbf{q}s\rangle \tag{D.32}$$

notice that the mass-weighted hessian $\tilde{K}$, has the dimension of frequency squared, thus it is dimensionally consistent with right-hand side. We now insert a complete set of states between $\tilde{K}$ and $|\mathbf{q}s\rangle$ then project onto $\langle 0b\alpha|$,

$$\sum_{l'b'\alpha'} \langle 0b\alpha|\tilde{K}|l'b'\alpha'\rangle\langle l'b'\alpha'|\mathbf{q},s\rangle = \omega^2_{\mathbf{q}s}\langle 0b\alpha|\mathbf{q}s\rangle \tag{D.33}$$

Then use the fact that,

$$\langle lb\alpha|\mathbf{q}s\rangle = \langle 0b\alpha|\tilde{T}(-\mathbf{R}_l)|\mathbf{q}s\rangle = e^{+i\mathbf{q}\cdot\mathbf{R}_l}\langle 0b\alpha|\mathbf{q}s\rangle \tag{D.34}$$

thus we have,

$$\sum_{l'b'\alpha'} K_{0b\alpha,l'b'\alpha'}e^{+i\mathbf{q}\cdot\mathbf{R}_{l'}}\langle 0b'\alpha'|\mathbf{q}s\rangle = \omega^2_{\mathbf{q}s}\langle 0b\alpha|\mathbf{q}s\rangle \tag{D.35}$$

we now define polarisation vector $\epsilon_{s,b\alpha}(\mathbf{q})$ via,

$$\epsilon_{s,b\alpha}(\mathbf{q}) \equiv \sqrt{N_{\mathbf{q}}}e^{-i\mathbf{q}\cdot\boldsymbol{\tau}_b}\langle 0b\alpha|\mathbf{q}s\rangle \tag{D.36}$$

thus,

$$\langle lb\alpha|\mathbf{q}s\rangle = \frac{1}{\sqrt{N_{\mathbf{q}}}}e^{i\mathbf{q}\cdot(\mathbf{R}_l+\boldsymbol{\tau}_b)}\epsilon_{s,b\alpha}(\mathbf{q}). \tag{D.37}$$

Consider,

$$\begin{aligned}
\langle \mathbf{q}s|\mathbf{q}s'\rangle &= \sum_{lb\alpha}\langle \mathbf{q}s|lb\alpha\rangle\langle lb\alpha|\mathbf{q}s'\rangle \\
&= \sum_{lb\alpha}\langle lb\alpha|\mathbf{q}s\rangle^* e^{+i\mathbf{q}\cdot\mathbf{R}_l}\langle 0b\alpha|\mathbf{q}s'\rangle \\
&= \sum_{lb\alpha}\Big(e^{+i\mathbf{q}\cdot\mathbf{R}_l}\langle 0b\alpha|\mathbf{q}s\rangle\Big)^* e^{+i\mathbf{q}\cdot\mathbf{R}_l}\langle 0b\alpha|\mathbf{q}s'\rangle \\
&= \sum_{lb\alpha}\langle \mathbf{q}s|0b\alpha\rangle e^{-i\mathbf{q}\cdot\mathbf{R}_l}e^{+i\mathbf{q}\cdot\mathbf{R}_l}\langle 0b\alpha|\mathbf{q}s'\rangle \\
&= \sum_{b\alpha}\epsilon^*_{s,b\alpha}(\mathbf{q})\epsilon_{s',b\alpha}(\mathbf{q})\frac{1}{N_{\mathbf{q}}}\sum_l 1 \\
&= \sum_{b\alpha}\epsilon^*_{s,b\alpha}(\mathbf{q})\epsilon_{s',b\alpha}(\mathbf{q})
\end{aligned} \tag{D.38}$$

where we have used $N_l = N_{\mathbf{q}}$ and since both sides equals to $\delta_{ss'}$, we have

$$\delta_{ss'} = \sum_{b\alpha} \epsilon^*_{s,b\alpha}(\mathbf{q})\epsilon_{s',b\alpha}(\mathbf{q}) \tag{D.39}$$

thus, $\boldsymbol{\epsilon}_{s,b\alpha}$ vector is orthonormal within a given $\mathbf{q}$-block in the full Hessian Φ. Substituting the polarisation vector yields,

$$\sum_{b'\alpha'}\sum_{l'} \frac{1}{\sqrt{m_b m_{b'}}} \Phi_{0b\alpha,l'b'\alpha'} e^{+i\mathbf{q}\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'}-\boldsymbol{\tau}_b)} \epsilon_{s,b'\alpha'}(\mathbf{q}) = \omega^2_{\mathbf{q}s}\epsilon_{s,b\alpha}(\mathbf{q}) \tag{D.40}$$

defining the dynamical matrix as,

$$D_{b\alpha,b'\alpha'}(\mathbf{q}) \equiv \frac{1}{\sqrt{m_b m_{b'}}} \sum_{l'} \Phi_{0b\alpha,l'b'\alpha'} e^{+i\mathbf{q}\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'}-\boldsymbol{\tau}_b)} \tag{D.41}$$

substitute into previous equation yields,

$$\sum_{b'\alpha'} D_{b\alpha,b'\alpha'}(\mathbf{q})\epsilon_{s,b'\alpha'}(\mathbf{q}) = \omega^2_{\mathbf{q}s}\epsilon_{s,b\alpha}(\mathbf{q}) \tag{D.42}$$

in vector notation,

$$D(\mathbf{q})\boldsymbol{\epsilon}_s(\mathbf{q}) = \omega^2_{\mathbf{q}s}\boldsymbol{\epsilon}_s(\mathbf{q}) \tag{D.43}$$

III

In this subsection, we shall derive the form of the normal mode coordinates and momenta, then deduce some of their properties,

$$\begin{aligned}
\hat{Q}_{\mathbf{q}s} &\equiv \langle \mathbf{q}s|\hat{\mu}\rangle \\
&= \sum_{lb\alpha} \langle \mathbf{q}s|lb\alpha\rangle\langle lb\alpha|\hat{\mu}\rangle \\
&= \sum_{lb\alpha} \langle lb\alpha|\mathbf{q}s\rangle^* \sqrt{m_b}\hat{u}_{lb\alpha} \\
&= \sum_{lb\alpha} \left(\langle 0b\alpha|\mathbf{q}s\rangle e^{+i\mathbf{q}\cdot\mathbf{R}_l}\right)^* \sqrt{m_b}\hat{u}_{lb\alpha} \\
&= \frac{1}{\sqrt{N_{\mathbf{q}}}} \sum_{lb\alpha} \left(\epsilon_{s,b\alpha}(\mathbf{q}) e^{+i\mathbf{q}\cdot\boldsymbol{\tau}_b} e^{+i\mathbf{q}\cdot\mathbf{R}_l}\right)^* \sqrt{m_b}\hat{u}_{lb\alpha} \\
&= \frac{1}{\sqrt{N_{\mathbf{q}}}} \sum_{lb\alpha} \sqrt{m_b}\hat{u}_{lb\alpha} e^{-i\mathbf{q}\cdot(\mathbf{R}_l+\boldsymbol{\tau}_b)} \epsilon^*_{s,b\alpha}(\mathbf{q})
\end{aligned} \tag{D.44}$$

and normal mode momentum is,

$$\begin{aligned}\hat{P}_{\mathbf{q}s} &\equiv \langle \mathbf{q}s|\hat{\pi}\rangle \\ &= \sum_{lb\alpha}\langle \mathbf{q}s|lb\alpha\rangle\langle lb\alpha|\hat{\pi}\rangle \\ &= \sum_{lb\alpha}\langle lb\alpha|\mathbf{q}s\rangle^{*}\frac{\hat{p}_{lb\alpha}}{\sqrt{m_b}} \\ &= \frac{1}{\sqrt{N_{\mathbf{q}}}}\sum_{lb\alpha}\frac{\hat{p}_{lb\alpha}}{\sqrt{m_b}}e^{-i\mathbf{q}\cdot(\mathbf{R}_l+\boldsymbol{\tau}_b)}\epsilon^{*}_{s,b\alpha}(\mathbf{q})\end{aligned} \tag{D.45}$$

Similarly,

$$\hat{u}_{lb\alpha} = \frac{1}{\sqrt{m_b}}\sum_{\mathbf{q}s}\langle lb\alpha|\mathbf{q}s\rangle\langle \mathbf{q}s|\hat{\mu}\rangle = \frac{1}{\sqrt{N_{\mathbf{q}}m_b}}\sum_{\mathbf{q}s}\hat{Q}_{\mathbf{q}s}e^{i\mathbf{q}\cdot(\mathbf{R}_l+\boldsymbol{\tau}_b)}\epsilon_{s,b\alpha}(\mathbf{q}) \tag{D.46}$$

$$\hat{p}_{lb\alpha} = \sqrt{m_b}\sum_{\mathbf{q}s}\langle lb\alpha|\mathbf{q}s\rangle\langle \mathbf{q}s|\hat{\pi}\rangle = \sqrt{\frac{m_b}{N_{\mathbf{q}}}}\sum_{\mathbf{q}s}\hat{P}_{\mathbf{q}s}e^{i\mathbf{q}\cdot(\mathbf{R}_l+\boldsymbol{\tau}_b)}\epsilon_{s,b\alpha}(\mathbf{q}) \tag{D.47}$$

Now consider the Bloch theorem we have used many times,

$$\langle lb\alpha|\mathbf{q}s\rangle = e^{i\mathbf{q}\cdot\mathbf{R}_l}\langle 0b\alpha|\mathbf{q}s\rangle \tag{D.48}$$

take complex conjugate of both sides,

$$\langle \mathbf{q}s|lb\alpha\rangle = e^{-i\mathbf{q}\cdot\mathbf{R}_l}\langle \mathbf{q}s|0b\alpha\rangle \tag{D.49}$$

Now define $|\chi_{\mathbf{q}s}\rangle$ via,

$$\langle lb\alpha|\chi_{\mathbf{q}s}\rangle \equiv \langle \mathbf{q}s|lb\alpha\rangle \tag{D.50}$$

substitute into the previous equation yields,

$$\langle lb\alpha|\chi_{\mathbf{q}s}\rangle = e^{-i\mathbf{q}\cdot\mathbf{R}_l}\langle 0b\alpha|\chi_{\mathbf{q}s}\rangle \tag{D.51}$$

comparison with the original Bloch theorem suggests we should relabel our ket $|\chi_{\mathbf{q}s}\rangle \rightarrow |-\mathbf{q}s\rangle$. Thus we have,

$$\langle lb\alpha|\mathbf{q}s\rangle^{*} = \langle lb\alpha|-\mathbf{q}s\rangle \tag{D.52}$$

one can choose the phases, and within degenerate subspaces choose the basis, so that this relation holds. and it directly follows from the expansion of $\hat{Q}_{\mathbf{q}s}$ and $\hat{P}_{\mathbf{q}s}$ in $\{|lb\alpha\rangle\}$ basis that,

$$\hat{Q}^{\dagger}_{\mathbf{q}s} = \hat{Q}_{-\mathbf{q}s} \tag{D.53}$$

$$\hat{P}^{\dagger}_{\mathbf{q}s} = \hat{P}_{-\mathbf{q}s} \tag{D.54}$$

Now consider,

$$\langle \mathbf{q}s|\tilde{K} = \omega^2_{\mathbf{q}s}\langle \mathbf{q}s| \tag{D.55}$$

right multiply by $|lb\alpha\rangle$ then insert a complete set of states,

$$\sum_{l'b'\alpha'} \langle \mathbf{q}s|l'b'\alpha'\rangle\langle l'b'\alpha'|\tilde{K}|lb\alpha\rangle = \omega^2_{\mathbf{q}s}\langle \mathbf{q}s|lb\alpha\rangle \tag{D.56}$$

Use the symmetry of mixed second derivative $\tilde{K}$ in atomic representation, $\langle l'b'\alpha'|\tilde{K}|lb\alpha\rangle = \langle lb\alpha|\tilde{K}|l'b'\alpha'\rangle$, we have,

$$\sum_{l'b'\alpha'} \langle \mathbf{q}s|l'b'\alpha'\rangle\langle lb\alpha|\tilde{K}|l'b'\alpha'\rangle = \omega^2_{\mathbf{q}s}\langle \mathbf{q}s|lb\alpha\rangle \tag{D.57}$$

use the original definition of $|\chi_{\mathbf{q}s}\rangle$ yields,

$$\sum_{l'b'\alpha'} \langle lb\alpha|\tilde{K}|l'b'\alpha'\rangle\langle l'b'\alpha'|-\mathbf{q}s\rangle = \omega^2_{\mathbf{q}s}\langle lb\alpha|-\mathbf{q}s\rangle \tag{D.58}$$

de-project from $\langle lb\alpha|$ and take out the unit operator yields,

$$\tilde{K}|-\mathbf{q}s\rangle = \omega^2_{\mathbf{q}s}|-\mathbf{q}s\rangle \tag{D.59}$$

thus $|-\mathbf{q}s\rangle$ has the same frequency as $|\mathbf{q}s\rangle$. Now consider the commutator,

$$[\hat{Q}_{\mathbf{q}s}, \hat{P}_{-\mathbf{q}'s'}] = \sum_{lb\alpha}\sum_{l'b'\alpha'} \langle \mathbf{q}s|lb\alpha\rangle\sqrt{m_b}\,[\hat{u}_{lb\alpha}, \hat{p}_{l'b'\alpha'}]\frac{1}{\sqrt{m'_b}}\langle -\mathbf{q}'s'|l'b'\alpha'\rangle \tag{D.60}$$

use the canonical commutation and the definition of $|\chi_{\mathbf{q}s}\rangle$ yields,

$$[\hat{Q}_{\mathbf{q}s}, \hat{P}_{-\mathbf{q}'s'}] = i\hbar\sum_{lb\alpha}\sum_{l'b'\alpha'} \langle \mathbf{q}s|lb\alpha\rangle\delta_{ll'}\delta_{bb'}\delta_{\alpha\alpha'}\langle l'b'\alpha'|\mathbf{q}'s'\rangle \tag{D.61}$$

contract the sum,

$$[\hat{Q}_{\mathbf{q}s}, \hat{P}_{-\mathbf{q}'s'}] = i\hbar \sum_{lb\alpha} \langle \mathbf{q}s|lb\alpha\rangle\langle lb\alpha|\mathbf{q}'s'\rangle \tag{D.62}$$

take out the unit operator,

$$[\hat{Q}_{\mathbf{q}s}, \hat{P}_{-\mathbf{q}'s'}] = i\hbar \langle \mathbf{q}s|\mathbf{q}'s'\rangle \tag{D.63}$$

thus,

$$[\hat{Q}_{\mathbf{q}s}, \hat{P}_{-\mathbf{q}'s'}] = i\hbar\delta_{\mathbf{q}\mathbf{q}'}\delta_{ss'} \tag{D.64}$$

Now consider,

$$\begin{aligned}
\mathrm{Im}\langle\hat{\mu}|\tilde{\omega}|\hat{\pi}\rangle &= \frac{1}{2i}\Big(\langle\hat{\mu}|\tilde{\omega}|\hat{\pi}\rangle - \langle\hat{\pi}|\tilde{\omega}|\hat{\mu}\rangle\Big) \\
&= \frac{1}{2i}\sum_{\mathbf{q}s}\langle\hat{\mu}|\mathbf{q}s\rangle\omega_{\mathbf{q}s}\langle\mathbf{q}s|\hat{\pi}\rangle - \frac{1}{2i}\sum_{\mathbf{q}s}\langle\hat{\pi}|\mathbf{q}s\rangle\omega_{\mathbf{q}s}\langle\mathbf{q}s|\hat{\mu}\rangle \\
&= \frac{1}{2i}\sum_{\mathbf{q}s}\langle-\mathbf{q}s|\hat{\mu}\rangle\omega_{\mathbf{q}s}\langle\mathbf{q}s|\hat{\pi}\rangle - \frac{1}{2i}\sum_{\mathbf{q}s}\langle-\mathbf{q}s|\hat{\pi}\rangle\omega_{\mathbf{q}s}\langle\mathbf{q}s|\hat{\mu}\rangle \\
&= \frac{1}{2i}\sum_{\mathbf{q}s}\omega_{\mathbf{q}s}\hat{Q}_{-\mathbf{q}s}\hat{P}_{\mathbf{q}s} - \frac{1}{2i}\sum_{\mathbf{q}s}\omega_{\mathbf{q}s}\hat{P}_{-\mathbf{q}s}\hat{Q}_{\mathbf{q}s}
\end{aligned} \tag{D.65}$$

then use the relabelling $\mathbf{q}' = -\mathbf{q}$ for the first sum we have,

$$\mathrm{Im}\langle\hat{\mu}|\tilde{\omega}|\hat{\pi}\rangle = \frac{1}{2i}\sum_{\mathbf{q}'s}\omega_{-\mathbf{q}'s}\hat{Q}_{\mathbf{q}'s}\hat{P}_{-\mathbf{q}'s} - \frac{1}{2i}\sum_{\mathbf{q}s}\omega_{\mathbf{q}s}\hat{P}_{-\mathbf{q}s}\hat{Q}_{\mathbf{q}s} \tag{D.66}$$

then use $\omega_{\mathbf{q}s} = \omega_{-\mathbf{q}s}$ and $\mathbf{q}'$ is a dummy index to combine the two sums,

$$\mathrm{Im}\langle\hat{\mu}|\tilde{\omega}|\hat{\pi}\rangle = \frac{1}{2i}\sum_{\mathbf{q}s}\omega_{\mathbf{q}s}\Big(\hat{Q}_{\mathbf{q}s}\hat{P}_{-\mathbf{q}s} - \hat{P}_{-\mathbf{q}s}\hat{Q}_{\mathbf{q}s}\Big) \tag{D.67}$$

then use the commutation relation in normal mode yields,

$$\mathrm{Im}\langle\hat{\mu}|\tilde{\omega}|\hat{\pi}\rangle = \frac{1}{2i}\sum_{\mathbf{q}s}\omega_{\mathbf{q}s}i\hbar \tag{D.68}$$

thus,

$$\mathrm{Im}\langle\hat{\mu}|\tilde{\omega}|\hat{\pi}\rangle = \frac{1}{2}\hbar\,\mathrm{tr}\,\tilde{\omega} \tag{D.69}$$

finally,

$$\hat{H}_0 = \hbar\langle\hat{a}|\tilde{\omega}|\hat{a}\rangle + \frac{1}{2}\hbar\,\mathrm{tr}\,\tilde{\omega} \tag{D.70}$$

insert a complete set of normal modes to obtain the more familiar expression,

$$\hat{H}_0 = \sum_{\mathbf{q}s} \hbar\omega_{\mathbf{q}s} \left(\hat{a}^\dagger_{\mathbf{q}s} \hat{a}_{\mathbf{q}s} + \frac{1}{2} \right) \tag{D.71}$$

Now we derive another useful expression of the normal mode coordinate in terms of ladder operators, consider,

$$\langle \mathbf{q}s|\hat{a}\rangle = \hat{a}_{\mathbf{q}s} = \frac{1}{\sqrt{2\hbar}} \left(\omega^{1/2}_{\mathbf{q}s} \hat{Q}_{\mathbf{q}s} + i\omega^{-1/2}_{\mathbf{q}s} \hat{P}_{\mathbf{q}s} \right) \tag{D.72}$$

and,

$$\langle -\mathbf{q}s|\hat{a}\rangle = \hat{a}_{-\mathbf{q}s} = \frac{1}{\sqrt{2\hbar}} \left(\omega^{1/2}_{\mathbf{q}s} \hat{Q}_{-\mathbf{q}s} + i\omega^{-1/2}_{\mathbf{q}s} \hat{P}_{-\mathbf{q}s} \right) \tag{D.73}$$

therefore,

$$\hat{a}^\dagger_{-\mathbf{q}s} = \frac{1}{\sqrt{2\hbar}} \left(\omega^{1/2}_{\mathbf{q}s} \hat{Q}^\dagger_{-\mathbf{q}s} - i\omega^{-1/2}_{\mathbf{q}s} \hat{P}^\dagger_{-\mathbf{q}s} \right) \tag{D.74}$$

using $\hat{Q}_{-\mathbf{q}s} = \hat{Q}^\dagger_{\mathbf{q}s}$, and $\hat{P}_{-\mathbf{q}s} = \hat{P}^\dagger_{\mathbf{q}s}$,

$$\hat{a}^\dagger_{-\mathbf{q}s} = \frac{1}{\sqrt{2\hbar}} \left(\omega^{1/2}_{\mathbf{q}s} \hat{Q}_{\mathbf{q}s} - i\omega^{-1/2}_{\mathbf{q}s} \hat{P}_{\mathbf{q}s} \right) \tag{D.75}$$

thus,

$$\hat{a}_{\mathbf{q}s} + \hat{a}^\dagger_{-\mathbf{q}s} = \sqrt{\frac{2}{\hbar}} \omega^{1/2}_{\mathbf{q}s} \hat{Q}_{\mathbf{q}s} \tag{D.76}$$

after a slight rearrangement, we obtain,

$$\hat{Q}_{\mathbf{q}s} = \left(\frac{\hbar}{2\omega_{\mathbf{q}s}} \right)^{1/2} \left(\hat{a}_{\mathbf{q}s} + \hat{a}^\dagger_{-\mathbf{q}s} \right). \tag{D.77}$$

IV

In this section, we demonstrate the dynamical matrix block-diagonalises the full mass-weighted hessian, where each small block of size $3 \times n_b$ has a well-defined

crystal momentum $\mathbf{q}$. Consider,

$$\begin{aligned}
D^*_{b\alpha,b'\alpha'}(\mathbf{q}) &= \frac{1}{\sqrt{m_b m_{b'}}} \sum_{l'} \Phi_{0b\alpha,l'b'\alpha'} e^{-i\mathbf{q}\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'}-\boldsymbol{\tau}_b)} \\
&= \frac{1}{\sqrt{m_b m_{b'}}} \sum_{l'} \Phi_{l'b'\alpha',0b\alpha} e^{-i\mathbf{q}\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'}-\boldsymbol{\tau}_b)} \\
&= \frac{1}{\sqrt{m_b m_{b'}}} \sum_{l'} \Phi_{0b'\alpha',(-l')b\alpha} e^{-i\mathbf{q}\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'}-\boldsymbol{\tau}_b)} \\
&= \frac{1}{\sqrt{m_b m_{b'}}} \sum_{l} \Phi_{0b'\alpha',lb\alpha} e^{-i\mathbf{q}\cdot(\mathbf{R}_{-l}+\boldsymbol{\tau}_{b'}-\boldsymbol{\tau}_b)} \\
&= \frac{1}{\sqrt{m_b m_{b'}}} \sum_{l} \Phi_{0b'\alpha',lb\alpha} e^{+i\mathbf{q}\cdot(\mathbf{R}_{l}-\boldsymbol{\tau}_{b'}+\boldsymbol{\tau}_b)} \\
&= \frac{1}{\sqrt{m_b m_{b'}}} \sum_{l} \Phi_{0b'\alpha',lb\alpha} e^{+i\mathbf{q}\cdot(\mathbf{R}_{l}+\boldsymbol{\tau}_{b}-\boldsymbol{\tau}_{b'})} \\
&= D_{b'\alpha',b\alpha}(\mathbf{q})
\end{aligned} \tag{D.78}$$

thus the dynamical matrix is Hermitian, the eigenvalue equation can then be written as,

$$\bar{D}(\mathbf{q})|s\rangle = \omega^2_{\mathbf{q}s}|s\rangle \tag{D.79}$$

where we have suppressed the parametric dependence of $\mathbf{q}$ of $|s\rangle$, but it is left as understood. Where, the overbar denotes the dynamical operator $\bar{D}$ operates in the $3n_b$ subspace for a given $\mathbf{q}$. The orthonormality of the polarisation vector can now be written as,

$$\langle s|s'\rangle = \delta_{ss'} \tag{D.80}$$

and completeness relation in the $\mathbf{q}$-subspace is,

$$\bar{1} = \sum_s |s\rangle\langle s| \tag{D.81}$$

Now, introduce a new basis for the full-mode space $\{|\mathbf{q}b\alpha\rangle\}$ via the relation,

$$\langle lb\alpha|\mathbf{q}b'\alpha'\rangle \equiv \frac{1}{\sqrt{N_{\mathbf{q}}}} e^{i\mathbf{q}\cdot(\mathbf{R}_l+\boldsymbol{\tau}_b)} \delta_{bb'}\delta_{\alpha\alpha'} \tag{D.82}$$

which satisfies the orthogonality condition,

$$\langle \mathbf{q}b\alpha|\mathbf{q}'b'\alpha'\rangle = \delta_{\mathbf{q}\mathbf{q}'}\delta_{bb'}\delta_{\alpha\alpha'} \tag{D.83}$$

and completeness relation,

$$\tilde{1} = \sum_{\mathbf{q}b\alpha} |\mathbf{q}b\alpha\rangle\langle\mathbf{q}b\alpha| \tag{D.84}$$

Now consider the matrix element of $\tilde{K}$ in this new basis,

$$\begin{aligned}
\langle\mathbf{q}b\alpha|\tilde{K}|\mathbf{q}'b'\alpha'\rangle &= \sum_{lb''\alpha''} \langle\mathbf{q}b\alpha|\tilde{K}|lb''\alpha''\rangle\langle lb''\alpha''|\mathbf{q}'b'\alpha'\rangle \\
&= \sum_{lb''\alpha''} \langle\mathbf{q}b\alpha|\tilde{K}|lb''\alpha''\rangle \frac{1}{\sqrt{N_{\mathbf{q}}}} e^{i\mathbf{q}'\cdot(\mathbf{R}_l+\boldsymbol{\tau}_{b''})} \delta_{b''b'}\delta_{\alpha''\alpha'} \\
&= \sum_{l} \langle\mathbf{q}b\alpha|\tilde{K}|lb'\alpha'\rangle \frac{1}{\sqrt{N_{\mathbf{q}}}} e^{i\mathbf{q}'\cdot(\mathbf{R}_l+\boldsymbol{\tau}_{b'})} \\
&= \frac{1}{\sqrt{N_{\mathbf{q}}}} \sum_{l'} \langle\mathbf{q}b\alpha|\tilde{K}|l'b'\alpha'\rangle e^{i\mathbf{q}'\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'})} \\
&= \frac{1}{\sqrt{N_{\mathbf{q}}}} \sum_{lb''\alpha''} \sum_{l'} \langle\mathbf{q}b\alpha|lb''\alpha''\rangle\langle lb''\alpha''|\tilde{K}|l'b'\alpha'\rangle e^{i\mathbf{q}'\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'})} \\
&= \frac{1}{N_{\mathbf{q}}} \sum_{lb''\alpha''} \sum_{l'} e^{-i\mathbf{q}\cdot(\mathbf{R}_l+\boldsymbol{\tau}_{b''})} \delta_{b''b}\delta_{\alpha''\alpha} \langle lb''\alpha''|\tilde{K}|l'b'\alpha'\rangle e^{i\mathbf{q}'\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'})} \\
&= \frac{1}{N_{\mathbf{q}}} \sum_{l} \sum_{l'} e^{-i\mathbf{q}\cdot(\mathbf{R}_l+\boldsymbol{\tau}_{b})} \langle lb\alpha|\tilde{K}|l'b'\alpha'\rangle e^{i\mathbf{q}'\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'})} \\
&= \frac{1}{N_{\mathbf{q}}} \sum_{l} \sum_{l'} e^{-i\mathbf{q}\cdot(\mathbf{R}_l+\boldsymbol{\tau}_{b})} \langle 0b\alpha|\tilde{K}|(l'-l)b'\alpha'\rangle e^{i\mathbf{q}'\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'})} \\
&= \frac{1}{N_{\mathbf{q}}} \sum_{l} \sum_{l''} e^{-i\mathbf{q}\cdot(\mathbf{R}_l+\boldsymbol{\tau}_{b})} \langle 0b\alpha|\tilde{K}|l''b'\alpha'\rangle e^{i\mathbf{q}'\cdot(\mathbf{R}_l+\mathbf{R}_{l''}+\boldsymbol{\tau}_{b'})} \\
&= \frac{1}{N_{\mathbf{q}}} \sum_{l} e^{i(\mathbf{q}'-\mathbf{q})\cdot\mathbf{R}_l} \sum_{l''} e^{-i\mathbf{q}\cdot\boldsymbol{\tau}_b} \langle 0b\alpha|\tilde{K}|l''b'\alpha'\rangle e^{i\mathbf{q}'\cdot(\mathbf{R}_{l''}+\boldsymbol{\tau}_{b'})} \\
&= \delta_{\mathbf{q}\mathbf{q}'} \sum_{l''} e^{-i\mathbf{q}\cdot\boldsymbol{\tau}_b} \langle 0b\alpha|\tilde{K}|l''b'\alpha'\rangle e^{i\mathbf{q}'\cdot(\mathbf{R}_{l''}+\boldsymbol{\tau}_{b'})} \\
&= \delta_{\mathbf{q}\mathbf{q}'} \sum_{l'} e^{-i\mathbf{q}\cdot\boldsymbol{\tau}_b} \langle 0b\alpha|\tilde{K}|l'b'\alpha'\rangle e^{i\mathbf{q}\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'})} \\
&= \delta_{\mathbf{q}\mathbf{q}'} \sum_{l'} \langle 0b\alpha|\tilde{K}|l'b'\alpha'\rangle e^{i\mathbf{q}\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'}-\boldsymbol{\tau}_b)} \\
&= \delta_{\mathbf{q}\mathbf{q}'} \frac{1}{\sqrt{m_b m_{b'}}} \sum_{l'} \Phi_{0b\alpha,l'b'\alpha'} e^{i\mathbf{q}\cdot(\mathbf{R}_{l'}+\boldsymbol{\tau}_{b'}-\boldsymbol{\tau}_b)} \\
&= \delta_{\mathbf{q}\mathbf{q}'} D_{b\alpha,b'\alpha'}(\mathbf{q})
\end{aligned} \tag{D.85}$$

Thus, the dynamical matrix does indeed block-diagonalise the full mass-weighted Hessian. In our new notation, this is,

$$\tilde{K} = \bigoplus_{\mathbf{q}} \bar{D}(\mathbf{q}). \tag{D.86}$$

V

We will now derive the form of the group velocity in a convenient form for calculation, which is defined as,

$$\mathbf{v}_{\mathbf{q}s} \equiv \nabla_{\mathbf{q}} \omega_{\mathbf{q}s} \tag{D.87}$$

However, this is not practical, as it is difficult to take gradient in reciprocal space directly. The eigenvalue equation of the dynamical operator in $\mathbf{q}$-subspace is,

$$\bar{D}(\mathbf{q})|\mathbf{q}s\rangle = \omega_{\mathbf{q}s}^2 |\mathbf{q}s\rangle \tag{D.88}$$

Now differentiate both sides,

$$\frac{\partial}{\partial q_\alpha}\Big(\bar{D}(\mathbf{q})|\mathbf{q}s\rangle\Big) = \frac{\partial}{\partial q_\alpha}\Big(\omega_{\mathbf{q}s}^2|\mathbf{q}s\rangle\Big) \tag{D.89}$$

$$\frac{\partial \bar{D}(\mathbf{q})}{\partial q_\alpha}|\mathbf{q}s\rangle + \bar{D}(\mathbf{q})\frac{\partial|\mathbf{q}s\rangle}{\partial q_\alpha} = 2\omega_{\mathbf{q}s}\frac{\partial \omega_{\mathbf{q}s}}{\partial q_\alpha}|\mathbf{q}s\rangle + \omega_{\mathbf{q}s}^2\frac{\partial|\mathbf{q}s\rangle}{\partial q_\alpha} \tag{D.90}$$

project onto $\langle \mathbf{q}s|$ yields,

$$\langle \mathbf{q}s|\frac{\partial \bar{D}(\mathbf{q})}{\partial q_\alpha}|\mathbf{q}s\rangle + \langle \mathbf{q}s|\bar{D}(\mathbf{q})\frac{\partial|\mathbf{q}s\rangle}{\partial q_\alpha} = 2\omega_{\mathbf{q}s}\frac{\partial \omega_{\mathbf{q}s}}{\partial q_\alpha}\langle \mathbf{q}s|\mathbf{q}s\rangle + \omega_{\mathbf{q}s}^2\langle \mathbf{q}s|\frac{\partial|\mathbf{q}s\rangle}{\partial q_\alpha} \tag{D.91}$$

since we have already shown $\bar{D}(\mathbf{q})$ is Hermitian, it can operate back into $\langle \mathbf{q}s|$,

$$\langle \mathbf{q}s|\frac{\partial \bar{D}(\mathbf{q})}{\partial q_\alpha}|\mathbf{q}s\rangle + \omega_{\mathbf{q}s}^2\langle \mathbf{q}s|\frac{\partial|\mathbf{q}s\rangle}{\partial q_\alpha} = 2\omega_{\mathbf{q}s}\frac{\partial \omega_{\mathbf{q}s}}{\partial q_\alpha} + \omega_{\mathbf{q}s}^2\langle \mathbf{q}s|\frac{\partial|\mathbf{q}s\rangle}{\partial q_\alpha} \tag{D.92}$$

cancel the same term on both sides,

$$\langle \mathbf{q}s|\frac{\partial \bar{D}(\mathbf{q})}{\partial q_\alpha}|\mathbf{q}s\rangle = 2\omega_{\mathbf{q}s}\frac{\partial \omega_{\mathbf{q}s}}{\partial q_\alpha} \tag{D.93}$$

thus,

$$\begin{aligned}
\frac{\partial \omega_{\mathbf{q}s}}{\partial q_{\alpha''}} &= \frac{1}{2\omega_{\mathbf{q}s}}\langle \mathbf{q}s|\frac{\partial \bar{D}(\mathbf{q})}{\partial q_{\alpha''}}|\mathbf{q}s\rangle \\
&= \frac{1}{2\omega_{\mathbf{q}s}}\sum_{\mathbf{q}'b\alpha}\sum_{\mathbf{q}''b'\alpha'}\langle \mathbf{q}s|\mathbf{q}'b\alpha\rangle\langle \mathbf{q}'b\alpha|\frac{\partial \bar{D}(\mathbf{q})}{\partial q_{\alpha''}}|\mathbf{q}''b'\alpha'\rangle\langle \mathbf{q}''b'\alpha'|\mathbf{q}s\rangle \\
&= \frac{1}{2\omega_{\mathbf{q}s}}\sum_{\mathbf{q}'b\alpha}\sum_{\mathbf{q}''b'\alpha'}\delta_{\mathbf{q}\mathbf{q}'}\langle s|b\alpha\rangle\langle \mathbf{q}'b\alpha|\frac{\partial \bar{D}(\mathbf{q})}{\partial q_{\alpha''}}|\mathbf{q}''b'\alpha'\rangle\langle b'\alpha'|s\rangle\delta_{\mathbf{q}''\mathbf{q}} \\
&= \frac{1}{2\omega_{\mathbf{q}s}}\sum_{b\alpha}\sum_{b'\alpha'}\langle s|b\alpha\rangle\langle \mathbf{q}b\alpha|\frac{\partial \bar{D}(\mathbf{q})}{\partial q_{\alpha''}}|\mathbf{q}b'\alpha'\rangle\langle b'\alpha'|s\rangle
\end{aligned} \tag{D.94}$$

then we recognise,

$$\langle b\alpha|s\rangle = \epsilon_{s,b\alpha}(\mathbf{q}) \tag{D.95}$$

thus,

$$v_{s\alpha}(\mathbf{q}) = \frac{1}{2\omega_{\mathbf{q}s}} \sum_{b'\alpha'} \sum_{b''\alpha''} \epsilon^*_{s,b'\alpha'}(\mathbf{q}) \frac{\partial D_{b'\alpha',b''\alpha''}(\mathbf{q})}{\partial q_\alpha} \epsilon_{s,b''\alpha''}(\mathbf{q}) \tag{D.96}$$

which is the expression we used in theory section. This also suggests we should define the group velocity operator in mode space as,

$$\tilde{\mathbf{v}} = \frac{1}{2}\tilde{\omega}^{-1/2}\Big[\nabla_{\mathbf{q}}\bar{D}(\mathbf{q})\Big]\tilde{\omega}^{-1/2} \tag{D.97}$$

where the zero frequency modes are excluded.

E
Scattering Matrix

In this section, we shall derive the explicit expression for the phonon scattering matrix. The scattering matrix Ω is obtained from the anharmonic part V' of the crystal Hamiltonian. We shall only consider three-phonon processes, as it is sufficient for our purpose. Thus, we only keep the leading anharmonic term

$$V' \simeq V^{(3)} = \frac{1}{3!} \sum_{lb\alpha} \sum_{l'b'\alpha'} \sum_{l''b''\alpha''} \Phi^{(3)}_{lb\alpha,l'b'\alpha',l''b''\alpha''} u_{lb\alpha} u_{l'b'\alpha'} u_{l''b''\alpha''}, \tag{E.1}$$

where

$$\Phi^{(3)}_{lb\alpha,l'b'\alpha',l''b''\alpha''} = \left. \frac{\partial^3 V}{\partial u_{lb\alpha} \partial u_{l'b'\alpha'} \partial u_{l''b''\alpha''}} \right|_{u=0}. \tag{E.2}$$

After transforming the displacements into phonon normal coordinates,

$$V^{(3)} = \frac{1}{3!} \sum_{kk'k''} \Phi^{(3)}_{kk'k''} Q_k Q_{k'} Q_{k''} \tag{E.3}$$

where,

$$\Phi^{(3)}_{kk'k''} = \left. \frac{\partial^3 V}{\partial Q_k \partial Q_{k'} \partial Q_{k''}} \right|_{u=0}. \tag{E.4}$$

$$V^{(3)} = \frac{1}{3!} \sum_{kk'k''} \Phi^{(3)}_{kk'k''} \left(\frac{\hbar^3}{8\omega_k \omega_{k'} \omega_{k''}} \right)^{1/2} (a_k + a^\dagger_{-k})(a_{k'} + a^\dagger_{-k'})(a_{k''} + a^\dagger_{-k''}) \tag{E.5}$$

To calculate the transition rate we use Fermi's golden rule,

$$\Gamma_{i \to f} = \frac{2\pi}{\hbar^2} |\langle f | V' | i \rangle|^2 \delta(\omega_f - \omega_i) \tag{E.6}$$

where $|i\rangle$ and $|f\rangle$ are both eigenstates of H_0, and V' is a weak perturbation. Consider a three phonon coalescence process,

$$k + k' \to k'' \tag{E.7}$$

equivalently as $|i\rangle \to |f\rangle$,

$$|N_k N_{k'} N_{k''}\rangle \to |(N_k - 1)(N_{k'} - 1)(N_{k''} + 1)\rangle \tag{E.8}$$

clearly, the only non-vanishing matrix element is,

$$|\langle f|\hat{a}_k \hat{a}_{k'} \hat{a}^\dagger_{k''}|i\rangle|^2 = N_k N_{k'}(N_{k''} + 1) \tag{E.9}$$

there are 3! permutations, the sum over $(k, k', -k'')$ is the same as the sum over $(k', k, -k'')$ and so forth, this exactly cancels with the prefactor of 1/3!. Averaging gives,

$$\langle N_k N_{k'}(N_{k''} + 1)\rangle = n_k n_{k'}(n_{k''} + 1) \tag{E.10}$$

where the average is done in the harmonic approximation, so there is no correlation between different normal modes. Thus,

$$\langle |\langle f|V^{(3)}|i\rangle|^2\rangle = |\Phi^{(3)}_{kk'-k''}|^2 \frac{\hbar^3}{8\omega_k\omega_{k'}\omega_{k''}} n_k n_{k'}(n_{k''} + 1) \tag{E.11}$$

and the scattering rate $W^{k''}_{kk'}$ is,

$$W^{k''}_{kk'} = \frac{\pi\hbar}{4\omega_k\omega_{k'}\omega_{k''}} |\Phi^{(3)}_{kk'-k''}|^2 n_k n_{k'}(n_{k''} + 1)\delta(\omega_{k''} - \omega_{k'} - \omega_k) \tag{E.12}$$

similarly, the rate for the reverse $k'' \to k + k'$ is,

$$W^{kk'}_{k''} = \frac{\pi\hbar}{4\omega_k\omega_{k'}\omega_{k''}} |\Phi^{(3)}_{-k-k'k''}|^2 (n_k + 1)(n_{k'} + 1)n_{k''}\delta(\omega_{k''} - \omega_{k'} - \omega_k) \tag{E.13}$$

clearly,

$$(\Phi^{(3)}_{kk'-k''})^* = \Phi^{(3)}_{-k-k'k''} \tag{E.14}$$

thus,

$$W^{kk'}_{k''} = \frac{\pi\hbar}{4\omega_k\omega_{k'}\omega_{k''}} |\Phi^{(3)}_{kk'-k''}|^2 (n_k + 1)(n_{k'} + 1)n_{k''}\delta(\omega_{k''} - \omega_{k'} - \omega_k) \tag{E.15}$$

so the net rate of destruction of phonon k, k' and creation of phonon k'' is,

$$R_{kk'}^{k''} = W_{kk'}^{k''} - W_{k''}^{kk'} \tag{E.16}$$

Define

$$F_{kk'k''}^{(3)} \equiv \left(\frac{\hbar^3}{8\omega_k\omega_{k'}\omega_{k''}}\right)^{1/2} \Phi_{kk'k''}^{(3)} \tag{E.17}$$

and,

$$\mathcal{G}_{kk'}^{k''} \equiv \frac{2\pi}{\hbar^2}\left|F_{kk'-k''}^{(3)}\right|^2 \delta(\omega_{k''} - \omega_k - \omega_{k'}). \tag{E.18}$$

Then the net rate for the process $k + k' \rightleftharpoons k''$ is

$$R_{kk'}^{k''} = \mathcal{G}_{kk'}^{k''}\Big[n_k n_{k'}(n_{k''}+1) - (n_k+1)(n_{k'}+1)n_{k''}\Big]. \tag{E.19}$$

At equilibrium,

$$\frac{n_k^0+1}{n_k^0} = e^{\hbar\omega_k/k_BT} \tag{E.20}$$

phonon scattering conserves energy,

$$\hbar\omega_k + \hbar\omega_{k'} = \hbar\omega_{k''} \tag{E.21}$$

$$e^{\hbar\omega_k/k_BT}e^{\hbar\omega_{k'}/k_BT} = e^{\hbar\omega_{k''}/k_BT} \tag{E.22}$$

$$\frac{n_k^0+1}{n_k^0}\frac{n_{k'}^0+1}{n_{k'}^0} = \frac{n_{k''}^0+1}{n_{k''}^0} \tag{E.23}$$

$$(n_k^0+1)(n_{k'}^0+1)n_{k''}^0 = n_k^0 n_{k'}^0(n_{k''}^0+1) \tag{E.24}$$

thus at equilibrium,

$$R_{kk'}^{k''} = 0 \tag{E.25}$$

For a linear deviation of phonon occupation number from local thermal equilibrium, we have

$$n_k = n_k^0 + \delta n_k \tag{E.26}$$

and keep only terms linear in δn of $n_k n_{k'}(n_{k''}+1)$, we have,

$$n^0_{k'}(n^0_{k''}+1)\delta n_k + n^0_k(n^0_{k''}+1)\delta n_{k'} + n^0_k n^0_{k'}\delta n_{k''} \tag{E.27}$$

and for $k'' \rightarrow k + k'$,

$$(n^0_{k'}+1)n^0_{k''}\delta n_k + (n^0_k+1)n^0_{k''}\delta n_{k'} + (n^0_k+1)(n^0_{k'}+1)\delta n_{k''} \tag{E.28}$$

Expanding the net rate gives,

$$R^{k''}_{kk'} = \mathcal{G}^{k''}_{kk'}\Big[(n^0_{k'}-n^0_{k''})\delta n_k + (n^0_k - n^0_{k''})\delta n_{k'} - (1+n^0_k+n^0_{k'})\delta n_{k''}\Big]. \tag{E.29}$$

Introduce the scaled deviation

$$\delta\hat{n}_k \equiv \frac{\delta n_k}{n^0_k(n^0_k+1)}. \tag{E.30}$$

Thus,

$$R^{k''}_{kk'} = \mathcal{G}^{k''}_{kk'}\Big[(n^0_{k'}-n^0_{k''})n^0_k(n^0_k+1)\delta\hat{n}_k + (n^0_k-n^0_{k''})n^0_{k'}(n^0_{k'}+1)\delta\hat{n}_{k'} - (1+n^0_k+n^0_{k'})n^0_{k''}(n^0_{k''}+1)\delta\hat{n}_{k''}\Big]. \tag{E.31}$$

Now introduce the new notation to reduce clutter,

$$A \equiv n^0_k, \qquad B \equiv n^0_{k'}, \qquad C \equiv n^0_{k''}, \tag{E.32}$$

so Eq. (E.24) becomes,

$$(A+1)(B+1)C = AB(C+1) \tag{E.33}$$

expand and cancel the common ABC term,

$$AC + BC + C = AB \tag{E.34}$$

$$(1+A+B)C = AB \tag{E.35}$$

we may also have,

$$AB - AC = CB + C \tag{E.36}$$

$$A(B-C) = C(B+1). \tag{E.37}$$

Similarly,

$$BA - BC = CA + C \tag{E.38}$$

$$B(A-C) = C(A+1), \tag{E.39}$$

The first term in the bracket of Eq. (E.31) is,

$$(B-C)A(A+1) = (A+1)(B+1)C = AB(C+1) \tag{E.40}$$

where we used Eq. (E.37) for the first equality then use Eq. (E.33) for the second equality. Now we move onto the second term in the bracket of Eq. (E.31),

$$(A-C)B(B+1) = (A+1)(B+1)C = AB(C+1) \tag{E.41}$$

where we used Eq. (E.39) for the first equality then use Eq. (E.33) for the second equality. Now we move onto the third term in the bracket of Eq. (E.31),

$$-(1+A+B)C(C+1) = -AB(C+1) \tag{E.42}$$

where we used Eq. (E.35). Therefore, by combining these three results, we have,

$$R^{k''}_{kk'} = \mathcal{P}^{k''}_{kk'} \left(\delta\hat{n}_k + \delta\hat{n}_{k'} - \delta\hat{n}_{k''}\right), \tag{E.43}$$

where

$$\mathcal{P}^{k''}_{kk'} \equiv \mathcal{G}^{k''}_{kk'} n^0_k n^0_{k'} (n^0_{k''} + 1). \tag{E.44}$$

$$\mathcal{P}^{k''}_{kk'} = \frac{2\pi}{\hbar^2} \left|F^{(3)}_{kk'-k''}\right|^2 n^0_k n^0_{k'} (n^0_{k''} + 1)\delta(\omega_{k''} - \omega_k - \omega_{k'}). \tag{E.45}$$

The collision integral for mode k receives negative contributions from processes in which k is destroyed, and positive contributions from processes in which k is created. Therefore

$$\left(\frac{\partial n_k}{\partial t}\right)_{\text{scat}} = -\sum_{k'k''} R^{k''}_{kk'} + \frac{1}{2}\sum_{k'k''} R^{k}_{k'k''}. \tag{E.46}$$

The factor 1/2 avoids double counting the pair k' and k'' in the creation process $k' + k'' \to k$. Substituting the linearised form of R, we obtain

$$\left(\frac{\partial n_k}{\partial t}\right)_{\text{scat}} = -\sum_{k'k''} \mathcal{P}^{k''}_{kk'} \left(\delta\hat{n}_k + \delta\hat{n}_{k'} - \delta\hat{n}_{k''}\right) + \frac{1}{2}\sum_{k'k''} \mathcal{P}^{k}_{k'k''} \left(\delta\hat{n}_{k'} + \delta\hat{n}_{k''} - \delta\hat{n}_k\right) \tag{E.47}$$

We now introduce the scaled collision operator $\hat{\Omega}$ by

$$\left(\frac{\partial n_k}{\partial t}\right)_{\text{scat}} = -\sum_{k'} \hat{\Omega}_{kk'} \delta\hat{n}_{k'} \tag{E.48}$$

Then,

$$-\sum_{k'} \hat{\Omega}_{kk'} \delta\hat{n}_{k'} = -\sum_{k'k''} \mathcal{P}^{k''}_{kk'} \left(\delta\hat{n}_k + \delta\hat{n}_{k'} - \delta\hat{n}_{k''}\right) + \frac{1}{2}\sum_{k'k''} \mathcal{P}^{k}_{k'k''} \left(\delta\hat{n}_{k'} + \delta\hat{n}_{k''} - \delta\hat{n}_k\right) \tag{E.49}$$

$$
\begin{aligned}
\sum_{k'}\hat{\Omega}_{kk'}\delta\hat{n}_{k'} &= \sum_{k'k''}\mathcal{P}^{k''}_{kk'}\delta\hat{n}_{k} + \sum_{k'k''}\mathcal{P}^{k''}_{kk'}\delta\hat{n}_{k'} - \sum_{k'k''}\mathcal{P}^{k''}_{kk'}\delta\hat{n}_{k''} \\
&\quad - \frac{1}{2}\sum_{k'k''}\mathcal{P}^{k}_{k'k''}\delta\hat{n}_{k'} - \frac{1}{2}\sum_{k'k''}\mathcal{P}^{k}_{k'k''}\delta\hat{n}_{k''} + \frac{1}{2}\sum_{k'k''}\mathcal{P}^{k}_{k'k''}\delta\hat{n}_{k} \\
&= \sum_{ab}\mathcal{P}^{b}_{ka}\delta\hat{n}_{k} + \sum_{ab}\mathcal{P}^{b}_{ka}\delta\hat{n}_{a} - \sum_{ab}\mathcal{P}^{b}_{ka}\delta\hat{n}_{b} \\
&\quad - \frac{1}{2}\sum_{ab}\mathcal{P}^{k}_{ab}\delta\hat{n}_{a} - \frac{1}{2}\sum_{ab}\mathcal{P}^{k}_{ab}\delta\hat{n}_{b} + \frac{1}{2}\sum_{ab}\mathcal{P}^{k}_{ab}\delta\hat{n}_{k} \\
&= \sum_{ab}\mathcal{P}^{b}_{ka}\delta\hat{n}_{k} + \sum_{k'b}\mathcal{P}^{b}_{kk'}\delta\hat{n}_{k'} - \sum_{ak'}\mathcal{P}^{k'}_{ka}\delta\hat{n}_{k'} \\
&\quad - \frac{1}{2}\sum_{k'b}\mathcal{P}^{k}_{k'b}\delta\hat{n}_{k'} - \frac{1}{2}\sum_{ak'}\mathcal{P}^{k}_{ak'}\delta\hat{n}_{k'} + \frac{1}{2}\sum_{ab}\mathcal{P}^{k}_{ab}\delta\hat{n}_{k} \\
&= \sum_{ab}\mathcal{P}^{b}_{ka}\delta\hat{n}_{k} + \sum_{k'b}\mathcal{P}^{b}_{kk'}\delta\hat{n}_{k'} - \sum_{ak'}\mathcal{P}^{k'}_{ka}\delta\hat{n}_{k'} \\
&\quad - \frac{1}{2}\sum_{k'b}\mathcal{P}^{k}_{k'b}\delta\hat{n}_{k'} - \frac{1}{2}\sum_{bk'}\mathcal{P}^{k}_{bk'}\delta\hat{n}_{k'} + \frac{1}{2}\sum_{ab}\mathcal{P}^{k}_{ab}\delta\hat{n}_{k} \\
&= \sum_{ab}\mathcal{P}^{b}_{ka}\delta\hat{n}_{k} + \sum_{k'b}\mathcal{P}^{b}_{kk'}\delta\hat{n}_{k'} - \sum_{ak'}\mathcal{P}^{k'}_{ka}\delta\hat{n}_{k'} - \sum_{k'b}\mathcal{P}^{k}_{k'b}\delta\hat{n}_{k'} + \frac{1}{2}\sum_{ab}\mathcal{P}^{k}_{ab}\delta\hat{n}_{k} \\
&= \sum_{ab}\mathcal{P}^{b}_{ka}\delta\hat{n}_{k} + \frac{1}{2}\sum_{ab}\mathcal{P}^{k}_{ab}\delta\hat{n}_{k} + \sum_{k'b}\mathcal{P}^{b}_{kk'}\delta\hat{n}_{k'} - \sum_{ak'}\mathcal{P}^{k'}_{ka}\delta\hat{n}_{k'} - \sum_{k'b}\mathcal{P}^{k}_{k'b}\delta\hat{n}_{k'} \\
&= \sum_{ab}\mathcal{P}^{b}_{ka}\delta\hat{n}_{k} + \frac{1}{2}\sum_{ab}\mathcal{P}^{k}_{ab}\delta\hat{n}_{k} + \sum_{k'a}\mathcal{P}^{a}_{kk'}\delta\hat{n}_{k'} - \sum_{k'a}\mathcal{P}^{k'}_{ka}\delta\hat{n}_{k'} - \sum_{k'a}\mathcal{P}^{k}_{k'a}\delta\hat{n}_{k'} \\
&= \sum_{ab}\mathcal{P}^{b}_{ka}\delta\hat{n}_{k} + \frac{1}{2}\sum_{ab}\mathcal{P}^{k}_{ab}\delta\hat{n}_{k} + \sum_{k'}\sum_{k''}\left(\mathcal{P}^{k''}_{kk'} - \mathcal{P}^{k'}_{kk''} - \mathcal{P}^{k}_{k'k''}\right)\delta\hat{n}_{k'} \\
&= \sum_{ab}\left(\mathcal{P}^{b}_{ka} + \frac{1}{2}\mathcal{P}^{k}_{ab}\right)\delta\hat{n}_{k} + \sum_{k'}\sum_{k''}\left(\mathcal{P}^{k''}_{kk'} - \mathcal{P}^{k'}_{kk''} - \mathcal{P}^{k}_{k'k''}\right)\delta\hat{n}_{k'} \\
&= \sum_{k'}\sum_{ab}\left(\mathcal{P}^{b}_{ka} + \frac{1}{2}\mathcal{P}^{k}_{ab}\right)\delta_{kk'}\delta\hat{n}_{k'} + \sum_{k'}\sum_{k''}\left(\mathcal{P}^{k''}_{kk'} - \mathcal{P}^{k'}_{kk''} - \mathcal{P}^{k}_{k'k''}\right)\delta\hat{n}_{k'} \\
&= \sum_{k'}\left[\sum_{ab}\left(\mathcal{P}^{b}_{ka} + \frac{1}{2}\mathcal{P}^{k}_{ab}\right)\delta_{kk'} + \sum_{k''}\left(\mathcal{P}^{k''}_{kk'} - \mathcal{P}^{k'}_{kk''} - \mathcal{P}^{k}_{k'k''}\right)\right]\delta\hat{n}_{k'} \\
&= \sum_{k'}\left[\sum_{k''k'''}\left(\mathcal{P}^{k''}_{kk'''} + \frac{1}{2}\mathcal{P}^{k}_{k''k'''}\right)\delta_{kk'} + \sum_{k''}\left(\mathcal{P}^{k''}_{kk'} - \mathcal{P}^{k'}_{kk''} - \mathcal{P}^{k}_{k'k''}\right)\right]\delta\hat{n}_{k'}
\end{aligned}
\tag{E.50}
$$

Then by comparing coefficients of $\delta\hat{n}_{k'}$, we have,

$$
\hat{\Omega}_{kk'} = \sum_{k''k'''}\left(\mathcal{P}^{k''}_{kk'''} + \frac{1}{2}\mathcal{P}^{k}_{k''k'''}\right)\delta_{kk'} + \sum_{k''}\left(\mathcal{P}^{k''}_{kk'} - \mathcal{P}^{k'}_{kk''} - \mathcal{P}^{k}_{k'k''}\right) \tag{E.51}
$$

Finally, since

$$
\sum_{k'}\hat{\Omega}_{kk'}\delta\hat{n}_{k'} = \sum_{k'}\Omega_{kk'}\delta n_{k'} = \sum_{k'}\Omega_{kk'}n^{0}_{k'}(n^{0}_{k'}+1)\delta\hat{n}_{k'} \tag{E.52}
$$

we obtain,

$$\Omega_{kk'} = \frac{1}{n^0_{k'}(n^0_{k'}+1)}\left[\sum_{k''k'''}\left(\mathcal{P}^{k''}_{kk'''} + \frac{1}{2}\mathcal{P}^{k}_{k''k'''}\right)\delta_{kk'} + \sum_{k''}\left(\mathcal{P}^{k''}_{kk'} - \mathcal{P}^{k'}_{kk''} - \mathcal{P}^{k}_{k'k''}\right)\right]. \tag{E.53}$$

F

Boltzmann Transport Equation

I

The linearised Boltzmann transport equation (LBTE) under steady state in phonon representation is,

$$\frac{\partial n_k^0}{\partial T}\mathbf{v}_k \cdot \nabla T = -\sum_{k'} \Omega_{kk'} \delta n_{k'} \tag{F.1}$$

The symbols have their usual meaning, the scattering matrix and phonon deviation population number can be symmetrised via,

$$\tilde{\Omega}_{kk'} = \Omega_{kk'} \frac{\sqrt{n_{k'}^0(n_{k'}^0+1)}}{\sqrt{n_k^0(n_k^0+1)}} \tag{F.2}$$

$$\delta\tilde{n}_k = \frac{\delta n_k}{\sqrt{n_k^0(n_k^0+1)}} \tag{F.3}$$

the divide both sides of LBTE by $\sqrt{n_k^0(n_k^0+1)}$ yields,

$$\frac{1}{\sqrt{n_k^0(n_k^0+1)}}\frac{\partial n_k^0}{\partial T}\mathbf{v}_k \cdot \nabla T = -\sum_{k'} \frac{1}{\sqrt{n_k^0(n_k^0+1)}} \Omega_{kk'} \frac{\sqrt{n_{k'}^0(n_{k'}^0+1)}}{\sqrt{n_{k'}^0(n_{k'}^0+1)}} \delta n_{k'} \tag{F.4}$$

$$\frac{1}{\sqrt{n_k^0(n_k^0+1)}}\frac{\partial n_k^0}{\partial T}\mathbf{v}_k \cdot \nabla T = -\sum_{k'} \tilde{\Omega}_{kk'} \delta\tilde{n}_{k'} \tag{F.5}$$

use,

$$C_k = \hbar\omega_k \frac{\partial n_k^0}{\partial T} \tag{F.6}$$

to obtain,

$$\frac{C_k}{\hbar\omega_k\sqrt{n_k^0(n_k^0+1)}}\mathbf{v}_k\cdot\nabla T=-\sum_{k'}\tilde{\Omega}_{kk'}\delta\tilde{n}_{k'} \tag{F.7}$$

use,

$$C_k=k_B\left(\frac{\hbar\omega_k}{k_BT}\right)^2 n_k^0(n_k^0+1) \tag{F.8}$$

$$\sqrt{C_k}=\sqrt{k_B}\left(\frac{\hbar\omega_k}{k_BT}\right)\sqrt{n_k^0(n_k^0+1)} \tag{F.9}$$

therefore,

$$\frac{C_k}{\sqrt{k_BT^2}\sqrt{C_k}}\mathbf{v}_k\cdot\nabla T=-\sum_{k'}\tilde{\Omega}_{kk'}\delta\tilde{n}_{k'} \tag{F.10}$$

$$\sqrt{\frac{C}{k_BT^2}}\sqrt{\frac{C_k}{C}}\mathbf{v}_k\cdot\nabla T=-\sum_{k'}\tilde{\Omega}_{kk'}\delta\tilde{n}_{k'} \tag{F.11}$$

finally introduce,

$$\theta_k^0=\sqrt{\frac{C_k}{C}} \tag{F.12}$$

yields,

$$\sqrt{\frac{C}{k_BT^2}}\theta_k^0\mathbf{v}_k\cdot\nabla T=-\sum_{k'}\tilde{\Omega}_{kk'}\delta\tilde{n}_{k'} \tag{F.13}$$

Since, the symmetrised scattering matrix is real and symmetric, its eigenvectors form a complete set in the mode space, we proceed by borrowing Dirac's bra-ket, using α as relaxon mode index, we have,

$$\tilde{1}=\sum_k|k\rangle\langle k|=\sum_\alpha|\alpha\rangle\langle\alpha| \tag{F.14}$$

which just means both phonon and relaxon modes forms a complete set within linearised scattering space, furthermore, they are orthonormal,

$$\langle k|k'\rangle=\delta_{kk'} \tag{F.15}$$

$$\langle\alpha|\alpha'\rangle=\delta_{\alpha\alpha'} \tag{F.16}$$

which allows us to expand the group velocity operator and the scattering operator in their spectral representation,

$$\tilde{\mathbf{v}} = \sum_k |k\rangle \mathbf{v}_k \langle k| \tag{F.17}$$

$$\tilde{\Omega} = \sum_\alpha |\alpha\rangle \tau_\alpha^{-1} \langle \alpha| \tag{F.18}$$

we also recognise,

$$\theta_k^0 = \langle k|0\rangle \tag{F.19}$$

which is real $\langle 0|k\rangle = \langle k|0\rangle^*$, and normalised $\langle 0|0\rangle = 1$, and,

$$\delta\tilde{n}_k = \langle k|\delta\rangle \tag{F.20}$$

then we have,

$$\sqrt{\frac{C}{k_B T^2}} \langle k|\tilde{\mathbf{v}}|0\rangle \cdot \nabla T = -\sum_{k'} \langle k|\tilde{\Omega}|k'\rangle \langle k'|\delta\rangle \tag{F.21}$$

take out the unit operator and de-project out of k-representation yields LBTE in its general form,

$$\sqrt{\frac{C}{k_B T^2}} \tilde{\mathbf{v}}|0\rangle \cdot \nabla T = -\tilde{\Omega}|\delta\rangle \tag{F.22}$$

which can be re-projected into the relaxon representation,

$$\sqrt{\frac{C}{k_B T^2}} \langle \alpha|\tilde{\mathbf{v}}|0\rangle \cdot \nabla T = -\sum_{\alpha'} \langle \alpha|\tilde{\Omega}|\alpha'\rangle \langle \alpha'|\delta\rangle \tag{F.23}$$

using the relaxon velocity,

$$\mathbf{V}_\alpha \equiv \langle \alpha|\tilde{\mathbf{v}}|0\rangle \tag{F.24}$$

and relaxon population number,

$$f_\alpha \equiv \langle \alpha|\delta\rangle \tag{F.25}$$

yields LBTE in the relaxon representation,

$$\sqrt{\frac{C}{k_B T^2}} \mathbf{V}_\alpha \cdot \nabla T = -\frac{f_\alpha}{\tau_\alpha} \tag{F.26}$$

II

We are now in a position for finding an expression for thermal conductivity. The harmonic heat flux density in k representation is,

$$\begin{aligned}
\mathbf{J} &= \frac{1}{\mathcal{V}}\sum_k \hbar\omega_k \mathbf{v}_k \delta n_k \\
&= \frac{1}{\mathcal{V}}\sum_k \hbar\omega_k \mathbf{v}_k \sqrt{n_k^0(n_k^0+1)}\delta\tilde{n}_k \\
&= \frac{1}{\mathcal{V}}\frac{\sqrt{k_BT^2}}{\sqrt{k_BT^2}}\sum_k \hbar\omega_k \mathbf{v}_k \sqrt{n_k^0(n_k^0+1)}\delta\tilde{n}_k \\
&= \frac{1}{\mathcal{V}}\sqrt{k_BT^2}\sum_k \frac{\hbar\omega_k}{\sqrt{k_BT^2}}\sqrt{n_k^0(n_k^0+1)}\mathbf{v}_k\delta\tilde{n}_k \\
&= \frac{1}{\mathcal{V}}\sqrt{k_BT^2}\sum_k \sqrt{k_B}\frac{\hbar\omega_k}{k_BT}\sqrt{n_k^0(n_k^0+1)}\mathbf{v}_k\delta\tilde{n}_k \\
&= \frac{1}{\mathcal{V}}\sqrt{k_BT^2}\sum_k \sqrt{C_k}\mathbf{v}_k\delta\tilde{n}_k \\
&= \frac{C}{\mathcal{V}}\frac{1}{\sqrt{C^2}}\sqrt{k_BT^2}\sum_k \sqrt{C_k}\mathbf{v}_k\delta\tilde{n}_k \\
&= \frac{C}{\mathcal{V}}\sqrt{\frac{k_BT^2}{C}}\sum_k \sqrt{\frac{C_k}{C}}\mathbf{v}_k\delta\tilde{n}_k \\
&= \frac{C}{\mathcal{V}}\sqrt{\frac{k_BT^2}{C}}\sum_k \theta_k^0\mathbf{v}_k\delta\tilde{n}_k \\
&= \frac{C}{\mathcal{V}}\sqrt{\frac{k_BT^2}{C}}\sum_k \langle 0|\tilde{\mathbf{v}}|k\rangle\langle k|\delta\rangle \\
&= \frac{C}{\mathcal{V}}\sqrt{\frac{k_BT^2}{C}}\langle 0|\tilde{\mathbf{v}}|\delta\rangle \\
&= c\sqrt{\frac{k_BT^2}{C}}\langle 0|\tilde{\mathbf{v}}|\delta\rangle
\end{aligned} \tag{F.27}$$

where $c = C/\mathcal{V}$. We introduce the response vector $\mathbf{F}$ as

$$|\delta\rangle = |\mathbf{F}\rangle\cdot\nabla T \tag{F.28}$$

Substitute into heat flux expression yields,

$$\mathbf{J} = c\sqrt{\frac{k_BT^2}{C}}\langle 0|\tilde{\mathbf{v}}|\mathbf{F}^\intercal\rangle\nabla T \tag{F.29}$$

Where $\mathbf{v}$ is understood to be a column vector, $\mathbf{F}$ as a row vector and ∇T as a column vector. Comparison with Fourier's law,

$$\mathbf{J} = -\boldsymbol{\kappa} \cdot \nabla T \tag{F.30}$$

yields an expression for thermal conductivity,

$$\boldsymbol{\kappa} = -c\sqrt{\frac{k_B T^2}{C}}\langle 0|\tilde{\mathbf{v}}|\mathbf{F}^\intercal\rangle \tag{F.31}$$

substitute response vector into LBTE yields,

$$\sqrt{\frac{C}{k_B T^2}}\tilde{\mathbf{v}}|0\rangle = -\tilde{\Omega}|\mathbf{F}\rangle \tag{F.32}$$

after a slight rearrangement,

$$|\mathbf{F}\rangle = -\sqrt{\frac{C}{k_B T^2}}\tilde{\Omega}^{-1}\tilde{\mathbf{v}}|0\rangle \tag{F.33}$$

However, there is one loose end we need to tie up, in the next appendix F.III, we will show,

$$\tilde{\Omega}|0\rangle = 0 \tag{F.34}$$

due to the conservation of energy for all scattering process. This flags up a serious issue, the true inverse $\tilde{\Omega}^{-1}$ is not defined on the zero-eigenvalue subspace. We should therefore use Moore-Penrose pseudoinverse $\tilde{\Omega}^{\sim 1}$. Importantly, $\{|\alpha\rangle\}$ remains to be a complete set, but the expansion of the inverse operator in eigenstate must have this problematic state removed. It is now left understood that in our notation $\tilde{\Omega}^{-1} = \tilde{\Omega}^{\sim 1}$ as there is no possibility of ambiguity. Now consider,

$$\langle 0|\tilde{\mathbf{v}}|0\rangle = \sum_k \langle 0|k\rangle \mathbf{v}_k \langle k|0\rangle = \frac{1}{C}\sum_k C_k \mathbf{v}_k = 0 \tag{F.35}$$

which vanishes by symmetry of $C_k = C_{-k}$ but $\mathbf{v}_k = -\mathbf{v}_{-k}$. This is reassuring, because negative thermal conductivity would imply spontaneous heat flow from colder to hotter regions, and violate the second law of thermodynamics. Substitute $|\mathbf{F}\rangle$ back into the expression for thermal conductivity yields,

$$\boldsymbol{\kappa} = c\langle 0|\tilde{\mathbf{v}}\tilde{\Omega}^{-1}\tilde{\mathbf{v}}^\intercal|0\rangle \tag{F.36}$$

which is a rather succinct expression of thermal conductivity. We can then insert the spectral representation of $\tilde{\Omega}^{-1}$,

$$\boldsymbol{\kappa} = c \sum_{\alpha} \langle 0|\tilde{\mathbf{v}}|\alpha\rangle \tau_{\alpha} \langle \alpha|\tilde{\mathbf{v}}^{\intercal}|0\rangle \tag{F.37}$$

then use the definition $\mathbf{V}_{\alpha} \equiv \langle 0|\tilde{\mathbf{v}}|\alpha\rangle = \langle \alpha|\tilde{\mathbf{v}}|0\rangle$ to obtain the thermal conductivity expression in relaxon representation.

$$\boldsymbol{\kappa} = c \sum_{\alpha} \mathbf{V}_{\alpha} \mathbf{V}_{\alpha}^{\intercal} \tau_{\alpha} \tag{F.38}$$

G

Second Law of Thermodynamics

I

In this section we recover the second law of thermodynamics for positive semi-definite thermal conductivity tensor, and its local version for macroscopic heat transfer, independent of phonon picture. Despite the full system is not under thermal equilibrium, but under the assumption that the system is *close to local* thermal equilibrium, the exact meaning of close will be discussed in the next section, then by thermodynamic relation

$$ds(\mathbf{r},t) = \frac{1}{T(\mathbf{r},t)} du(\mathbf{r},t) \tag{G.1}$$

where,

$$\frac{1}{T(\mathbf{r},t)} \equiv \frac{\partial s(\mathbf{r},t)}{\partial u(\mathbf{r},t)} \tag{G.2}$$

then we have,

$$\frac{\partial s(\mathbf{r},t)}{\partial t} = \frac{1}{T(\mathbf{r},t)} \frac{\partial u(\mathbf{r},t)}{\partial t} \tag{G.3}$$

consider,

$$\nabla \cdot \left(\frac{\mathbf{J}}{T}\right) = \frac{1}{T} \nabla \cdot \mathbf{J} - \frac{1}{T^2} \mathbf{J} \cdot \nabla T \tag{G.4}$$

the continuity equation requires local conservation of energy,

$$\frac{\partial u(\mathbf{r},t)}{\partial t}+\nabla\cdot\mathbf{J}=0 \tag{G.5}$$

divide both sides by T yields,

$$\frac{1}{T}\frac{\partial u}{\partial t}+\frac{1}{T}\nabla\cdot\mathbf{J}=0 \tag{G.6}$$

substitute Eq. (G.3) and Eq. (G.4) into Eq. (G.6) yields,

$$\frac{\partial s}{\partial t}+\nabla\cdot\left(\frac{\mathbf{J}}{T}\right)=-\frac{1}{T^2}\mathbf{J}\cdot\nabla T \tag{G.7}$$

We now define,

$$\sigma\equiv\mathbf{J}\cdot\nabla\frac{1}{T}=-\frac{1}{T^2}\mathbf{J}\cdot\nabla T \tag{G.8}$$

Fourier's law is given by,

$$\mathbf{J}=-\boldsymbol{\kappa}\nabla T \tag{G.9}$$

substituting back into the definition of σ yields,

$$\sigma=\frac{1}{T^2}(\boldsymbol{\kappa}\nabla T)\cdot\nabla T=\frac{1}{T^2}\nabla T\cdot(\boldsymbol{\kappa}\nabla T)=\frac{1}{T^2}(\nabla T)^{\intercal}\boldsymbol{\kappa}\nabla T \tag{G.10}$$

then provided that $\boldsymbol{\kappa}$ is positive semidefinite in the sense that,

$$(\nabla T)^{\intercal}\boldsymbol{\kappa}\nabla T\geq 0 \tag{G.11}$$

we have,

$$\sigma\geq 0 \tag{G.12}$$

Now go back to Eq. (G.7) and integrate over the volume of the crystal,

$$\int_\Omega\frac{\partial s}{\partial t}\mathrm{d}V+\int_\Omega\nabla\cdot\frac{\mathbf{J}}{T}\mathrm{d}V=\int_\Omega\sigma\mathrm{d}V \tag{G.13}$$

using the definition for entropy density and divergence theorem yields,

$$\frac{\mathrm{d}S}{\mathrm{d}t}+\int_{\partial\Omega}\frac{\mathbf{J}}{T}\cdot\mathrm{d}\mathbf{S}=\int_\Omega\sigma\mathrm{d}V \tag{G.14}$$

now we use the previous result $\sigma \geq 0$, and move the flux integral to the other side,

$$\mathrm{d}S \geq -\int_{\partial\Omega} \frac{\mathbf{J}(\mathbf{r},t)}{T(\mathbf{r},t)} \cdot \mathrm{d}\mathbf{S}\mathrm{d}t \tag{G.15}$$

then provided that the boundary separating the system and the reservoir, is at a well-defined temperature, we can take it outside the integral, then using the definition of heat flux density, we obtain,

$$\mathrm{d}S \geq \frac{đq}{T} \tag{G.16}$$

which is Clausius inequality. Which is reassuring, because negative thermal conductivity implies heat would spontaneously flow from colder region to hotter region. Thus, thermal conductivity must be positive semidefinite, otherwise we would violate the second law of thermodynamics. Furthermore, like the continuity equation of local conservation of energy as the local form of the first law of thermodynamics, the analogous local form of second law of thermodynamics can be written in the form of,

$$\frac{\partial s}{\partial t} \geq -\nabla \cdot \left(\frac{\mathbf{J}}{T}\right) \tag{G.17}$$

II

In this section we show that, our expression for phonon distribution under the time evolution governed by LBTE for heat conduction is consistent with the second law of thermodynamics, provided that phonon scattering must increase the local entropy density.

First, we need to verify whether the phonon entropy from statistical mechanics expression is consistent with classical thermodynamic definition of entropy, The local entropy density of the phonon population distribution is,

$$s(\mathbf{r},t) = \frac{k_B}{\mathcal{V}} \sum_k \Big((n_k+1)\ln(n_k+1) - n_k \ln n_k \Big) \tag{G.18}$$

at global thermal equilibrium $n_k = n_k^0$, so we have,

$$S^0 = k_B \sum_k \Big((n_k^0+1)\ln(n_k^0+1) - n_k^0 \ln n_k^0 \Big) \tag{G.19}$$

since we have,

$$n_k^0 = \frac{1}{e^{\hbar\omega_k/k_BT} - 1} \tag{G.20}$$

and,

$$n_k^0 + 1 = \frac{e^{\hbar\omega_k/k_BT}}{e^{\hbar\omega_k/k_BT} - 1} \tag{G.21}$$

thus,

$$\frac{n_k^0 + 1}{n_k^0} = e^{\hbar\omega_k/k_BT} \tag{G.22}$$

then taking the differential of global entropy,

$$\begin{aligned}
dS^0 &= k_B \sum_k \frac{\partial}{\partial n_k^0}\Big((n_k^0+1)\ln(n_k^0+1) - n_k^0 \ln n_k^0\Big) dn_k^0 \\
&= k_B \sum_k \Big(\ln(n_k^0+1) + 1 - \ln n_k^0 - 1\Big) dn_k^0 \\
&= k_B \sum_k \ln\Big(\frac{n_k^0+1}{n_k^0}\Big) dn_k^0 \\
&= k_B \sum_k \ln\Big(e^{\hbar\omega_k/k_BT}\Big) dn_k^0 \\
&= k_B \sum_k \frac{\hbar\omega_k}{k_BT} dn_k^0 \\
&= \frac{1}{T}\sum_k \hbar\omega_k dn_k^0 \\
&= \frac{dU^0}{T}
\end{aligned} \tag{G.23}$$

At thermal equilibrium, heat transfer is reversible and since there is no work done, thus we have,

$$\mathrm{d}U^0 = đq_{\mathrm{rev}} \tag{G.24}$$

dropping the superscript 0, as it is left understood, we obtain,

$$\mathrm{d}S = \frac{đq_{\mathrm{rev}}}{T} \tag{G.25}$$

which is exactly thermodynamic definition of entropy. Subtlety, this not only holds for global thermodynamic equilibrium, but also holds for local thermodynamic

equilibrium. For there to be heat transfer, consider now local thermal equilibrium, where the evolution of phonon population is governed by the Boltzmann transport equation (BTE),

$$\frac{\partial n_k(\mathbf{r},t)}{\partial t} + \mathbf{v}_k \cdot \nabla n_k(\mathbf{r},t) = \left(\frac{\partial n_k}{\partial t}\right)_{\text{scat}} \tag{G.26}$$

The time evolution of local entropy density is then,

$$\frac{\partial s}{\partial t} = \frac{k_B}{\mathcal{V}} \sum_k \ln\!\left(\frac{n_k+1}{n_k}\right) \frac{\partial n_k}{\partial t} \tag{G.27}$$

using BTE,

$$\frac{\partial s}{\partial t} + \frac{k_B}{\mathcal{V}} \sum_k \ln\!\left(\frac{n_k+1}{n_k}\right) \mathbf{v}_k \cdot \nabla n_k = \frac{k_B}{\mathcal{V}} \sum_k \ln\!\left(\frac{n_k+1}{n_k}\right) \left(\frac{\partial n_k}{\partial t}\right)_{\text{scat}} \tag{G.28}$$

recognise phonon entropy flux density as,

$$\mathbf{J}_s \equiv \frac{k_B}{\mathcal{V}} \sum_k ((n_k+1)\ln(n_k+1) - n_k \ln n_k)\mathbf{v}_k \tag{G.29}$$

and,

$$\sigma = \left(\frac{\partial s}{\partial t}\right)_{\text{scat}} = \frac{k_B}{\mathcal{V}} \sum_k \ln\!\left(\frac{n_k+1}{n_k}\right) \left(\frac{\partial n_k}{\partial t}\right)_{\text{scat}} \tag{G.30}$$

substitute both back into Eq. (G.28) yields,

$$\frac{\partial s}{\partial t} + \nabla \cdot \mathbf{J}_s = \sigma \tag{G.31}$$

For a system near local thermal equilibrium, we use the approximation

$$n_k(\mathbf{r},t) \simeq n_k^0(T(\mathbf{r},t)) + \delta n_k(\mathbf{r},t) \tag{G.32}$$

where $\delta n_k(\mathbf{r},t)$ is a linear response due to a gentle perturbation ∇T, so that $\delta n_k \sim \nabla T$. Then we can Taylor series expand local entropy density to first order about local thermal equilibrium, for a single mode s_k, where $s = \sum_k s_k$ we have

$$\begin{aligned} s_k &= s_k(n_k^0) + \left.\frac{\partial s_k(n_k)}{\partial n_k}\right|_{n_k=n_k^0} (n_k - n_k^0) + \mathcal{O}(\delta n_k^2) \\ &= s_k(n_k^0) + \frac{k_B}{\mathcal{V}} \ln\!\left(\frac{n_k^0+1}{n_k^0}\right) \delta n_k + \mathcal{O}(\delta n_k^2) \\ &= s_k(n_k^0) + \frac{k_B}{\mathcal{V}} \frac{\hbar\omega_k}{k_B T} \delta n_k + \mathcal{O}(\delta n_k^2) \\ &= s_k(n_k^0) + \frac{1}{\mathcal{V}} \frac{\hbar\omega_k}{T} \delta n_k + \mathcal{O}(\delta n_k^2) \end{aligned} \tag{G.33}$$

thus we have,

$$ds(\mathbf{r},t) = \frac{1}{T(\mathbf{r},t)} du(\mathbf{r},t) \tag{G.34}$$

notice that now, both the local entropy and internal energy is not at local thermal equilibrium, but a linear deviation away from it, and *temperature remains well-defined*,

$$T(\mathbf{r},t) = \frac{\partial u(\mathbf{r},t)}{\partial s(\mathbf{r},t)} \tag{G.35}$$

The entropy flux density is,

$$\mathbf{J}_s = \sum_k s_k(n_k^0)\mathbf{v}_k + \frac{1}{T}\frac{1}{\mathcal{V}}\sum_k \hbar\omega_k \mathbf{v}_k \delta n_k + \sum_k \mathcal{O}(\delta n_k^2) \tag{G.36}$$

then recognise $s_k(n_k^0)$ is even under $k \to -k$, but $\mathbf{v}_k$ is odd under the same operation, thus the first sum vanishes by symmetry, then neglecting $\mathcal{O}(\delta n_k^2)$ yields, and finally recognise the middle sum as the heat flux density leads to,

$$\mathbf{J}_s = \frac{\mathbf{J}}{T} \tag{G.37}$$

which holds provided that the deviation of phonon population from local thermal equilibrium population is *linear*. Equating the microscopic and macroscopic expression of σ, leads to

$$\left(\frac{\partial s}{\partial t}\right)_{\text{scat}} = \mathbf{J} \cdot \nabla \frac{1}{T} \geq 0 \tag{G.38}$$

thus, phonon scattering must increase the local entropy density.

III

In this section, we show that the scattering operator must be positive semi-definite, in order to satisfy the second law.

Begin by considering the following Taylor series expansion up to first order in δn_k,

$$\begin{aligned}\ln\Big(\frac{n_k+1}{n_k}\Big) &= \ln\Big(\frac{n_k^0+1}{n_k^0}\Big) + \frac{\partial}{\partial n_k}\ln\Big(\frac{n_k+1}{n_k}\Big)\bigg|_{n_k=n_k^0}\delta n_k + \mathcal{O}(\delta n_k^2) \\ &= \frac{\hbar\omega_k}{k_BT} + \Big(\frac{1}{n_k^0+1} - \frac{1}{n_k^0}\Big)\delta n_k + \mathcal{O}(\delta n_k^2) \\ &= \frac{\hbar\omega_k}{k_BT} - \frac{\delta n_k}{n_k^0(n_k^0+1)} + \mathcal{O}(\delta n_k^2) \end{aligned} \tag{G.39}$$

then introduce the scattering matrix via,

$$\Big(\frac{\partial n_k}{\partial t}\Big)_{\text{scat}} \simeq -\sum_{k'}\Omega_{kk'}\delta n_{k'} \tag{G.40}$$

thus,

$$
\begin{aligned}
\sigma &= \Big(\frac{\partial s}{\partial t}\Big)_{\text{scat}} \\
&= \frac{k_B}{\mathcal{V}} \sum_k \ln\Big(\frac{n_k+1}{n_k}\Big)\Big(\frac{\partial n_k}{\partial t}\Big)_{\text{scat}} \\
&\simeq -\frac{k_B}{\mathcal{V}} \sum_k \ln\Big(\frac{n_k+1}{n_k}\Big) \sum_{k'} \Omega_{kk'}\delta n_{k'} \\
&\simeq -\frac{k_B}{\mathcal{V}} \sum_k \Big(\frac{\hbar\omega_k}{k_BT} - \frac{\delta n_k}{n_k^0(n_k^0+1)}\Big) \sum_{k'} \Omega_{kk'}\delta n_{k'} \\
&= -\frac{k_B}{\mathcal{V}} \sum_{kk'} \frac{\hbar\omega_k}{k_BT}\Omega_{kk'}\delta n_{k'} + \frac{k_B}{\mathcal{V}} \sum_{kk'} \frac{\delta n_k}{n_k^0(n_k^0+1)}\Omega_{kk'}\delta n_{k'} \\
&= -\frac{k_B}{\mathcal{V}} \sum_{kk'} \frac{\hbar\omega_k}{k_BT}\Omega_{kk'}\delta n_{k'} + \frac{k_B}{\mathcal{V}} \sum_{kk'} \frac{\delta n_k}{\sqrt{n_k^0(n_k^0+1)}}\Omega_{kk'}\frac{\sqrt{n_{k'}^0(n_{k'}^0+1)}}{\sqrt{n_k^0(n_k^0+1)}}\frac{\delta n_{k'}}{\sqrt{n_{k'}^0(n_{k'}^0+1)}} \\
&= -\frac{k_B}{\mathcal{V}} \sum_{kk'} \frac{\hbar\omega_k}{k_BT}\Omega_{kk'}\delta n_{k'} + \frac{k_B}{\mathcal{V}} \sum_{kk'} \delta\tilde{n}_k\tilde{\Omega}_{kk'}\delta\tilde{n}_{k'} \\
&= -\frac{k_B}{\mathcal{V}} \sum_{kk'} \frac{\hbar\omega_k}{k_BT}\Omega_{kk'}\delta n_{k'} + \frac{k_B}{\mathcal{V}} \sum_{kk'} \langle\delta|k\rangle\langle k|\tilde{\Omega}|k'\rangle\langle k'|\delta\rangle \\
&= -\frac{k_B}{\mathcal{V}} \sum_{kk'} \frac{\hbar\omega_k}{k_BT}\Omega_{kk'}\delta n_{k'} + \frac{k_B}{\mathcal{V}} \langle\delta|\tilde{\Omega}|\delta\rangle \\
&= -\frac{k_B}{\mathcal{V}} \sum_{kk'} \frac{\hbar\omega_k\sqrt{n_k^0(n_k^0+1)}}{k_BT}\Omega_{kk'}\frac{\sqrt{n_{k'}^0(n_{k'}^0+1)}}{\sqrt{n_k^0(n_k^0+1)}}\frac{\delta n_{k'}}{\sqrt{n_{k'}^0(n_{k'}^0+1)}} + \frac{k_B}{\mathcal{V}} \langle\delta|\tilde{\Omega}|\delta\rangle \\
&= -\frac{k_B}{\mathcal{V}} \sum_{kk'} \frac{\hbar\omega_k\sqrt{n_k^0(n_k^0+1)}}{k_BT}\tilde{\Omega}_{kk'}\delta\tilde{n}_{k'} + \frac{k_B}{\mathcal{V}} \langle\delta|\tilde{\Omega}|\delta\rangle \\
&= -\frac{k_B}{\mathcal{V}} \sum_{kk'} \frac{\sqrt{C_k}}{\sqrt{k_B}}\tilde{\Omega}_{kk'}\delta\tilde{n}_{k'} + \frac{k_B}{\mathcal{V}} \langle\delta|\tilde{\Omega}|\delta\rangle \\
&= -\frac{k_B}{\mathcal{V}} \sqrt{\frac{C}{k_B}} \sum_{kk'} \sqrt{\frac{C_k}{C}}\tilde{\Omega}_{kk'}\delta\tilde{n}_{k'} + \frac{k_B}{\mathcal{V}} \langle\delta|\tilde{\Omega}|\delta\rangle \\
&= -\frac{k_B}{\mathcal{V}} \sqrt{\frac{C}{k_B}} \sum_{kk'} \theta_k^0\tilde{\Omega}_{kk'}\delta\tilde{n}_{k'} + \frac{k_B}{\mathcal{V}} \langle\delta|\tilde{\Omega}|\delta\rangle \\
&= -\frac{k_B}{\mathcal{V}} \sqrt{\frac{C}{k_B}} \sum_{kk'} \langle 0|k\rangle\langle k|\tilde{\Omega}|k'\rangle\langle k'|\delta\rangle + \frac{k_B}{\mathcal{V}} \langle\delta|\tilde{\Omega}|\delta\rangle \\
&= -\frac{k_B}{\mathcal{V}} \sqrt{\frac{C}{k_B}} \langle 0|\tilde{\Omega}|\delta\rangle + \frac{k_B}{\mathcal{V}} \langle\delta|\tilde{\Omega}|\delta\rangle
\end{aligned}
\tag{G.41}
$$

but from first line we also have,

$$\begin{aligned}\sqrt{\frac{C}{k_B}}\langle 0|\tilde{\Omega}|\delta\rangle &= -\sum_k \frac{\hbar\omega_k}{k_B T}\Big(\frac{\partial n_k}{\partial t}\Big)_{\text{scat}} \\ &= -\frac{1}{k_B T}\frac{\partial}{\partial t}\Big(\sum_k \hbar\omega_k n_k\Big)_{\text{scat}} \\ &= -\frac{1}{k_B T}\Big(\frac{\partial U}{\partial t}\Big)_{\text{scat}} \\ &= 0 \end{aligned} \tag{G.42}$$

where we have used the fact that scattering conserves total energy, then,

$$0 = \langle 0|\tilde{\Omega}|\delta\rangle = \langle \delta|\tilde{\Omega}|0\rangle \tag{G.43}$$

since $|\delta\rangle$ is arbitrary we have,

$$\tilde{\Omega}|0\rangle = 0 \tag{G.44}$$

thus this is a consequence of conservation of energy. We now also have,

$$\sigma = \frac{k_B}{\mathcal{V}}\langle \delta|\tilde{\Omega}|\delta\rangle \geq 0 \tag{G.45}$$

thus, $\tilde{\Omega}$ is positive semi-definite, so that the second law is satisfied. Eq. (G.44) and Eq. (G.45) may be regarded as the first and the second law of thermodynamics respectively, in the linearised scattering space.

H

Symmetry of Thermal Conductivity Tensor

Thermal conductivity is a second-rank tensor. Under an orthogonal transformation represented by the matrix $\mathbf{R}$, its components transform according to

$$\boldsymbol{\kappa}' = \mathbf{R}\boldsymbol{\kappa}\mathbf{R}^{\intercal}. \tag{H.1}$$

If $\mathbf{R}$ is a symmetry operation of the crystal point group, then the material must be indistinguishable before and after the operation. Therefore,

$$\boldsymbol{\kappa} = \mathbf{R}\boldsymbol{\kappa}\mathbf{R}^{\intercal}. \tag{H.2}$$

This is an example of *Neumann's principle* [60]: the symmetry elements of any physical property of a crystal must include the symmetry elements of the crystal itself.

The full symmetry group of a crystal is its space group, which contains both translations and point-group operations. However, the thermal conductivity is a homogeneous bulk property, so a lattice translation cannot change its macroscopic components. The relevant constraints on $\boldsymbol{\kappa}$ therefore come from the point-group part of the space group.

Solid p-H_2 crystallises in the hcp structure, whose point group is D_{6h}. The corresponding Brillouin zone is a hexagonal prism, reflecting the same rotational symmetry.

The components of thermal conductivity tensor transform like quadratic functions. From the character table for D_{6h}, the quadratic functions

$$x^2 + y^2 \tag{H.3}$$

and

$$z^2 \tag{H.4}$$

transform as the totally symmetric irreducible representation A_{1g}. By contrast, the off-diagonal combinations transform as non-totally-symmetric representations. For example, (xz, yz) transform as E_{1g}, while $(x^2 - y^2, xy)$ transform as E_{2g}. These components are therefore forbidden in an invariant tensor. Consequently, the thermal conductivity tensor for hcp solid p-H_2 must have the form

$$\boldsymbol{\kappa} = \begin{pmatrix} \kappa_\perp & 0 & 0 \\ 0 & \kappa_\perp & 0 \\ 0 & 0 & \kappa_\parallel \end{pmatrix}. \tag{H.5}$$

Here $\kappa_\perp$ is the thermal conductivity in the basal plane, and $\kappa_\parallel$ is the thermal conductivity along the crystallographic c-axis.

The same result could also be obtained by applying each point-group operation directly to a general second-rank tensor using Eq. (H.2). Any tensor component that is not invariant under all symmetry operations must vanish, and the sixfold rotational symmetry enforces $\kappa_{xx} = \kappa_{yy}$.

D_{6h} $(6/mmm)$	E	$2C_6$	$2C_3$	C_2	$3C_2'$	$3C_2''$	i	$2S_3$	$2S_6$	σ_h	$3\sigma_d$	$3\sigma_v$		
A_{1g}	1	1	1	1	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	1	1	−1	−1	1	1	1	1	−1	−1	R_z	
B_{1g}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1		
B_{2g}	1	−1	1	−1	−1	1	1	−1	1	−1	−1	1		
E_{1g}	2	1	−1	−2	0	0	2	1	−1	−2	0	0	(R_x, R_y)	(yz, xz)
E_{2g}	2	−1	−1	2	0	0	2	−1	−1	2	0	0		(x^2-y^2, xy)
A_{1u}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1		
A_{2u}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1	z	
B_{1u}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1		
B_{2u}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1		
E_{1u}	2	1	−1	−2	0	0	−2	−1	1	2	0	0	(x, y)	
E_{2u}	2	−1	−1	2	0	0	−2	1	1	−2	0	0		

Figure H.1: The character table for D_{6h} was taken from the Oxford Honour School of Chemistry character tables. [61]